SpectraMaster

منظومة تحليل أطياف أشعة جاما

Gamma Spectrum Analysis System

User Manual

Gamma spectrum analysis system: analysis, calibration, activity, conversion and reports

Version 1.0.0 · Windows

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1. Introduction and installation

1.1 What SpectraMaster does

SpectraMaster is a **gamma spectrum analysis system** for the laboratory, for HPGe and NaI detectors. It analyses spectra (peaks, nuclides, activities and concentrations), calibrates them (energy, resolution, efficiency), keeps the analysis as a traceable session and prints the reports. It also converts spectrum files in the style of ORTEC *DataMaster* (A49-B32), with the same naming rule and the same command line (`-i input -j output` with the DataMaster format numbers), and its menus, keys and workflow follow **Aptec PCMCA/WIN**. In detail:

- **Formats:** reads **59 spectrum formats** (Aptec, ORTEC, Canberra/Genie, IAEA, N42, Excel and many legacy laboratory formats), whatever the file extension, and writes 35 of them. Every written file is read back and any difference is reported.
- **Viewing:** a fast plot with zoom history, an overview strip, ROIs, overlays, stepping through the found peaks and image export.
- **Two analysis methods:** the Aptec method on regions of interest (peak search, Multifit, identify, background subtract, quantify), and SpectraMaster's fitted-peak analysis with nuclide identification from a library built on IAEA ENSDF data.
- **Calibrations:** energy and FWHM calibration from a source, with every line-to-peak assignment shown for review; a calibration drift check; efficiency calibration with independent validation.
- **Activity:** activity and concentration (Bq, Bq/kg, Bq/L, Bq/m^3) with an uncertainty budget, decision threshold and detection limit (Currie or ISO 11929). Corrections: background, reagent blank, reference peak, true-coincidence summing (gamma-gamma and 511 keV annihilation) and self-attenuation.
- **Traceability:** analysis sessions with fingerprints of the files used, an undoable edit history, inspection and manual refit of the peak fits, review, and a JSON report record.
- **Many spectra at once:** batch analysis by measurement group, and reports for many files.
- **Archive search** across tens of thousands of old spectra (text, nuclides, date, detector).
- **Detector quality control** with control charts and Westgard rules (ISO/IEC 17025, 7.7).

The interface is fully bilingual (Arabic and English) and remembers your choices between sessions.

> **Responsibility.** SpectraMaster computes and documents; it does not certify. The results must be reviewed by a qualified analyst and the method validated by the laboratory. The checks against reference data (appendix E) are not a validation or a statement of compliance.

1.2 Installation

SpectraMaster consists of two programs. Python or any other runtime is **not** needed.

SpectraMaster.exe	The graphical program (windows, plots, analysis, reports). Started with the DataMaster arguments `-i in -j out` it converts silently without opening a window.
smconvert.exe	The command-line converter and tool box for batch files, scripts and the Windows task scheduler (chapter 4 and appendix B).

Installer (recommended)

1. Run `SpectraMaster_Setup_1.0.0.exe` (administrator rights are needed) and choose the language of the setup wizard (English or Arabic).
2. Accept the license, keep the folder `C:\Program Files\SpectraMaster` and choose the components: main program, command-line tool `smconvert` (optionally added to the system `PATH`), user manuals and sample spectra.
3. Choose the tasks: desktop icon and the file associations (`.s0 .chn .cnf .spe .spc .iec .n42`, each can be switched off). Associated files open in SpectraMaster with a double-click; all of them also get **Open with › SpectraMaster**.

The Start menu group **SpectraMaster** contains the program, both user manuals, the release notes, the sample spectra folder and the uninstaller. To pin SpectraMaster to the taskbar, right-click its Start-menu entry (or the running program's taskbar button) and choose **Pin to taskbar**. The licences of all bundled components are in the installation folder (`licenses\THIRD_PARTY_NOTICES.txt`).

Uninstall with **Settings › Apps** or the Start-menu entry. The uninstaller asks whether your own data (archive index, efficiency curves, detector profiles, QC tracks in `%APPDATA%\SpectraMaster` and the settings) should be deleted too; the default answer keeps them. It removes only SpectraMaster's own file associations and `PATH` entry.

Portable copy

The two programs are also available as single packed executables: copy them into any folder (even a USB stick) and start `SpectraMaster.exe`; nothing has to be installed. The first start after a reboot can take a few seconds while Windows unpacks the program.

> **Note.** The programs are not code-signed, so Windows SmartScreen may ask for confirmation the first time (**More info › Run anyway**, after checking with your IT department). If Windows still shows the old icon after an update, restart the computer: Windows keeps icons in a cache.

1.3 First start, splash screen and language

At start-up a splash screen shows the logo, the designer and the version for about 1.5 seconds (click it to close it at once). It can be switched off with **Help › Show Splash Screen at Startup**.



Figure 1 — Splash screen.

The program starts in **English** the first time. Switch the language with **View › Language › English / العربية**; the whole interface changes at once (including the right-to-left layout) and the choice is remembered. The spectrum plot always runs left to right (low energy on the left) in both languages.

1.4 Windows and dialogs

- Every window has **minimize, maximize and close** buttons. A window larger than the screen is fitted to the screen when it opens; if its contents cannot shrink that far, they scroll and the last row of buttons (OK, Calculate, Close …) stays visible below.
- The calibration windows (Quick Energy, Full Calibration) and Fit Inspection are **modal**, as in Aptec: while they are open the main window does not respond, so the spectrum cannot change in the middle of a calibration. Batch analysis and batch reports stay modeless because they run for minutes.
- Long operations run in the background with a busy message (analysis, efficiency measurement, activity calculation, calibration proposal, peak-shape measurement, the analysis before a report), so the window keeps responding.

2. The main window

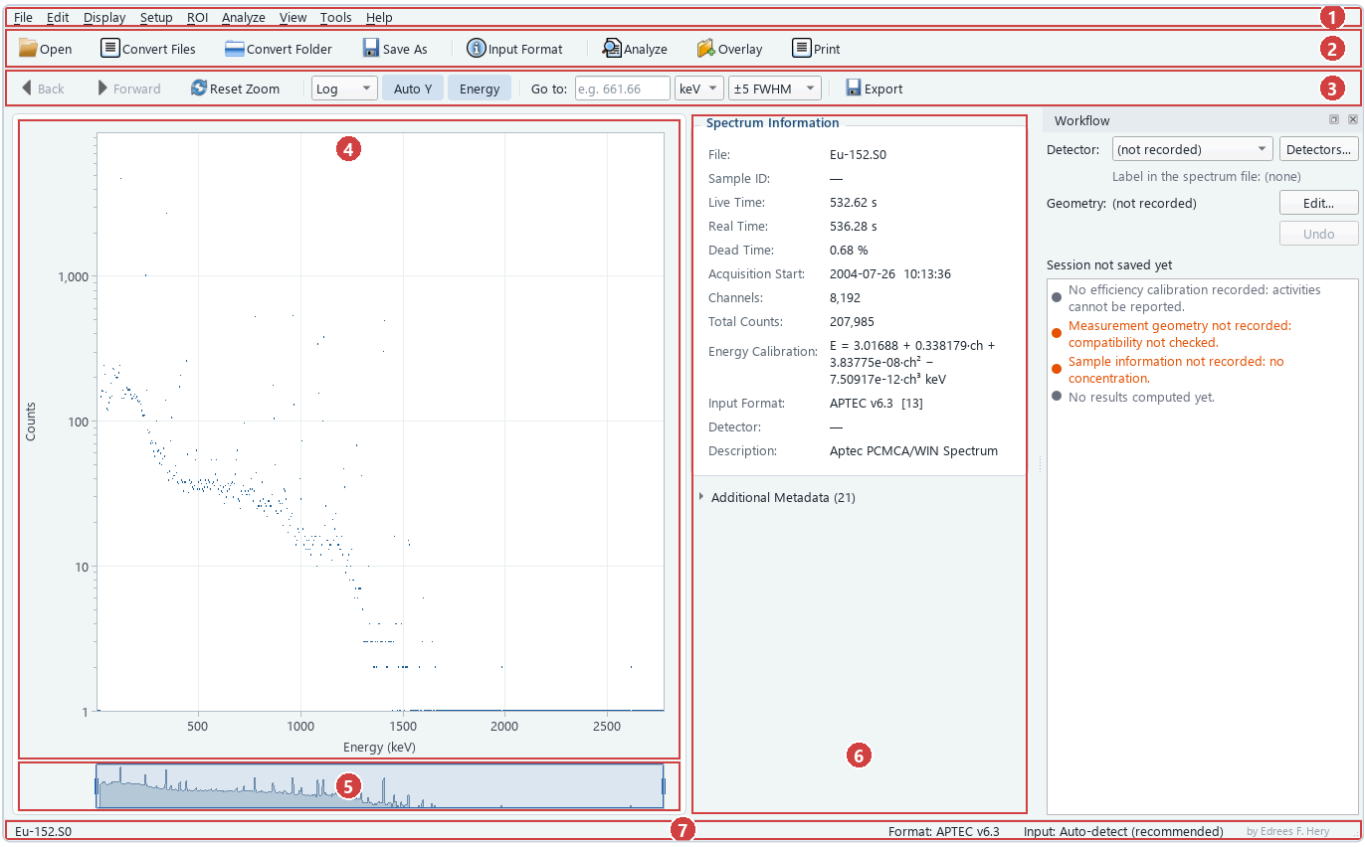


Figure 2 — The main window with a calibration spectrum (Eu-152, Aptec S0) on a logarithmic scale.

1. **Menu bar** — File, Edit, Display, Setup, ROI, Analyze, View, Tools, Help (Aptec's order, chapter 7).
2. **Main toolbar** — Open, Convert Files, Convert Folder, Save As, Input Format, Analyze, Overlay, Print.
3. **Navigation toolbar** — zoom Back / Forward, Reset Zoom, Y scale (Linear / Log / √), Auto Y, Energy axis, *Go to* energy or channel with window width, Export image.
4. **Spectrum plot** — counts versus energy (keV) or channel; mouse read-out in the status bar.
5. **Overview strip** — the whole spectrum with the visible window; drag it to pan.
6. **Spectrum information** — file, sample ID, live/real/dead time, start time, channels, total counts, energy calibration, input format, detector, description and all additional metadata. For files that contain several spectra a selector "Spectrum i of n" appears here.
7. **Status bar** — file path or messages, cursor position (channel, keV, counts), current format, input mode (auto-detect or a fixed format).

Further panels open from **View › More View Options**: *Regions of Interest*, *Analysis Results* and *Workflow Status* (section 6.6).

2.1 Menus

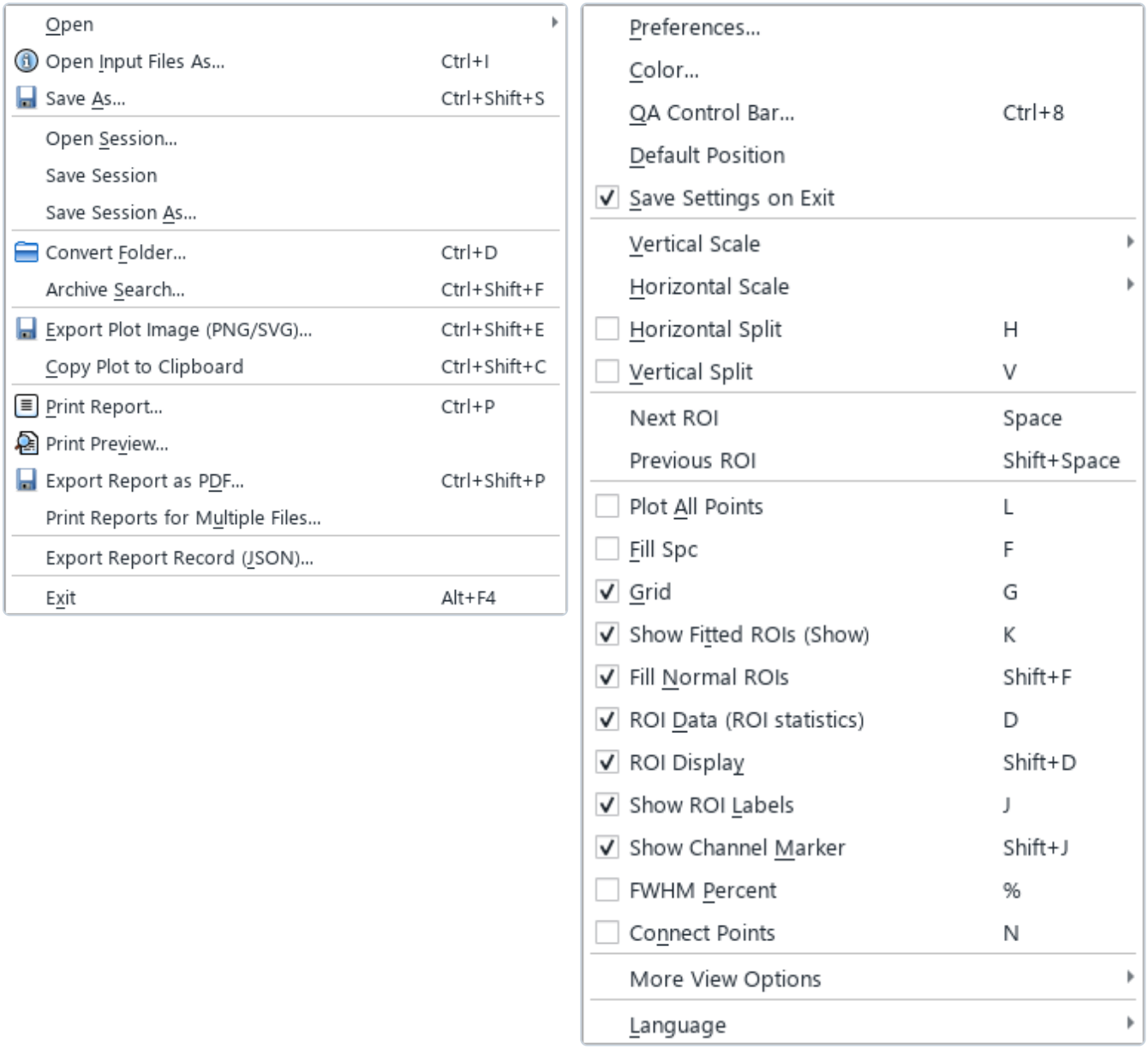


Figure 3 — The File and View menus.

File	Open (spectrum / multiple files to convert), Open Input Files As, Save As (Ctrl + Shift + S), Open / Save Session, Convert Folder, Archive Search, Export / Copy plot image, Print, Print Preview, Export Report as PDF, Print Reports for Multiple Files, Export Report Record (JSON), Exit.
Edit	Selection and Copy to ClipBrd, Sample Description (F5), Quantity Description (F6), Spectrum Data, Certificate, Library, Decay Correction Calculation, Load Header Information, Load File into MCArd, Reset Header (section 7.7).
Display	Normal, intermediate and stabilized spectrum; energy, resolution and efficiency calibration curves.
Setup	Quick Energy Calibration (F2), Full Energy / Resolution / Efficiency Calibration (Shift + F2), and **Advanced Calibration Tools**: Energy Calibration from a source, Check Calibration Drift, Efficiency Calibration (chapters 8 and 9).
ROI	Properties, Paint (F3), Edit (F4), Delete Indexed (Del), Delete All, Sort, Flag Definitions (section 7.3).
Analyze	Setup ›, Peak Search (P), Multifit (M), Identify (I), Background Subtract (B), Quantify (Q); Auto-analyze on Open, Peak Labels; **Activity Calculation**; **Fit Inspection / Manual Refit**; **Mark as Reviewed**, **Batch Analysis**; then the spectrum operations (section 7.5).
View	Preferences, Color, QA bar, split display, scales, marker and window movement, ROI stepping, spectrum and ROI drawing options, *More View Options* (zoom, Go to, overlays, panels), language (section 7.6).
Tools	Detector QC, acceptance test, chi-square counting test (chapter 12).
Help	User Manual (Arabic / English, F1), Mouse & Keyboard Shortcuts, GammaVision / MAESTRO Integration, splash-screen option, About SpectraMaster (version, analysis engine version, designer, data sources and the full list of formats).

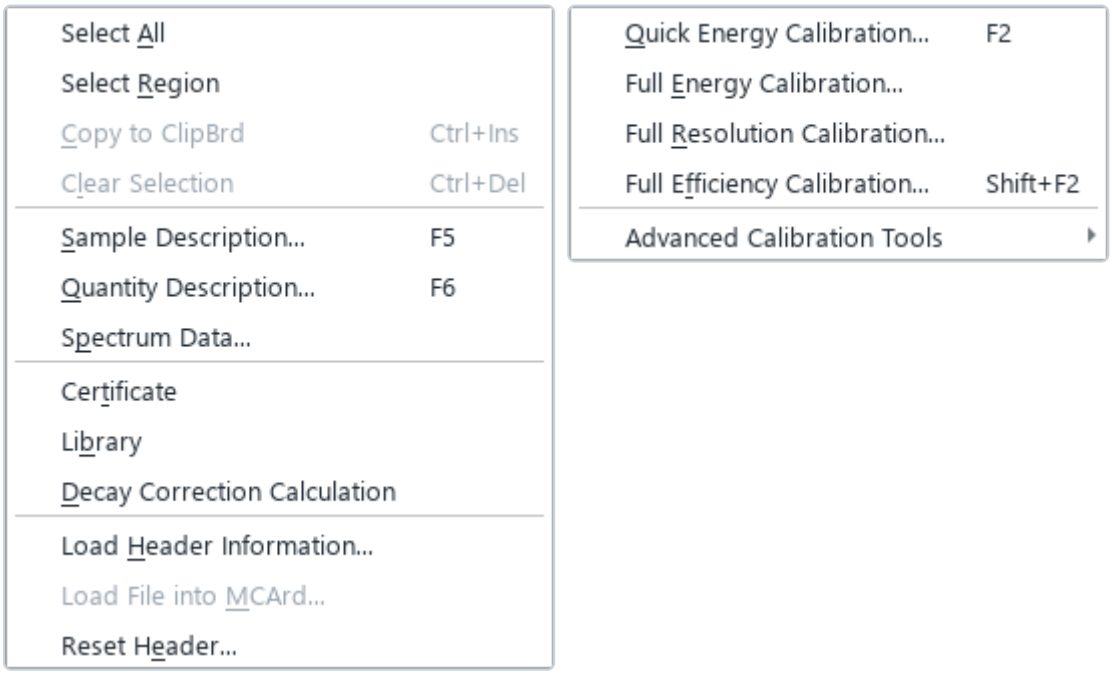


Figure 4 — The Edit and Setup menus.

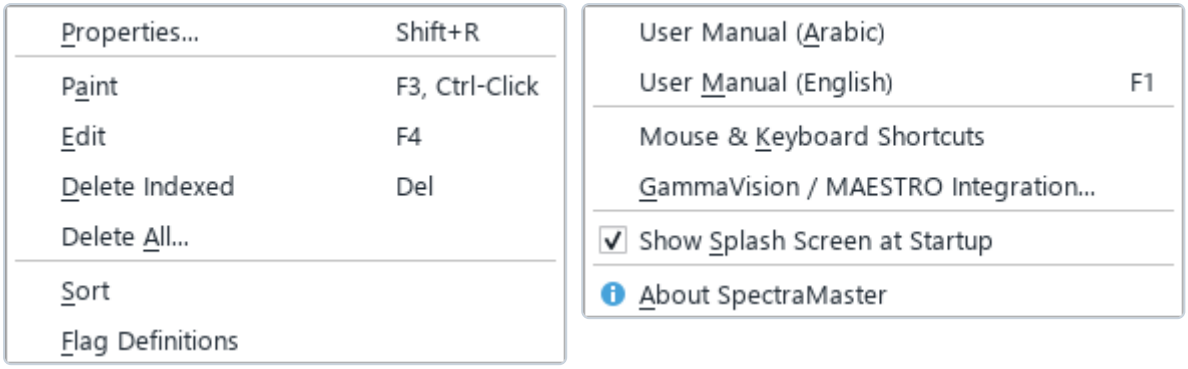


Figure 5 — The ROI and Help menus. The Analyze menu is in section 7.4.

A right click on the plot opens a context menu with the zoom history, Y scale, ROI deletion, analysis, labels, overlay and image export.



Figure 6 — Help › About SpectraMaster: program and analysis-engine versions, features and data sources.

3. Opening spectra

3.1 Open a spectrum

1. Choose **File** › **Open** › **Spectrum…** (Ctrl + O) or the **Open** button.
2. Select the file. With **Show description** ticked, the panel on the right shows the detected format, sample ID, description, channels, live/real time, start time and a thumbnail of the spectrum *before* you open it — useful when browsing archives with cryptic file names.
3. Click **Open**. The spectrum is drawn and its details appear in the information panel.

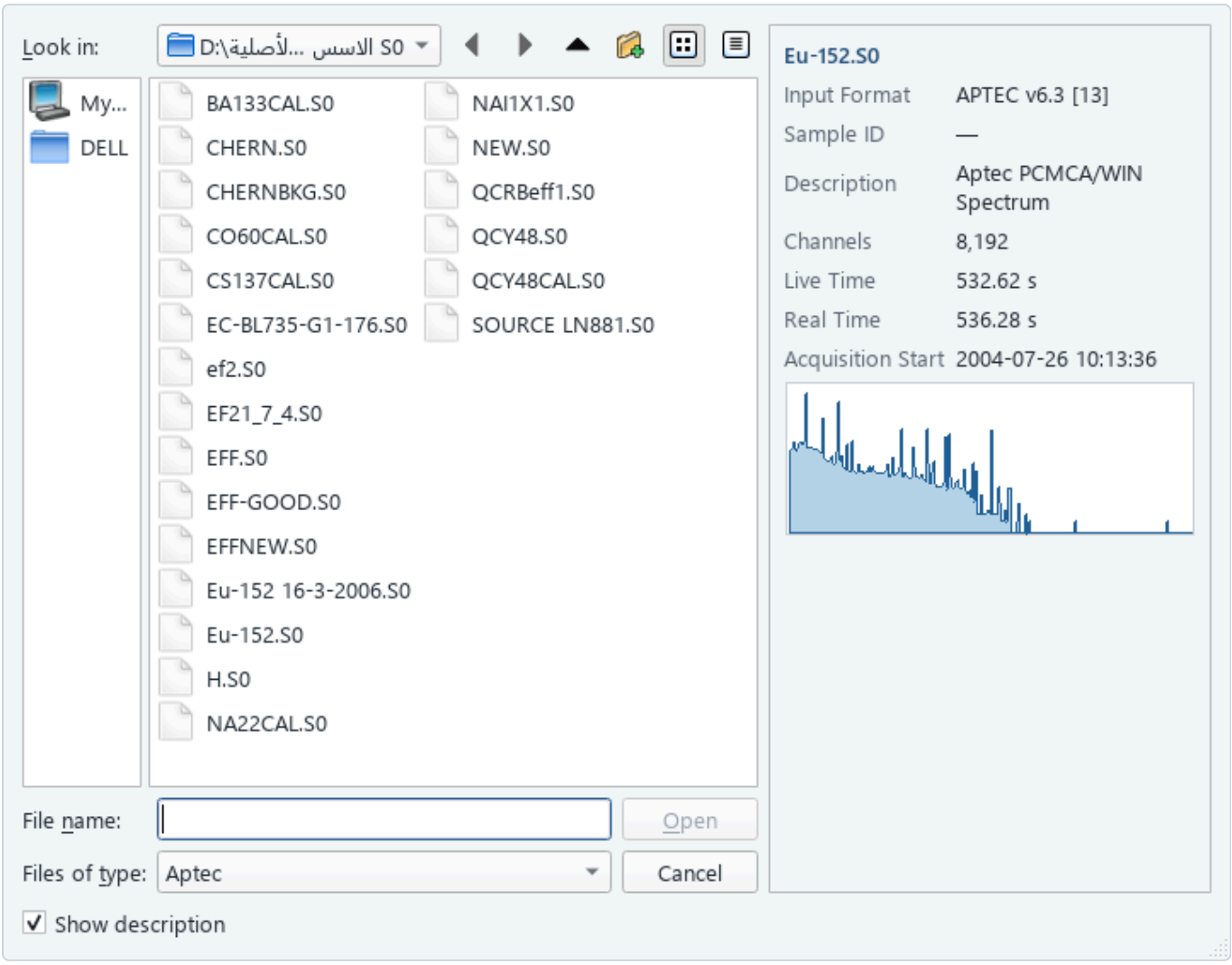


Figure 7 — Open dialog with the description preview.

The file filter lists all readable extensions; choose *All files (*)* for files with unusual extensions — the content is recognised, not the name.

3.2 Input format: auto-detect or fixed

By default the input format is **detected automatically** from the file content (binary signatures and header keywords), then from the extension. If neither identifies the file, a last stage looks at the content alone with the formats that can be recognised safely that way (ASCII, Channel Data CSV, EML, IAEA ASCII .dat, NPL,

Wallac, Laval, JCAC, TKA). A spectrum found this way carries a note in the status bar (*format recognised from the content*). If several formats fit, the message names them; choose one by hand (a TKA file under another name, for example, cannot be told from one-column ASCII, also with the extension .txt when its first two values could be a live and a real time). A file whose channels hold NaN or infinity opens, with a note in the status bar, but peak search and analysis refuse it.

File › Open Input Files As... (Ctrl + I) forces one format, exactly like DataMaster. The choice applies to Open, batch conversion and overlays and is shown in the status bar (*Input:* ...).

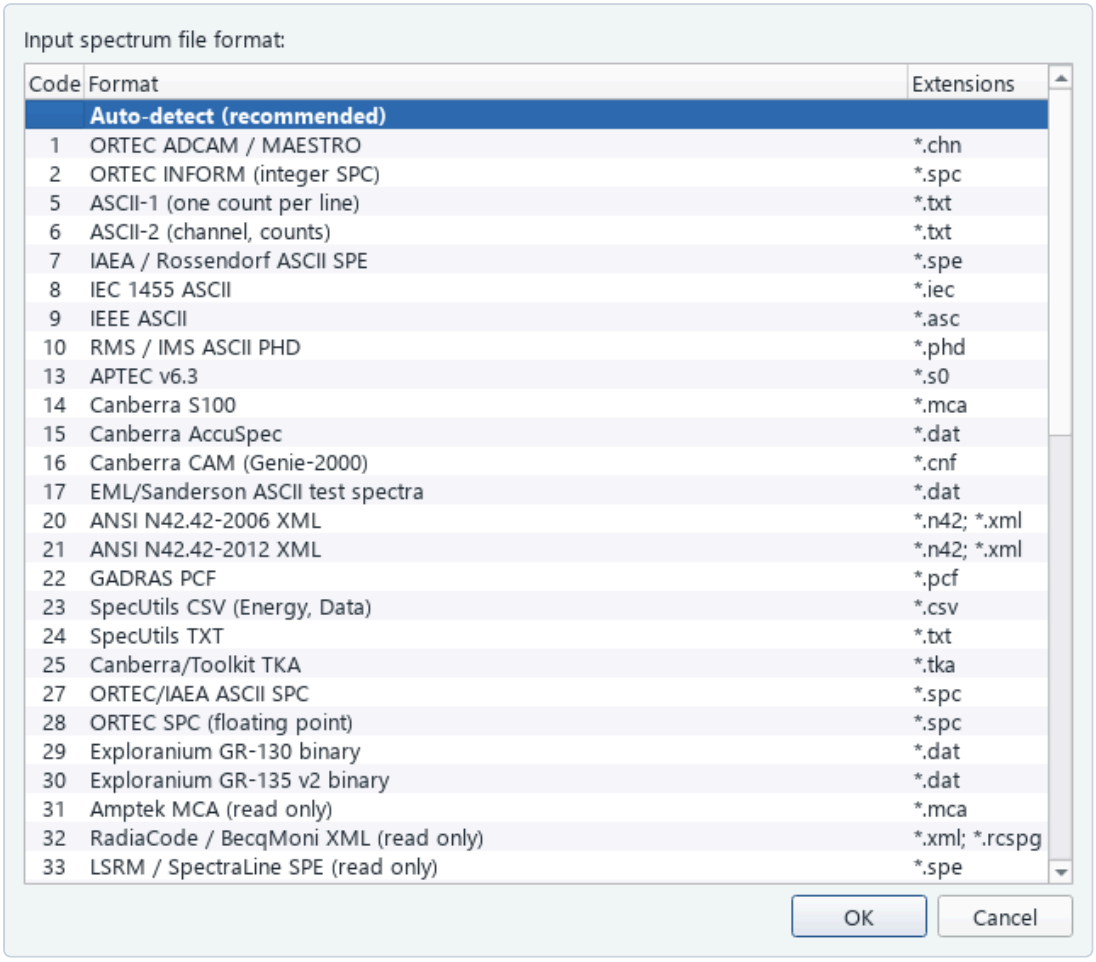


Figure 8 — Choosing the input format (keep *Auto-detect* unless a file is not recognised).

> **Tip.** Several formats share an extension (`.spc` , `.dat` , `.txt` , `.xml` , `.s0`). Auto-detect distinguishes them by content. Force a format only when auto-detect fails; if a forced format cannot read the file, the error message suggests returning to auto-detect.

> **Uncalibrated MCA files.** Some ORTEC CHN, Canberra CNF and Amptek MCA files store the placeholder calibration E = 0 + 1 × channel. SpectraMaster reads it as **no energy calibration**; the status bar says so and the stored values are kept in the metadata. Calibrate the spectrum (chapter 8) before identifying nuclides.

3.3 Drag and drop, command line, multi-spectrum files

- Drop **one file** on the window to open it; drop **a folder** to open the Convert Folder dialog; drop **several files** to open the Convert Files dialog.

- `SpectraMaster.exe file1.S0` opens the file at start-up (also used by "Open with…" in Windows Explorer).
- Files holding several spectra (e.g. N42, PCF, some CNF) show a selector *Spectrum i of n* in the information panel.

3.4 Supported formats

The complete table with codes is in appendix A and in **Help › About SpectraMaster**. The most used ones in our laboratory:

Code	Format	Notes
13	Aptec v6.3 `.S0`	Read and write; counts, times, energy and FWHM calibration, Aptec sample/source records and ROIs. Aptec v3.0 (78) and v4.3 (79) are read only.
1	ORTEC CHN	MAESTRO / GammaVision; 3 calibration terms.
7	IAEA SPE	Open text format; keeps Unicode text and the calibration.
16	Canberra CAM `.CNF` (Genie-2000)	3 calibration terms; text fields are Latin-1 only (see below).
20 / 21	ANSI N42.42 XML	Recommended long-term archive format.
50	Excel `.xlsx`	Sheets *Spectrum* (channel, energy, counts) and *Info*.
60–79	Aptec import/export family	Aptec ASCII, Channel-Data CSV, Davidson, IAEA DAT, NPL, Rump, Wallac, Laval, BTI, CI-90, JCAC, ND66, Norland, Oxford, …

Calibration terms. Some formats store fewer energy-calibration coefficients than the source (e.g. CHN, SPC, CNF, S100 and Channel-Data CSV keep 3 terms, IEC 1455 4, Laval and Norland 2). SpectraMaster then **refits** the calibration by least squares over the stored channels and warns: `… stores only 3 calibration terms: energy calibration refitted (max deviation 0.206 keV)`. Check the deviation: a few hundredths of a keV are harmless, larger values (cubic terms on 8k spectra) may shift high-energy peaks. The original coefficients are kept in the metadata (`original_energy_coeffs`) of formats that store metadata.

Arabic text in CNF, CHN and SPC. These binary formats store sample ID, description and detector as 8-bit Latin text. Arabic letters cannot be stored; a warning names the fields concerned. Use Latin sample IDs for files destined to Genie-2000 / MAESTRO, or keep an Aptec S0, IAEA SPE, N42, JSON or Excel copy, which preserve Arabic text.

Genie channel numbering. Genie-2000 numbers channels from 1, SpectraMaster (like ORTEC and Aptec) from 0. The calibration polynomial is shifted automatically on reading and writing CNF, so peak energies are identical in both programs; only the channel numbers differ by one.

4. Converting spectra

4.1 Save the displayed spectrum (Save As)

1. Open the spectrum (chapter 3).
2. **File › Save As…** (Ctrl + Shift + S): choose the **output format** in the list (code, name, extensions) and click OK.
3. Confirm the file name. The proposed name follows the DataMaster rule (below); the last output folder is remembered.

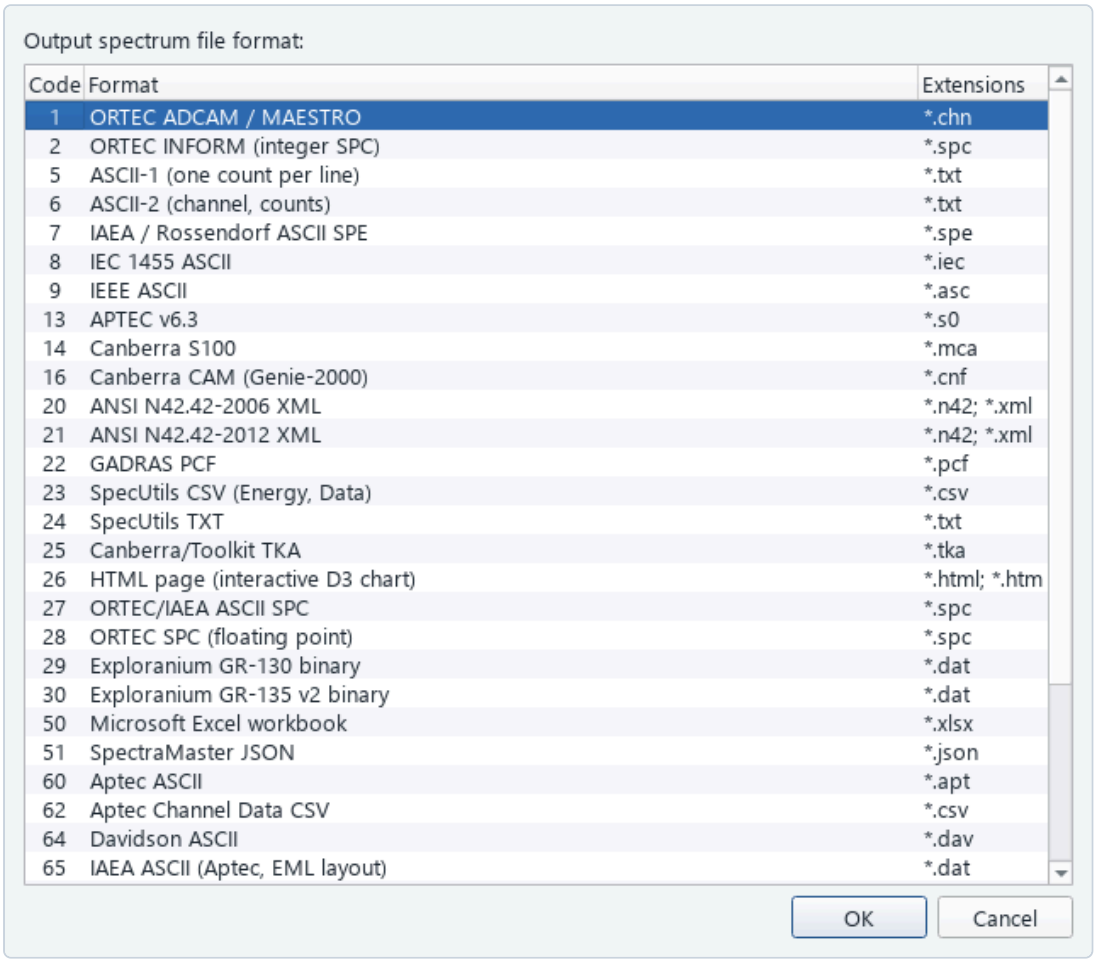


Figure 9 — Choosing the output format.

Warnings appear in the status bar after saving. **What a format cannot hold is never changed silently:** every written file is read back with the same format and every difference is reported — counts, number of channels, first channel, energies (more than 0.001 keV), live and real time, start time, sample ID, description, detector and FWHM calibration. Formats that can only be written say that they could not be checked. In particular:

- **Negative counts** (after a background subtraction or Strip) are refused by the formats that store only non-negative channels (CHN, SPC, CNF, Exploranium, NPL ASCII, SPC ASCII): choose a format that keeps them, e.g. IAEA SPE, N42, JSON or Excel.
- **Fractional counts** are rounded in integer formats, with a warning giving the largest change.
- An **unknown acquisition date** is reported for formats that always store one; CHN reads its placeholder date back as unknown.

- **Arabic-Indic digits** (١٢٣) in the sample ID and description are written as 0–9 in formats that cannot hold them, with a warning.
- A spectrum with NaN or infinite counts is not written.

4.2 The DataMaster naming rule

The output name is the input name with its extension appended after an underscore, plus the extension of the output format: `CHERN.S0` → `CHERN_S0.chn`. Files with several spectra get a running number: `RUN.n42` → `RUN_n42_1.spe`, `RUN_n42_2.spe`, … This avoids overwriting when `A.S0` and `A.CHN` are both converted to SPE.

4.3 Converting many files or whole folders

- **File › Open › Select Multiple Files to Convert…** (Ctrl + M): pick any number of files.
- **File › Convert Folder…** (Ctrl + D): choose a folder, optionally **Include sub-folders**, then **Find Files**. With *All files (ignore extensions)* every file is sniffed by content.



Figure 10 — Convert Folder: the Aptec spectra of a folder, output to CHN next to each input file.

Choose the output format, then the output folder: **Same folder as each input file**, or one folder — with **Keep sub-folder structure** the input tree is recreated inside it. **Overwrite existing files** controls replacement. Click **Convert**; each line shows OK, the number of outputs, a warning or the error. A failure never stops the batch. **Open Output Folder** opens Explorer at the result.

Input format in batches. In auto-detect mode only files whose content is recognised (or whose extension belongs to a plain-text format) are converted, so reports, Word files and pictures in the same folder are skipped; recognised spectra are added whatever their extension. With a fixed input format the extension decides, as in DataMaster.

4.4 Command line (smconvert.exe)

DataMaster syntax

```
smconvert -i InputFile -j OutputFile          (i, j = format codes, i = 0 auto-detects)

smconvert -13 CHERN.S0 -1 "C:\Spectra\Converted Chern.chn"
smconvert -0  SAMPLE.CNF -7 SAMPLE.SPE
SpectraMaster.exe -13 CHERN.S0 -16 CHERN.CNF  (silent, no window; on error the window opens)
```

The exit code is 0 on success and 1 on failure (2 for wrong arguments), so batch files can test `%ERRORLEVEL%`. A calibration refit and the read-back differences are printed as warnings.

Modern subcommands

```
smconvert formats                                  list all formats with codes
smconvert info CHERN.S0                            all metadata of a file
smconvert convert D:\Archive --to n42_2012 -r --out E:\N42 --keep-tree
smconvert convert *.S0 --to chn                    glob patterns are accepted
smconvert convert D:\In --from aptec_s0 --to 7 --no-overwrite -q
```

`--to` and `--from` accept the code (`7`), the key (`iaea_spe`) or an alias (`spe` , `chn` , `cnf`). Other subcommands: `peaks` , `identify` , `calibrate` , `sum` , `subtract` , `rebin` , `effcal` , `activity` , `index` , `rescue` — see the following chapters and appendix B.

Archive rescue. `smconvert rescue D:\OldArchive -o E:\Rescued --to n42_2012 --to ci_cnf --resume` converts a whole archive, re-reads every output to verify counts and times, writes a log workbook and can be resumed after an interruption. The archive itself is only read.
`smconvert index ROOT -o catalog.xlsx` writes a catalogue of all spectra.

4.5 GammaVision / MAESTRO integration

GammaVision and MAESTRO can call SpectraMaster as their external converter instead of DataMaster. **Help › GammaVision / MAESTRO Integration…** shows the exact **Program** and **Arguments** to enter in GammaVision (**File › Settings › Import/Export**) with **Copy** buttons.

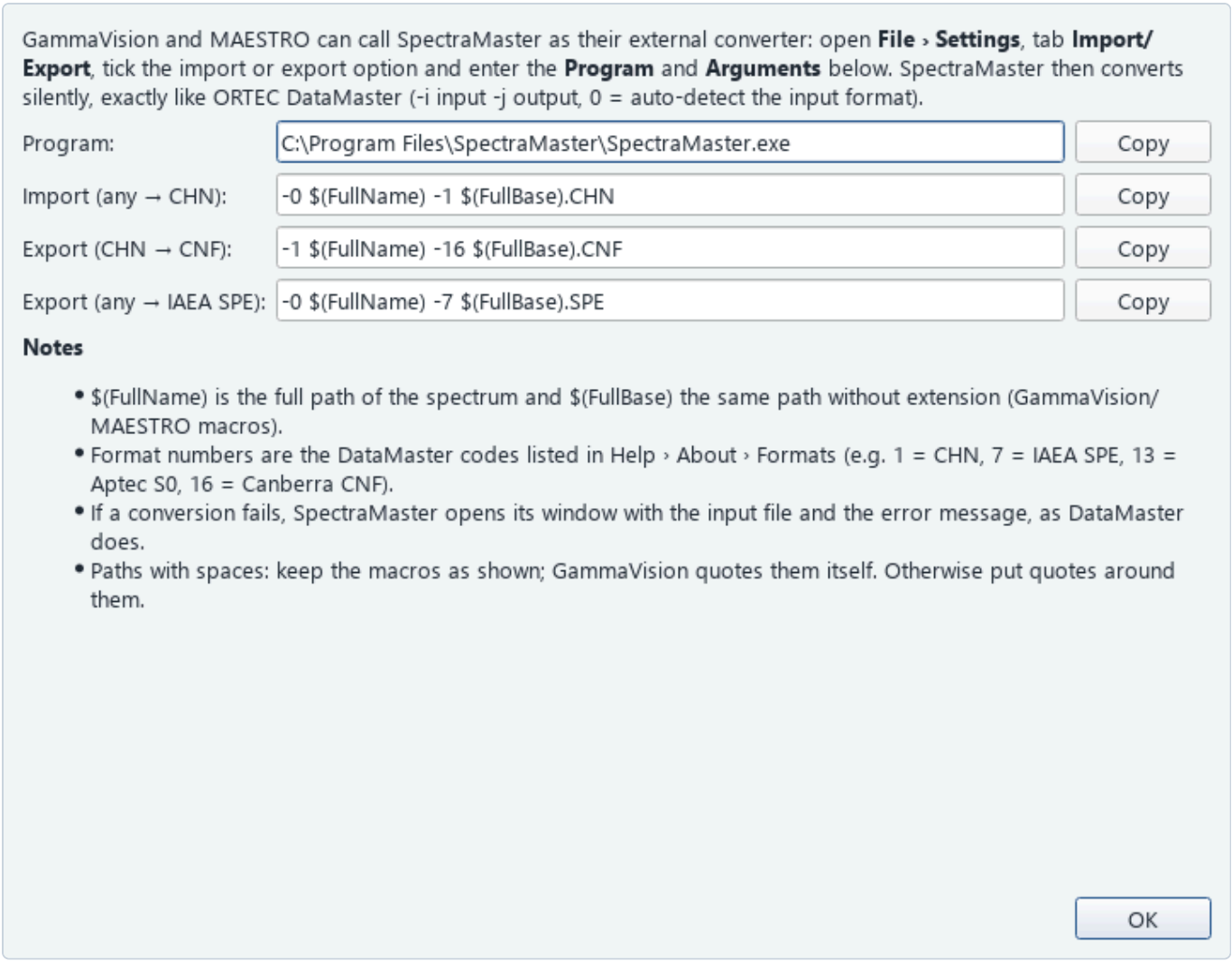


Figure 11 — Values for GammaVision / MAESTRO.

`$(FullName)` and `$(FullBase)` are GammaVision macros (full path, path without extension). If a conversion fails, SpectraMaster opens its window with the input file and the error message, as DataMaster does.

5. Viewing and zooming

5.1 Mouse and keyboard

The same list is available in the program under **Help › Mouse & Keyboard Shortcuts**.

Keys / mouse	Action
Wheel	Zoom X around the cursor
Ctrl + Wheel / ↑ ↓	Zoom Y (counts)
Shift + Wheel / ← →	Pan (Shift+← → : half a screen)
Drag horizontally	Zoom to the dragged channel range
Drag a box	Rubber-band zoom of X and Y
Ctrl + Drag	Define a region of interest (ROI)
Right / Middle drag, Shift + Drag	Pan (also Y when Y is zoomed)
Right click	Context menu
+ / −	Zoom in / out
Backspace, Alt + ←	Previous zoom
Alt + →	Next zoom
Home, Double-click, Ctrl+0	Full spectrum
Ctrl + ← / →	Previous / next found peak, with its information
Space, Tab (Shift: back)	Next ROI; the next found peak when there are no ROIs
Double-click a peak	Peak information (no ROIs)
Ctrl + G	Go to energy / channel
Ctrl + L	Linear / log counts axis
Ctrl + Shift + Y	Auto-scale Y to the view
Ctrl + E	Energy / channel axis
F5	Peak search & identify
Ctrl + Shift + L	Peak labels on/off
Ctrl + Shift + O	Add overlay spectrum
Ctrl + Shift + E	Export plot image (PNG / SVG)
Ctrl + Shift + C	Copy plot to clipboard
Delete (ROI table)	Delete the selected ROI
Ctrl + O	Open a spectrum (with preview)
Ctrl + M	Select multiple files to convert
Ctrl + D	Convert a folder
Ctrl + I	Open input files as (input format)

Keys / mouse	Action
Ctrl + S, Ctrl + Shift + S	Save As (choose the output format)
Ctrl + P	Print report
Ctrl + Shift + P	Export report as PDF
Ctrl + Shift + F	Archive search
F1	User manual in the interface language
Alt + F4	Exit

5.2 Zoom, history and overview strip

- **Horizontal drag** zooms to the dragged channel range (counts axis automatic); dragging a **box** zooms both axes.
- Every zoom is stored in a **history**: **Back** (Backspace) and **Forward** (Alt+→). **Reset Zoom** (Ctrl+0) or a double click on an empty part of the plot shows the full spectrum.
- The **overview strip** under the plot always shows the whole spectrum: drag the blue window to pan, drag its edges to change the width, click elsewhere to jump, use the wheel to zoom.

5.3 Stepping through the found peaks

After an analysis, the found peaks can be visited one by one, as in MAESTRO:

- Ctrl+← / Ctrl+→ move the marker to the previous / next found peak.
- Space / Tab (with Shift: back) index the found peaks and wrap around at the ends. When ROIs exist, Space indexes the ROIs first, as in Aptec.
- A **double click on a peak** indexes it.

The indexed peak is shown in the status bar (energy, channel, net area ± u and its nuclide, or *no library match*) and selected in the peaks table. When a nuclide is selected, its library lines are drawn **green and solid** where a found peak lies at the line, and **blue and dashed** where none does.

5.4 Go to an energy or channel

Type an energy in the **Go to** box (Ctrl+G), e.g. `661.66`, and press Enter. The unit box selects keV or channel; the width box selects ±3/5/10 FWHM (from the FWHM calibration or the detector model) or ±5…100 keV. Uncalibrated spectra accept channels only.

5.5 Counts axis and energy axis

The counts axis can be **Linear**, **Log** or **√ (square root)** (Ctrl+L toggles linear/log). **Auto Y** fits the counts axis to the visible part. **Energy** (Ctrl+E) switches the horizontal axis between keV and channel (keV only for calibrated spectra).

5.6 Regions of interest (ROI)

Paint an ROI with F3 or Ctrl+click left then right of the peak, or let Peak Search (P) set them. The *Regions of Interest* panel lists start, end, centroid, energy, FWHM (channels and keV), gross, background and net area with ±1σ, % error, net count rate and flags, then isotope, activity and MDA after Identify and Quantify. The background follows Aptec's rules, from end points outside the ROI (ROI Properties, Shift+R). Delete with Del, **Delete** or **Clear All**; **Export CSV…** saves the table. Details in section 7.3.

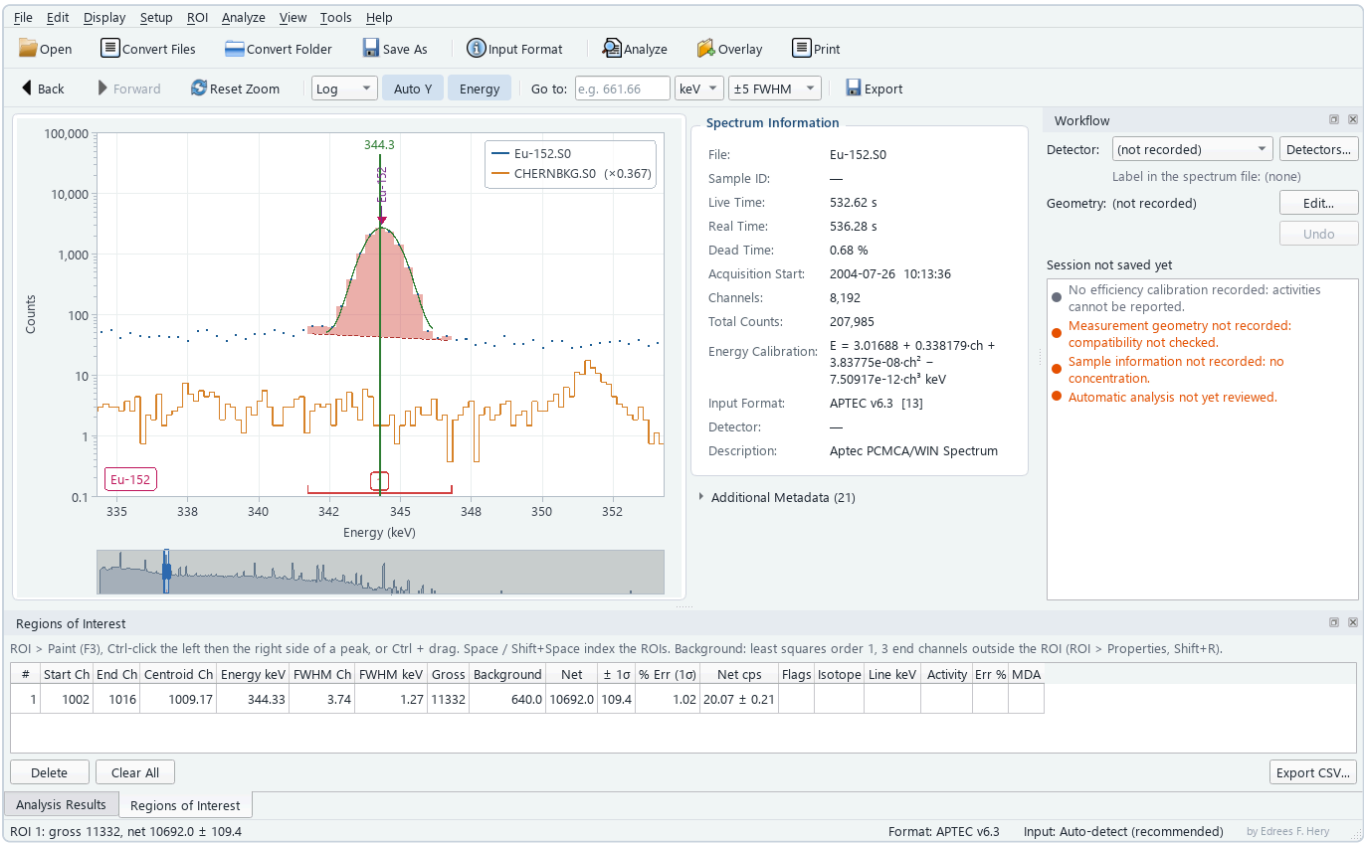


Figure 12 — Zoom on the 344.3 keV Eu-152 line with an ROI and the background spectrum CHERNBKG.S0 as a live-time normalised overlay.

5.7 Overlays

Overlay (Ctrl+Shift+O) draws other spectra — typically the background or a previous measurement — over the displayed one, each in its own colour with a legend.
View › More View Options › Overlays › Normalize Overlays to Live Time scales each overlay by $LT_{displayed}/LT_{overlay}$ (the factor is shown in the legend). Overlays of different calibration are aligned in energy. Remove them one by one or all at once from the right-click menu.

5.8 Exporting the plot

Export (Ctrl+Shift+E) saves the current view as a PNG image or as an SVG vector drawing (for reports and publications); Ctrl+Shift+C copies it to the clipboard for Word or PowerPoint.

6. SpectraMaster analysis: fitted peaks and nuclide identification

6.1 Running the analysis

This analysis is the alternative to the Aptec method (chapter 7). Choose it in **Analyze › Setup › Analysis Method › SpectraMaster**, then press P (**Analyze › Peak Search**) or the **Analyze** button. The analysis runs in the background (a few seconds for an 8k HPGe spectrum) and opens the *Analysis Results* panel. **Analyze › Auto-analyze on Open** analyses every spectrum as soon as it is opened.

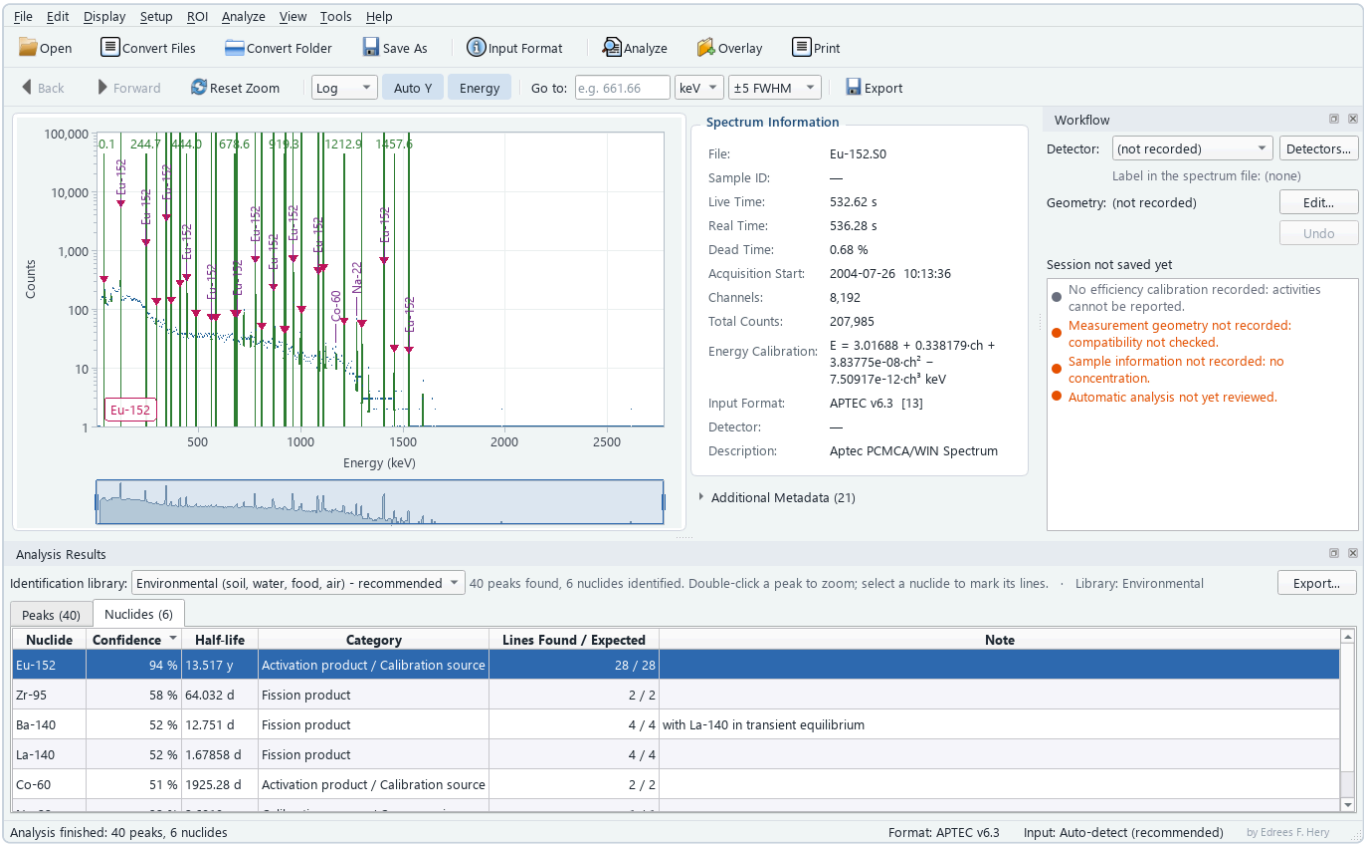


Figure 13 — Eu-152 calibration spectrum after P with the SpectraMaster method; Eu-152 selected, its lines marked on the plot.

6.2 The peaks table

For each peak: energy (keV), centroid (channel), FWHM (keV), net area ± 1σ, count rate (cps), significance (net area / its uncertainty) and the assigned nuclide. A double click zooms to the peak. Notes mark special peaks: *X-ray region*, *annihilation* (511 keV), *single / double escape*, *sum peak* (e.g. 1173+1332), *possible backscatter*.

How it works, in brief:

- **Width model.** The FWHM model comes from the FWHM calibration (or the energy calibration and a detector-type guess) and is rescaled to the strongest isolated peaks. For HPGe without a FWHM calibration its energy dependence is corrected too, between the lowest and highest of those peaks.
- **Implausible width model.** If the free-width fits of the strongest peaks show that the model is about twice or half the real width (for example a stored FWHM calibration of one channel, or a wrong detector guess), the model is scaled by 2 or ½ before the search and a note says so (*FWHM model implausible*).

- **Search and fit.** Peaks are searched with a Gaussian matched filter (second-difference method) and accepted above a significance of 3. Neighbouring peaks are fitted together as multiplets (Gaussians + background + step); areas and uncertainties come from the fit covariance. HPGe peaks sit on a straight background; scintillator (NaI) peaks on a log-quadratic continuum whose curvature is measured on a wider region that stops before the neighbouring peaks. When the detector's peak shape was measured (section 6.6), peaks are fitted with its low-energy tail.
- **Spectra that are not Poisson counts.** If the spectrum was already background subtracted, stripped, normalized or smoothed, a note in the results says that the uncertainties are not valid.

6.3 The nuclides table and the confidence value

Each candidate nuclide shows its **confidence**, half-life, category (natural, fission, activation, calibration, medical, industrial, actinide; nuclides with two common origins show both, e.g. Cs-137 "fission / calibration"), the number of lines found / expected and a note (series membership, "typical background", interferences such as U-235 185.7 keV under Ra-226 186.2 keV). Selecting a nuclide marks all its library lines on the plot (section 5.3).

> **The confidence is a heuristic score (0–100 %), not a probability.** It combines how much of the expected (detectable) line pattern is explained, the energy agreement, the statistical significance, penalties for a missing strongest line or inconsistent area ratios, and a greedy treatment of shared peaks. Weak lines below the local detection limit are not counted as missing. Results — especially low-confidence ones, single-line nuclides and short-lived nuclides in old spectra — **must be reviewed by an expert**. The identification does not replace the laboratory's validated analysis procedure.

Sampling date. When the session's sample information has a sampling date (end, else start; section 9.7), a nuclide that has decayed by more than 10 half-lives between sampling and the start of the measurement (less than 0.1 % left) gets a low plausibility; if it is still listed, its note says how many half-lives have elapsed. Natural-series members and a daughter whose parent is still present (Tc-99m with Mo-99) are not affected. Run the analysis again after entering the sample information. Radiopharmaceuticals (Mo-99, Tc-99m, F-18, Ga-67, I-123 …) are treated as common nuclides only with the *medical* library. A nuclide found from one line only, although it has other lines in the spectrum's range, carries the note *one line matched … confirm*.

Peak labels (Ctrl + Shift + L) write the nuclide names on the plot; **Export…** in the results panel saves peaks, nuclides and spectrum details to Excel or CSV.

6.4 Uncalibrated spectra

Without an energy calibration peaks are listed by channel only and no nuclide is identified (a message explains this). Calibrate the spectrum first (chapter 8), e.g. with Eu-152 or Co-60/Cs-137 lines.

6.5 The nuclide library

The built-in library contains 200 nuclides with 2,719 gamma and X-ray lines. Values are taken from the IAEA ENSDF evaluation (IAEA *ndlab* database behind LiveChart of Nuclides) and cross-checked against Sandia National Laboratories' *SandiaDecay* and the LNHB/DDEP recommended data. Lines that duplicated another evaluation were removed and lines lost from daughters restored; where IAEA and two other evaluations disagree, the DDEP value is used (for example Ac-228 674.75 keV: 0.105 %). Short-lived daughters in equilibrium (Ba-137m, Pr-144, Rh-106, La-140 …) are included in the parent's intensities; natural-series

members are separate entries. On the command line another library can be used: `smconvert identify FILE --lib mylib.csv` with lines `nuclide, energy_keV, intensity_%[, half_life, category]`.

Limitations

- The identification does not correct true-coincidence summing and uses no detailed efficiency model: the expected line ratios use a generic HPGe/NaI efficiency shape. (The activity calculation can correct summing, section 9.5.)
- Strongly overlapping multiplets and very low-statistics spectra can give false or missed identifications.
- Decay since the measurement is not considered: a nuclide is proposed if its lines are present, even if it is implausible for the sample history.

6.6 Detector profiles, peak shape and analysis sessions

The **Workflow** panel (**View › More View Options › Workflow Status**) shows the analysis session of the displayed spectrum: its detector, geometry, calibrations and the status of each step, coloured by level. **Detectors…** opens the detector profiles; **Edit…** describes the sample geometry. A profile keeps a permanent id, so renaming a detector keeps its calibrations and sessions. Choosing the detector of a measurement is an undoable edit, and an efficiency calibration of another detector is then reported as incompatible.

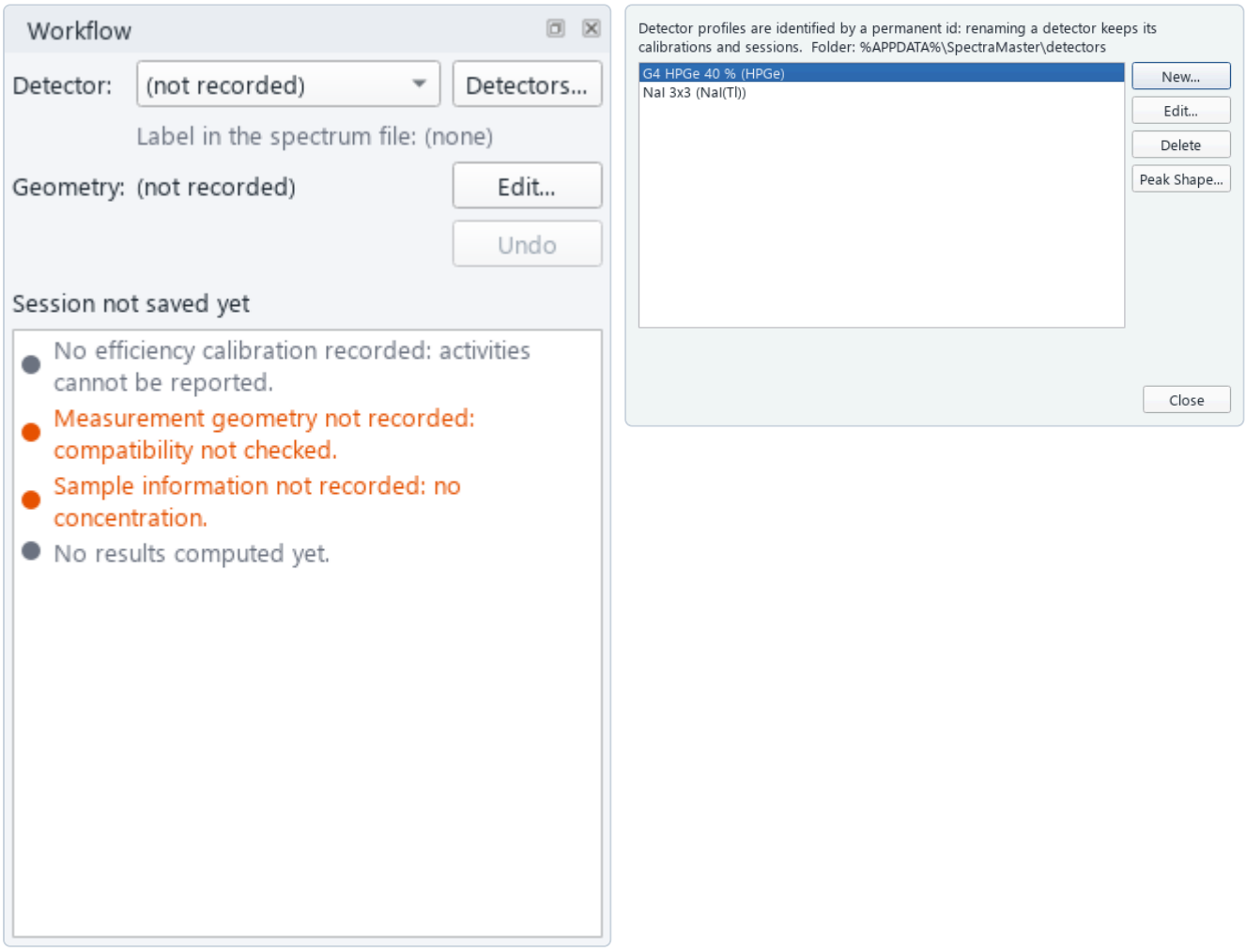


Figure 14 — The Workflow panel and the detector profiles.

Peak Shape… in the detector profiles measures the low-energy tail of that detector's peaks from calibration spectra with strong, well separated peaks. The spectra are read and measured in the background (*Measuring the peak shape …*); you confirm the result before it is recorded. From then on, the analysis of that detector's samples fits peaks with this shape. Without it, peaks are Gaussian.

File › Save Session stores the session (`.smsession`): the spectrum reference with its SHA-256 fingerprint, the calibrations, the background and reagent blank files with their fingerprints, the coincidence table used, the results, the review and the edit history. When a session is opened, the spectrum file is checked against the fingerprint. Results computed by an older analysis engine are shown as stored and marked as requiring recalculation; the engine version is shown in **Help › About**.

6.7 Fit inspection and manual refit

Analyze › Fit Inspection / Manual Refit… shows how every peak was fitted and lets you correct a fit, with every change recorded.

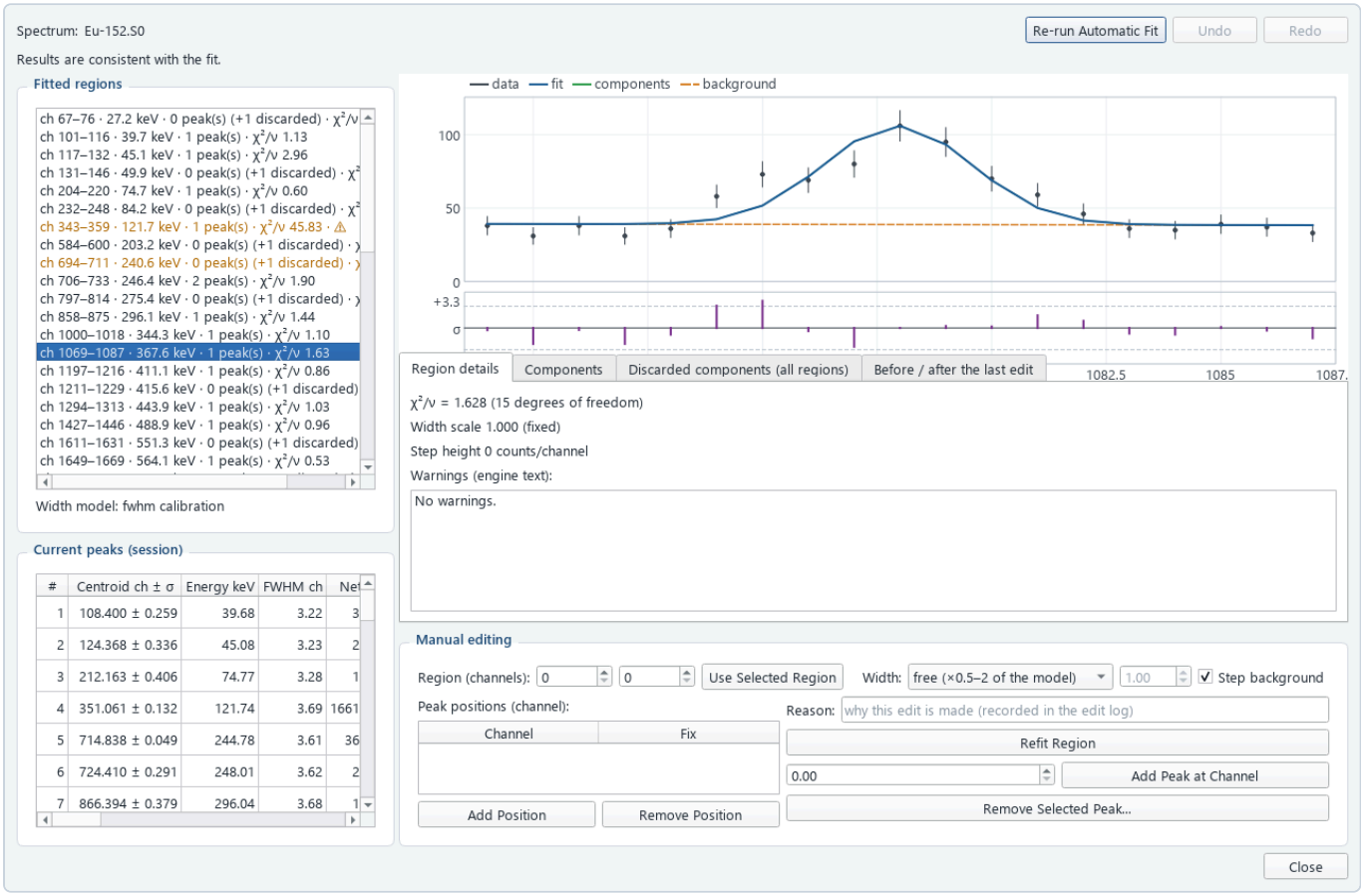


Figure 15 — Fit inspection of the Eu-152 spectrum: one fitted region with data, fit, components and background, and the normalised residuals below.

- **Fitted regions** (left): each region with its energy, number of peaks and χ^2/ν. Regions with a poor fit or a warning are coloured. The plot shows the data with error bars, the fit, its components and the background, and the residuals in units of σ.
- Tabs: **Region details** (χ^2/ν, width scale, step height, engine warnings), **Components** (centroid, energy, FWHM, net area, whether it is reported and why not), **Discarded components** (all regions) and **Before / after the last edit** (change of centroid and area in units of σ).

- **Manual editing:** choose a region (or **Use Selected Region**), the peak positions (fixed or free), the width (free ×0.5–2 of the model, held at the model, or a fixed multiple) and the step background, give a **reason**, then **Refit Region**. **Add Peak at Channel** and **Remove Selected Peak…** refit the neighbours. **Undo** / **Redo** step through the edits; **Re-run Automatic Fit** starts again (the edits stay in the edit log).
- The edited peaks are used by the activity calculation. A spectrum made in this session by background subtraction, or by the Aptec operations Strip, Combine, Normalize or Shift, carries its propagated variance (sample + k^2 × background) and is refitted with it, so its negative channels are kept and the uncertainties stay valid. Any other change of the counts (Smooth, editing a channel) makes the variance unknown. The variance is not saved in a file: a net spectrum opened from a file is refitted with the note that its uncertainties are not valid.

7. The Aptec method: menus and workflow

SpectraMaster follows the menus, keys and workflow of **Aptec PCMCA/WIN 6.31**, so that Aptec users need nothing new. The menu bar, in order: **File, Edit, Display, Setup, ROI, Analyze, View, Tools, Help**. The method works on **regions of interest (ROIs)**: ROIs are set on the peaks, identified, and quantified. **The Aptec method is the default**; the fitted-peak analysis of chapter 6 is chosen in **Analyze › Setup › Analysis Method**.

Aptec's algorithms were implemented as its help file describes them and compared with the results Aptec itself stored in its spectrum files (section 7.8). Aptec was not decompiled and none of its code was copied.

7.1 The workflow at a glance

1. Calibrate	F2 quick energy calibration, or Shift + F2 full energy, resolution and efficiency calibration (section 7.2).
2. Describe	F5 Sample Description, F6 Quantity Description (quantity, units, activity time).
3. ROIs	P Peak Search sets the ROIs, or F3 / Ctrl +click paints them.
4. Multiplets	M Multifit decomposes multiplets into Gaussian peaks.
5. Identify	I identifies the ROIs with the isotope library.
6. Background	B subtracts the background count rate of identified ROIs (optional).
7. Activity	Q Quantify: activity or concentration and MDA of every ROI.
8. Results	The *Regions of Interest* list, CSV export and reports (chapter 10).

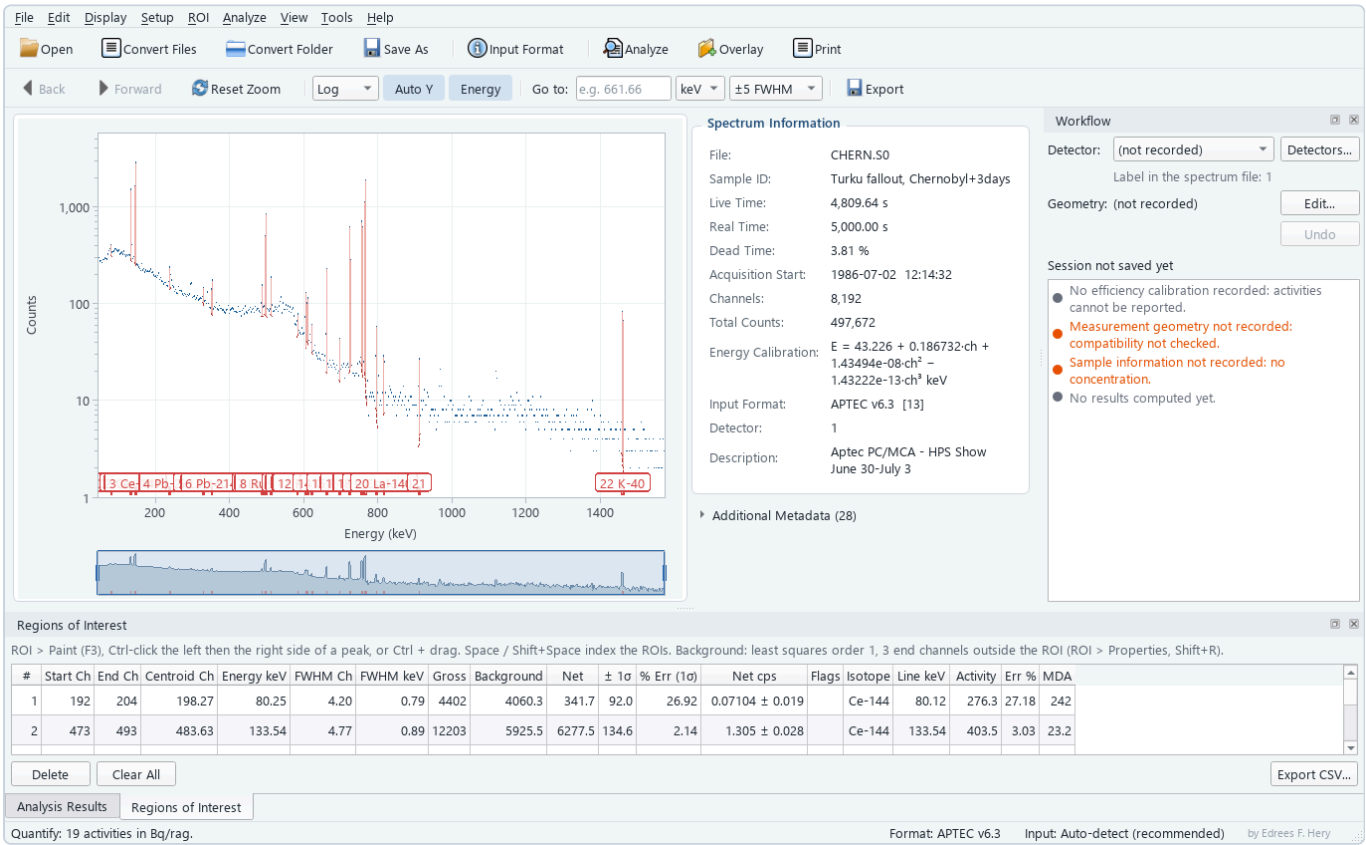


Figure 16 — A soil spectrum (CHERN.S0) after P, I with Aptec's isotope.lib, and Q: isotope names on the ROIs, activities and MDAs in the ROI list.

7.2 Setup menu: calibrations

- **Quick Energy Calibration** (F2), as in Aptec: put the channel marker on a peak, press F2, type its energy and press OK; then move the marker to the next peak and press F2 again. Up to 16 channel ↔ energy points; the points of the previous OK are kept when the window is reopened, and energies are kept at full precision. A first- or second-order polynomial is chosen from the number of points.
- **Full calibration** (Shift+F2, S_Calib style): energy and resolution (FWHM) with up to 256 points and degree 5, and efficiency with Aptec's types (5th or 3rd order ln-ln, polynomial, cubic spline). Points are loaded from the ROIs (*Load ROIs*), activities from a certificate (**Edit** › **Certificate**), decay-corrected to the measurement date.
- Both windows are modal: finish or cancel the calibration before working in the main window again.
- The **Advanced Calibration Tools** (energy calibration from a source, drift check, efficiency calibration with validation) are described in chapters 8 and 9.

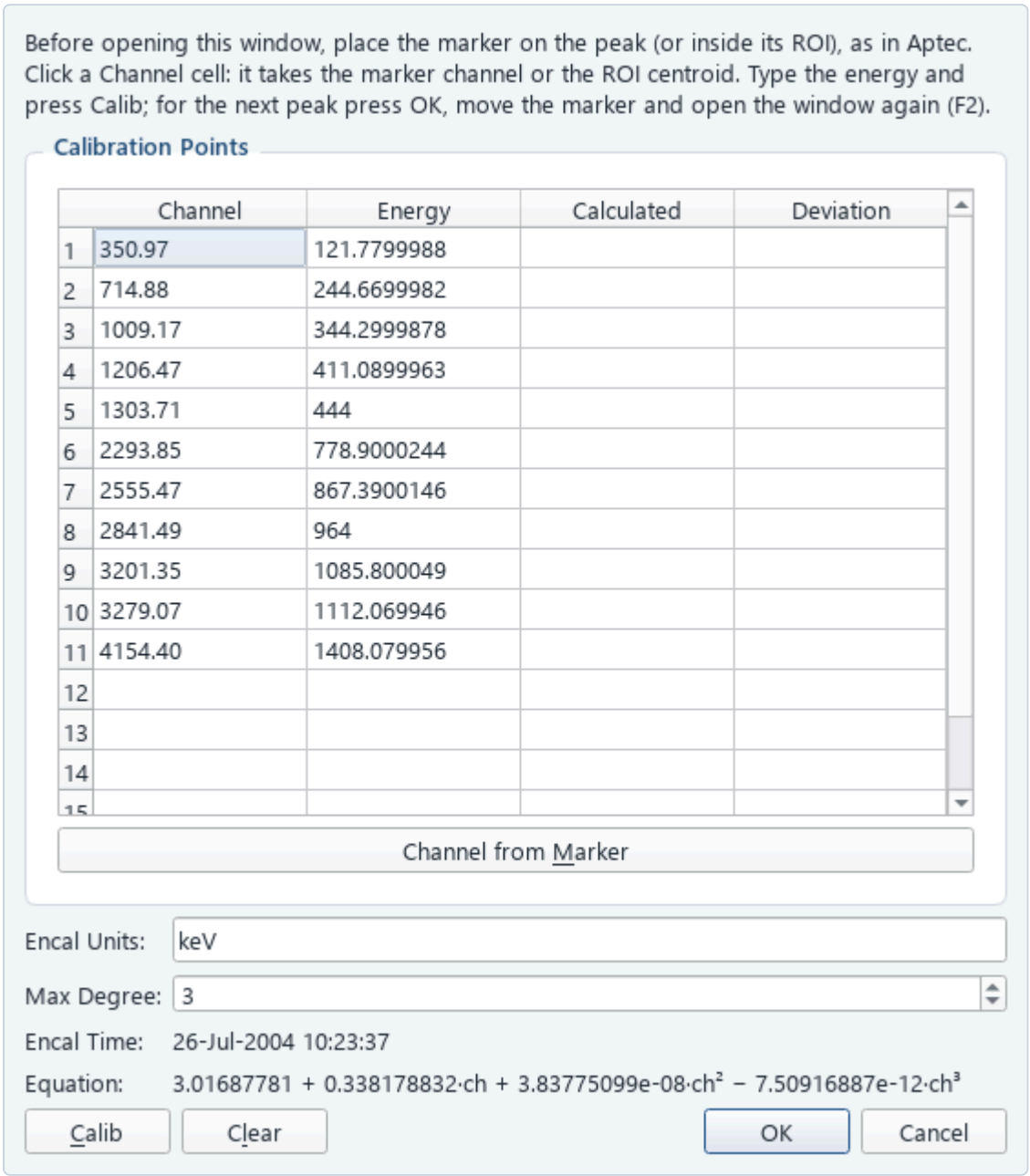


Figure 17 — Quick Energy Calibration (F2).

7.3 ROI menu

- **Properties** (Shift + R): background line (straight line between the end-point averages, or least squares of order 1–5), number of end points (fixed, or varying from a value at channel 0 to a value at the last channel), centroid range and weighting, MDA sigma (default 1.645) and error sigma.
- **Paint** (F3, or Ctrl +click left then right of the peak): the Paint bar at the bottom expands, shifts, sets (Enter) or cancels (Esc) the ROI; its statistics are shown while painting.
- **Edit** (F4) the indexed ROI, **Delete Indexed** (Del), **Delete All**, **Sort**, **Flag Definitions**. Space / Shift + Space index the next / previous ROI.

ROI statistics follow Aptec's rules: end points outside the ROI, touching ROIs share one background, net = gross − background, % error (1σ) = √(gross + background) / net × 100 and MDA = $(K^2 + 2K\sqrt{(2B)})$ / live time. They reproduce the values Aptec stored for 73 of 79 ROIs.

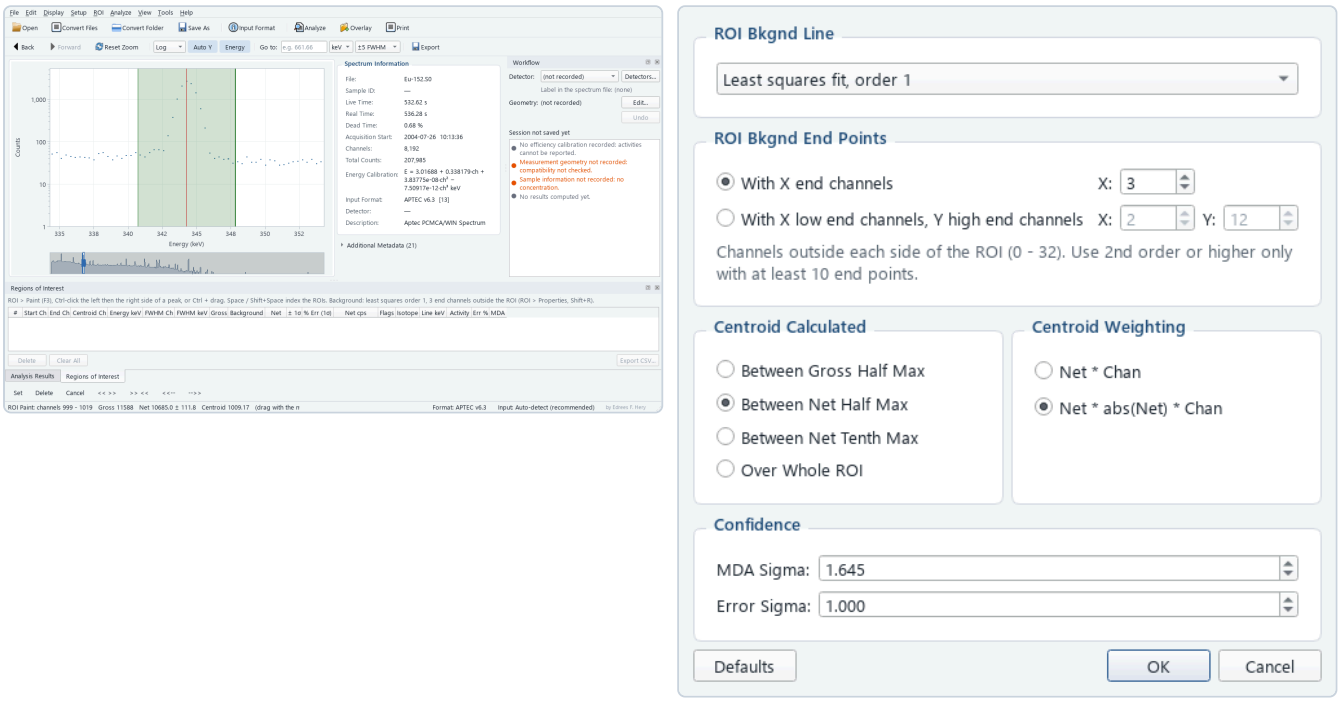


Figure 18 — Painting an ROI on the 344.3 keV line with the Paint bar, and ROI Properties.

7.4 Analyze menu

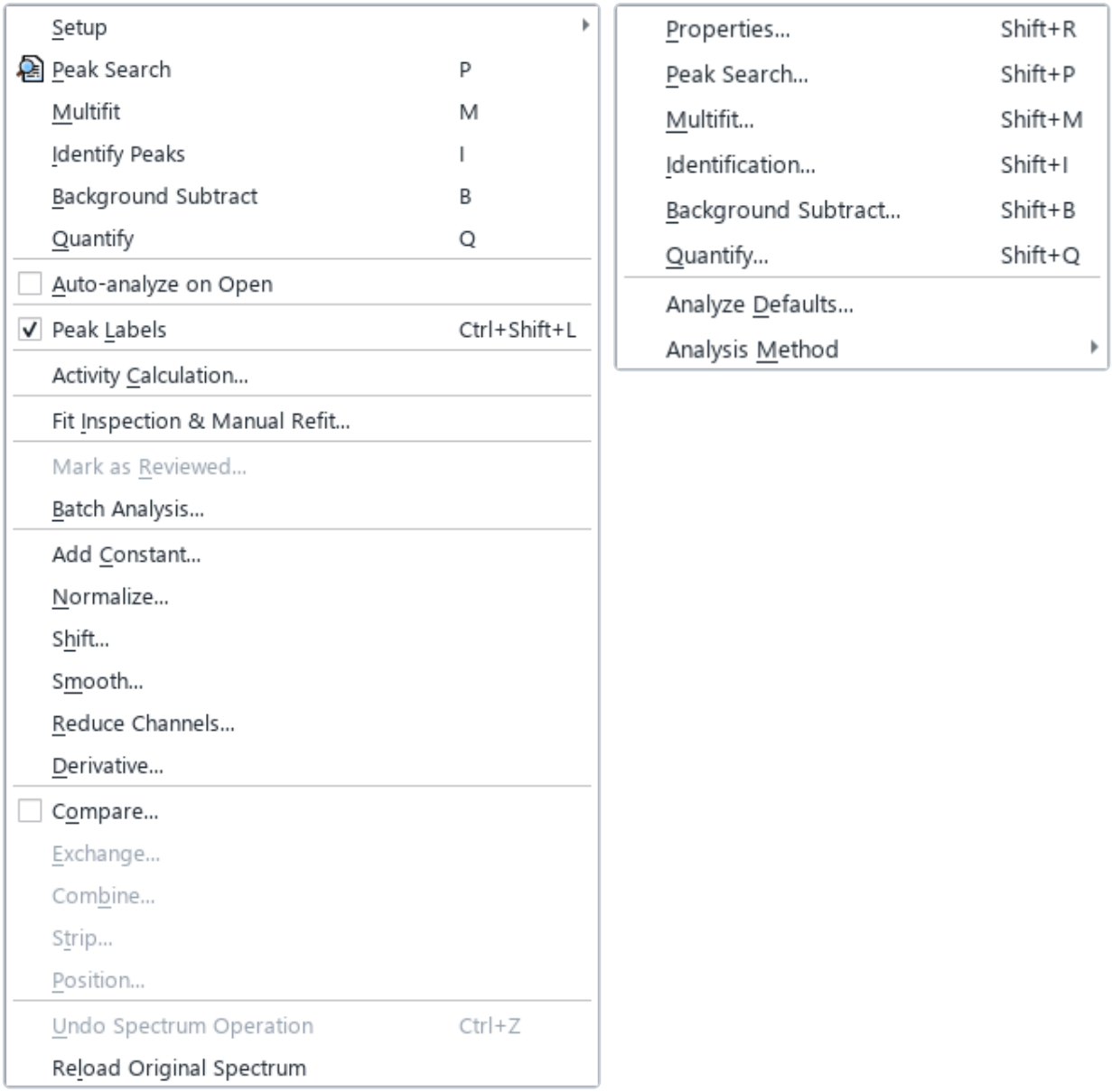


Figure 19 — The Analyze menu in Aptec's order and its Setup submenu.

Setup ›

ROI Properties (Shift + R), Peak Search (Shift + P), Multifit (Shift + M), Identification (Shift + I), Background Subtract (Shift + B), Quantify (Shift + Q), **Analyze Defaults** (factory defaults, or save and recall your own defaults for all of them at once) and **Analysis Method** (Aptec or SpectraMaster).

Peak Search (P)

Sets ROIs on the peaks, rejects every ROI whose 1-sigma net error is above the **Max Peak Error** and recalculates the backgrounds until the result is stable; the search can be limited to a channel range (*Partial*). **Note:** the search follows the Aptec help but does not reproduce Aptec's ROIs exactly: it finds about 79 % of them (section 7.8). Always review the ROIs and add or delete with F3 and Del.

Multifit (M)

Decomposes each ROI into Gaussian peaks by Marquardt-Levenberg least squares. An ROI is a multiplet when it is wider than the resolution calibration expects or shows several significant centroids. The width is fixed from

the resolution calibration (or varying) and varies automatically when 511 keV lies in the ROI; the background is fixed from the end points (or varying). A peak is kept when its amplitude ≥ K × its uncertainty (K = 1.645). Without a resolution calibration one is made from the ROI widths. Each fitted peak becomes its own ROI, flagged `m` (a fitted singlet `s`), so Identify and Quantify use the separated peaks; each Gaussian and their sum are drawn. Changing the ROI Properties removes the fits.

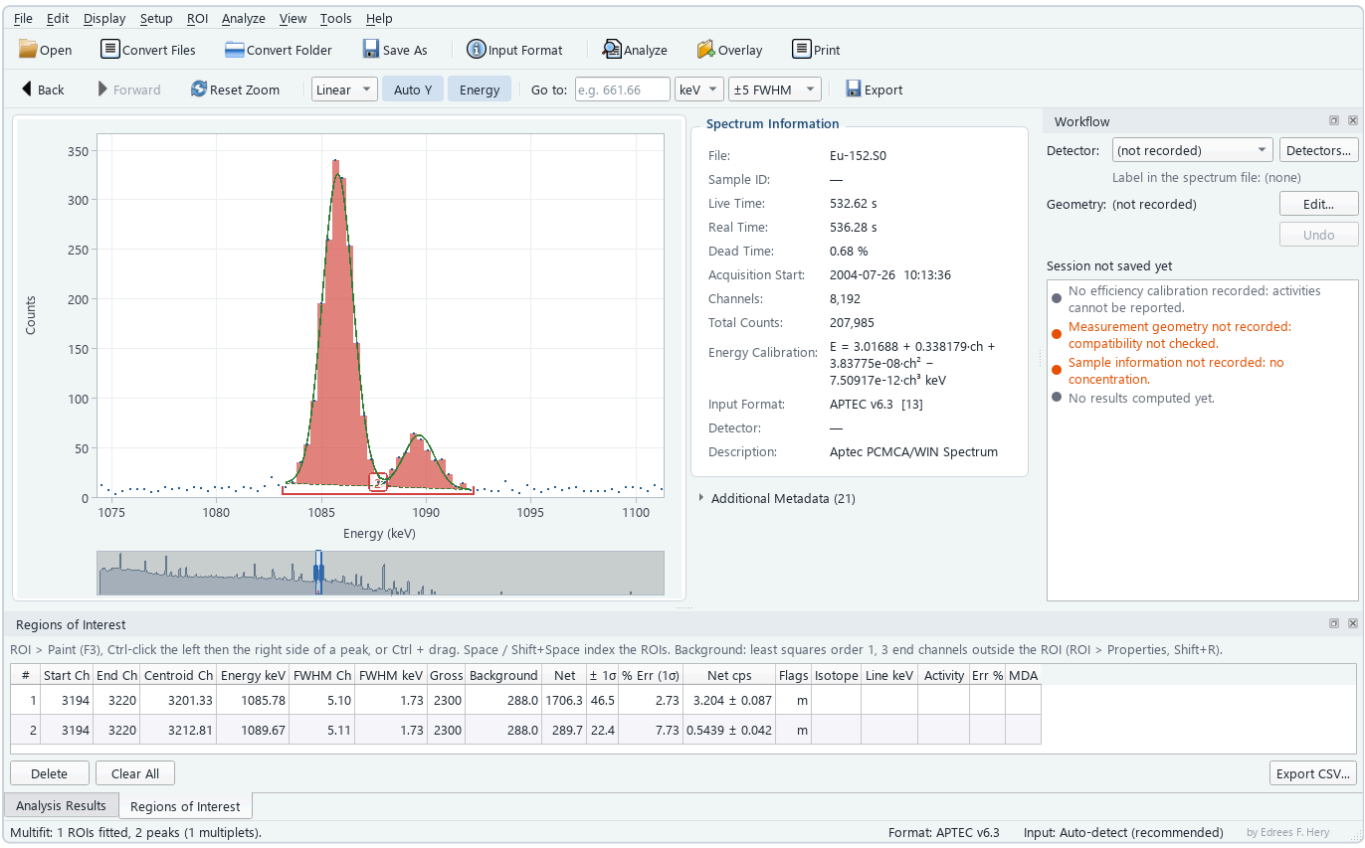


Figure 20 — The Eu-152 doublet 1085.8 / 1089.7 keV after `M`: two Gaussians (dashed) and their sum.

Identify Peaks (`I`)

With an Aptec library (`isotope.lib`) or SpectraMaster's bundled library. Library lines within the **tolerance** of the ROI centroid energy are candidates: a fraction of the ROI FWHM (default 0.5) or a fixed tolerance in keV. An isotope is accepted when its lines with an ROI carry at least the **Fraction Limit** (%) of the yield of all its lines inside the spectrum's energy range. Isotopes decayed beyond the **Half Life Limit** can be rejected.

Background Subtract (`B`)

Choose the background spectrum in `Shift` + `B`. It is searched and identified with the same setups, and the net count rate of the same isotope line is subtracted when it is above its MDA; uncertainties are combined in quadrature (flag `b`). No background-subtracted Aptec file was available to compare with.

Quantify (`Q`)

Activity = net rate / (efficiency × yield), decay-corrected from the count start to the **activity time** (`F6`) and converted to the chosen units; as **activity** in the sample, **concentration** (divided by the quantity) or **total activity** (concentration × total quantity), multiplied by the W/X/Y constants when selected. Emission units (gamma/s) are not divided by the yield. The efficiency comes from the spectrum's efficiency calibration or a saved curve, **at the ROI centroid energy** as Aptec does.

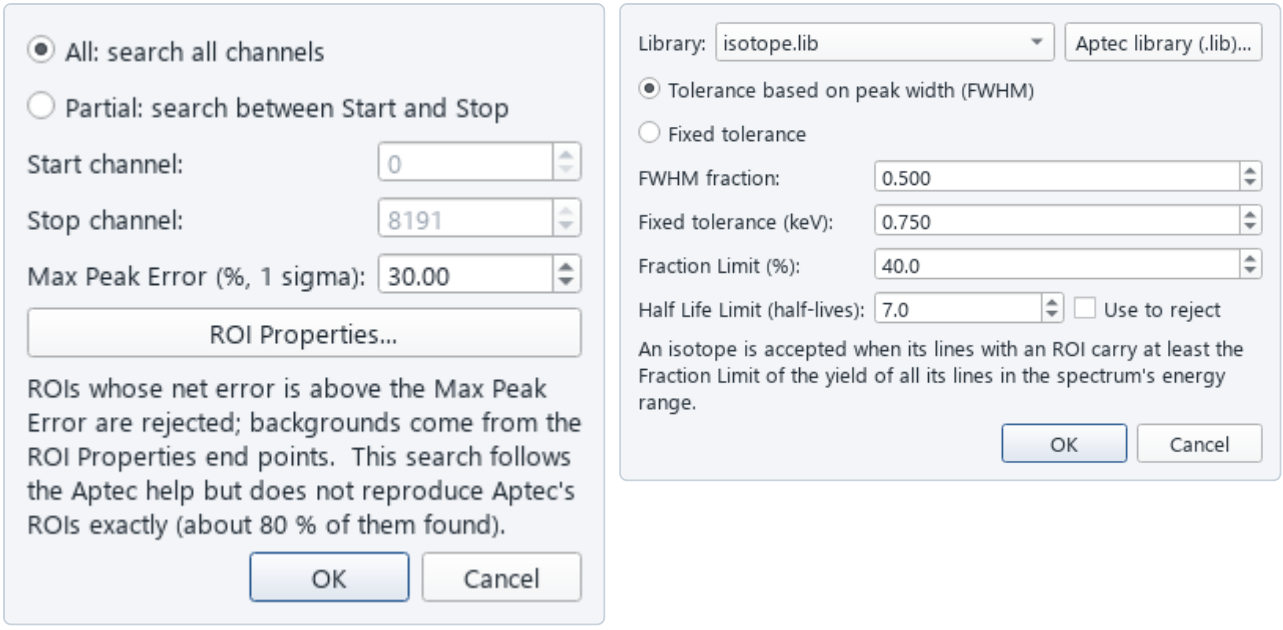


Figure 21 — Peak Search setup (Shift+P) and Identification setup (Shift+I).

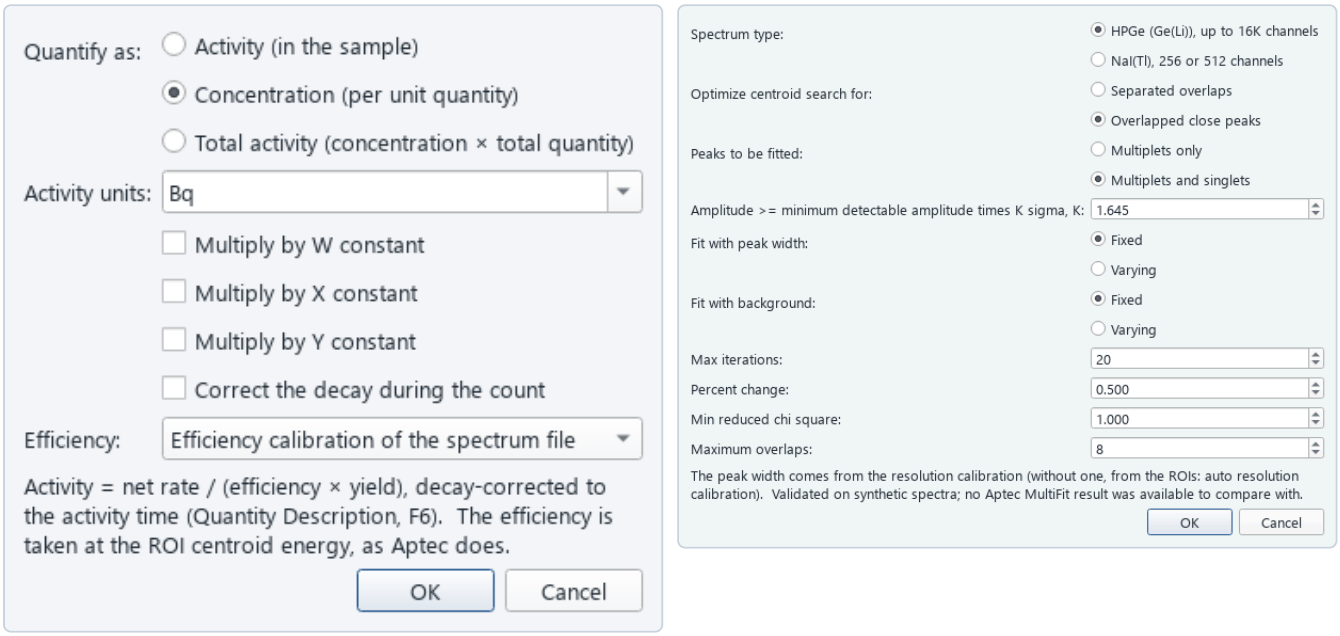


Figure 22 — Quantify setup (Shift+Q) and Multifit setup (Shift+M).

Results in the ROI list

The *Regions of Interest* list gains **Isotope**, **Line keV**, **Activity**, **Err %** and **MDA**, all exported to CSV. Aptec's flags in the Flags column:

`?`	Net rate (or activity) below the MDA.
`m` / `s`	Peak of a fitted multiplet / fitted singlet (Multifit).
`b`	Background subtracted.
`t`	Not decay-corrected over the count (count longer than 7 half-lives).
`>`	Decay correction to the activity time overflowed.

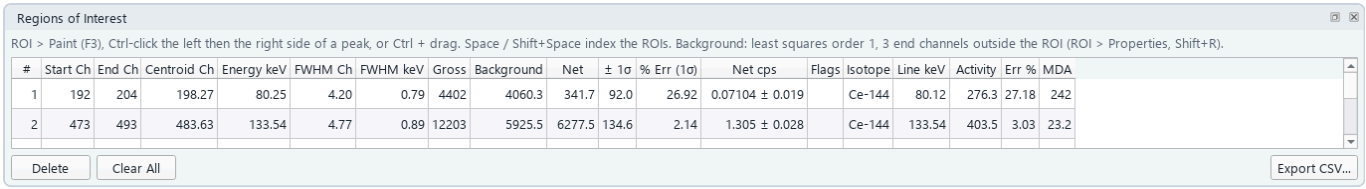


Figure 23 — The ROI list after Identify and Quantify.

7.5 Spectrum operations (lower Analyze menu)

- **Add Constant**, **Normalize** (counts and live / true times; with a compare spectrum the live-time ratio is proposed), **Shift** by whole channels, **Smooth** over 3, 5, 7 or 9 points, **Reduce Channels** (calibrations and ROIs follow), **Derivative** 1st / 2nd / 3rd ("Original data" returns).
- **Compare** another spectrum file over the displayed one, **Exchange** them, **Combine** / **Strip** the compare spectrum channel by channel (after a confirmation showing both live times) and **Position** it vertically (also Ctrl + ↑ / ↓).
- **Undo Spectrum Operation** (Ctrl + Z, 20 steps) and **Reload Original Spectrum**.

> The operations change only the spectrum on the screen; the file is unchanged until you save it (**File › Save As**). After smoothing, stripping, a derivative or a normalization the counts are no longer Poisson counts and uncertainties from √counts are not valid (the status bar warns). Aptec does not document its smoothing weights (binomial here) or derivative formula, so these were not matched to Aptec.

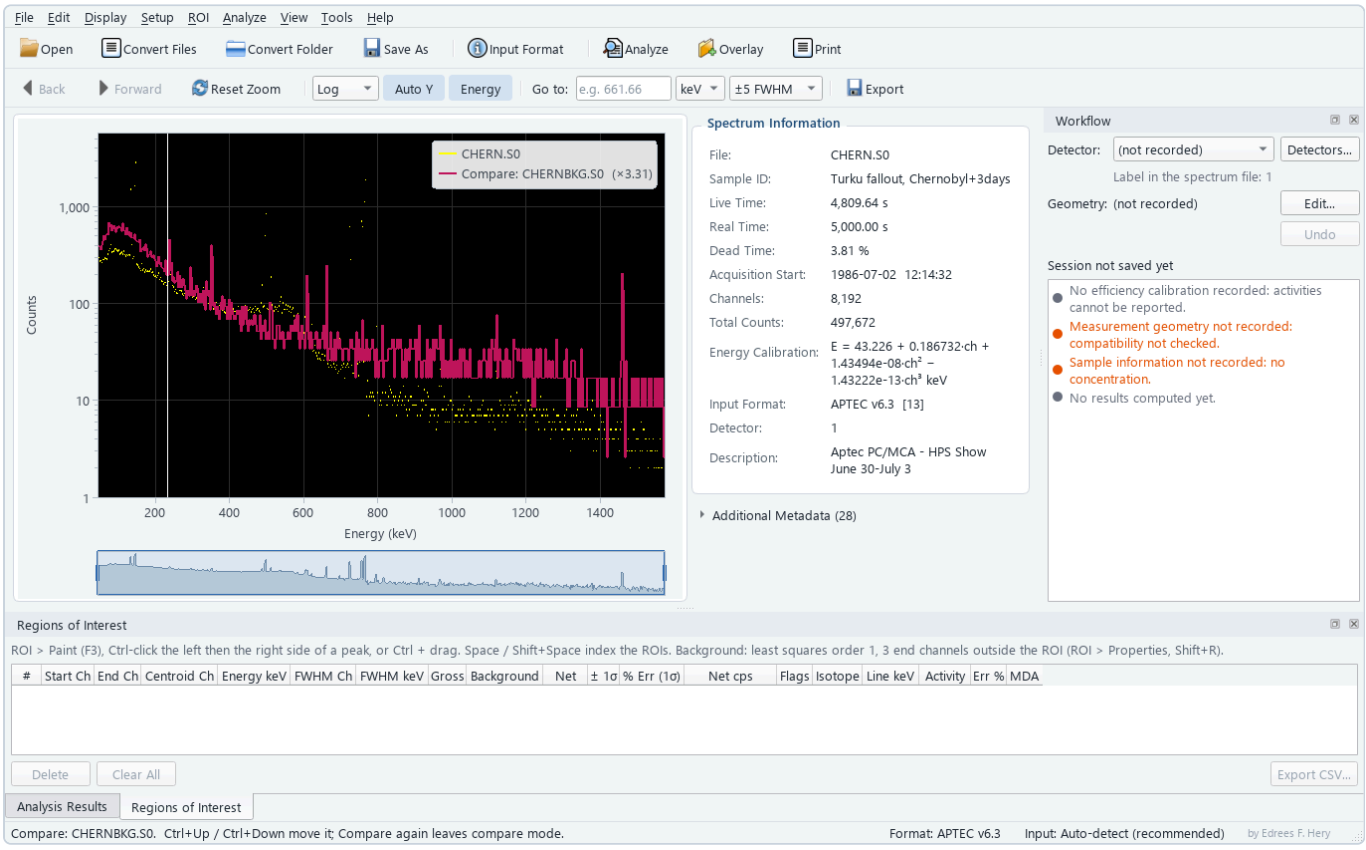


Figure 24 — Compare mode: the background spectrum CHERNBKG over the sample spectrum, with Aptec's colours (yellow on black).

7.6 View menu

Preferences, **Color** (default, black & white for printing, or **Aptec colors**), the **QA Control Bar** (Ctrl + 8), **Default Position**, **Save Settings on Exit** and the **split display**, horizontal (H) or vertical (V). The spectrum is drawn as **points** as in Aptec; toggles: Plot All Points (L), Fill Spc (F), Grid (G), Connect Points (N), Show Fitted ROIs (K), Fill ROIs (Shift + F), ROI Data box (D), ROI Display (Shift + D), ROI Labels (J), Channel Marker (Shift + J), FWHM Percent (%). Vertical scale: auto (A), auto zero (Z), next / previous type (S / Shift + S); horizontal: channels (C) or energy (E). The arrow keys move the channel marker, PgUp / PgDn the spectrum window, Esc shows the whole spectrum. **More View Options** holds SpectraMaster's own view commands (zoom history, Go to, overlays, peak labels and the panels). All keys: section 5.1.

7.7 Edit and Display menus

- **Edit**: Select All / Select Region of the display and Copy to ClipBrd as a picture (Ctrl + Ins, clear with Ctrl + Del); **Sample Description** (F5: title, IDs, sampling times); **Quantity Description** (F6: quantity and units, W/X/Y constants, activity units and time, total quantity); **Spectrum Data** (correct the acquisition times, with care); **Certificate** (calibration-source activities); **Library**; **Decay Correction Calculation**; **Load Header Information** from another spectrum file (calibrations, descriptions, ROIs); **Reset Header**.
- **Display**: the normal spectrum, or the energy, resolution or efficiency calibration with its points (Space indexes the points and shows the calculated value and deviation).

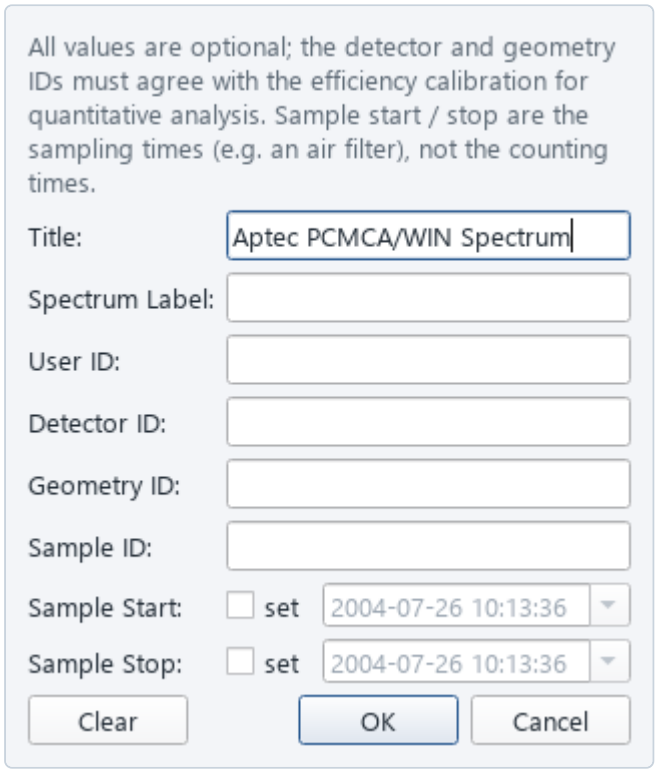


Figure 25 — Sample Description (F5).

7.8 Comparison with Aptec

Each step was compared with the results Aptec stored in its spectrum files (the read-only laboratory archive; only numbers were published), with success limits set **before** measuring and evaluated once on a separate test set:

Step	Result	Verdict
ROI statistics	gross 79/79; background, net, error 73/79	agree
Peak Search (4203 spectra)	finds 79 % of Aptec's ROIs, precision 62 %, edges 2 channels apart	**fail** (limits 90 %, 85 %, 1 channel)
Identify (736 ROIs)	agrees with Aptec for 88 %	**fail** (limit 95 %)
Quantify (49 ROIs)	49/49 within 0.003 % once the efficiency is taken at the centroid energy as Aptec does (45/49 with the pre-registered method)	agree (correction found afterwards)
Multifit	no Multifit result in any file; synthetic validation: bias ≤ 2.8 % for doublets ≥ 1 FWHM apart	pass (synthetic)
Background Subtract, smoothing, derivative	no Aptec data to compare with	not verified

Details in `docs/baseline/aptec_*` of the program's repository. Agreement with Aptec does not prove either program correct and does not replace the laboratory's validation procedures.

8. Energy calibration, drift check and spectrum arithmetic

The input files are never modified: a calibration is applied to the displayed spectrum (saved only with **File › Save As** or in the session), and the command-line tools write new files.

8.1 Energy and FWHM calibration from a source

Setup › Advanced Calibration Tools › Energy Calibration… proposes a calibration from the lines of a known source in the displayed spectrum.

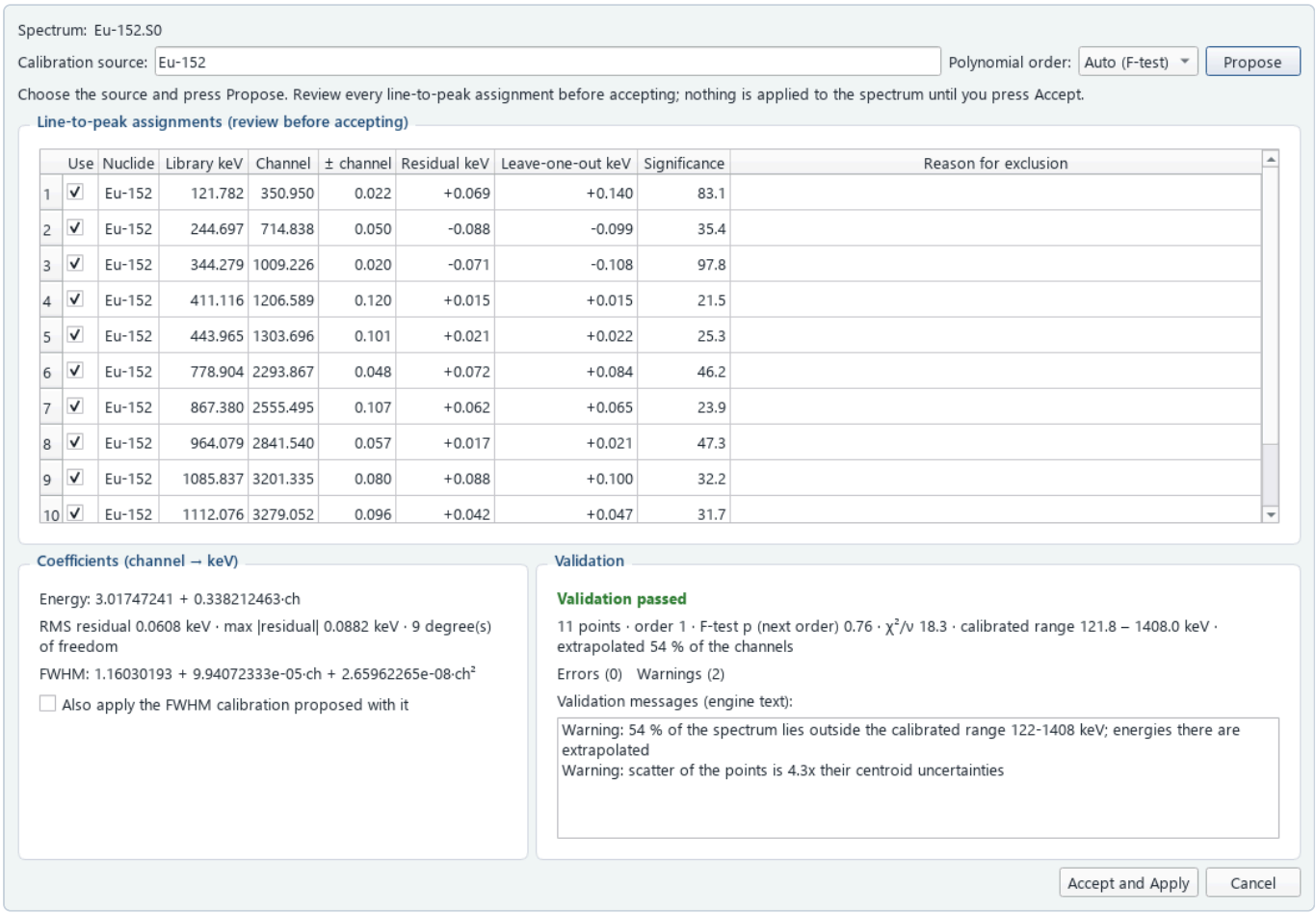


Figure 26 — Energy calibration of the Eu-152 spectrum: 11 lines assigned, order 1, validation passed with two warnings.

1. Type the **calibration source**: one nuclide (`Eu-152`), several (`Co-60,Cs-137`), `mixed` or `background` (natural lines). Choose the **polynomial order** (*Auto (F-test)* or 1–3) and press **Propose**. The search runs in the background.
2. **Review every line-to-peak assignment**: library energy, channel ± its uncertainty, residual, the leave-one-out residual (the line predicted by a fit without it) and the significance. Untick a wrong assignment and give the **reason for exclusion**; the calibration is refitted.
3. Read the **validation**: number of points, order, F-test, χ^2/ν, the calibrated range and how much of the spectrum lies outside it, with errors and warnings in the engine's words. **Accept and Apply** is disabled while there are errors.
4. Optionally tick **Also apply the FWHM calibration proposed with it**. The FWHM calibration is fitted through the **measured** widths of the lines (each line fitted with a free width, its neighbours together): the

511 keV annihilation line is left out, outliers more than 10 % off are removed, and a second order is used only if it rises with energy and stays positive over the whole spectrum. A first-order fit that falls with energy or reaches zero (often a doublet among the lines) gives no FWHM calibration; the reason is shown as the tooltip of the FWHM line. Fewer than two measured widths give no FWHM calibration. When you exclude a line or change the order, the FWHM calibration is fitted again through the widths of the lines still used.

5. **Accept and Apply**. Nothing is changed before this step.

8.2 Calibration drift check

Setup › Advanced Calibration Tools › Check Calibration Drift... measures the energies of reference lines in the displayed spectrum and compares them with the current calibration. Empty reference lines mean the natural background lines (K-40, U and Th series), which every long measurement contains.

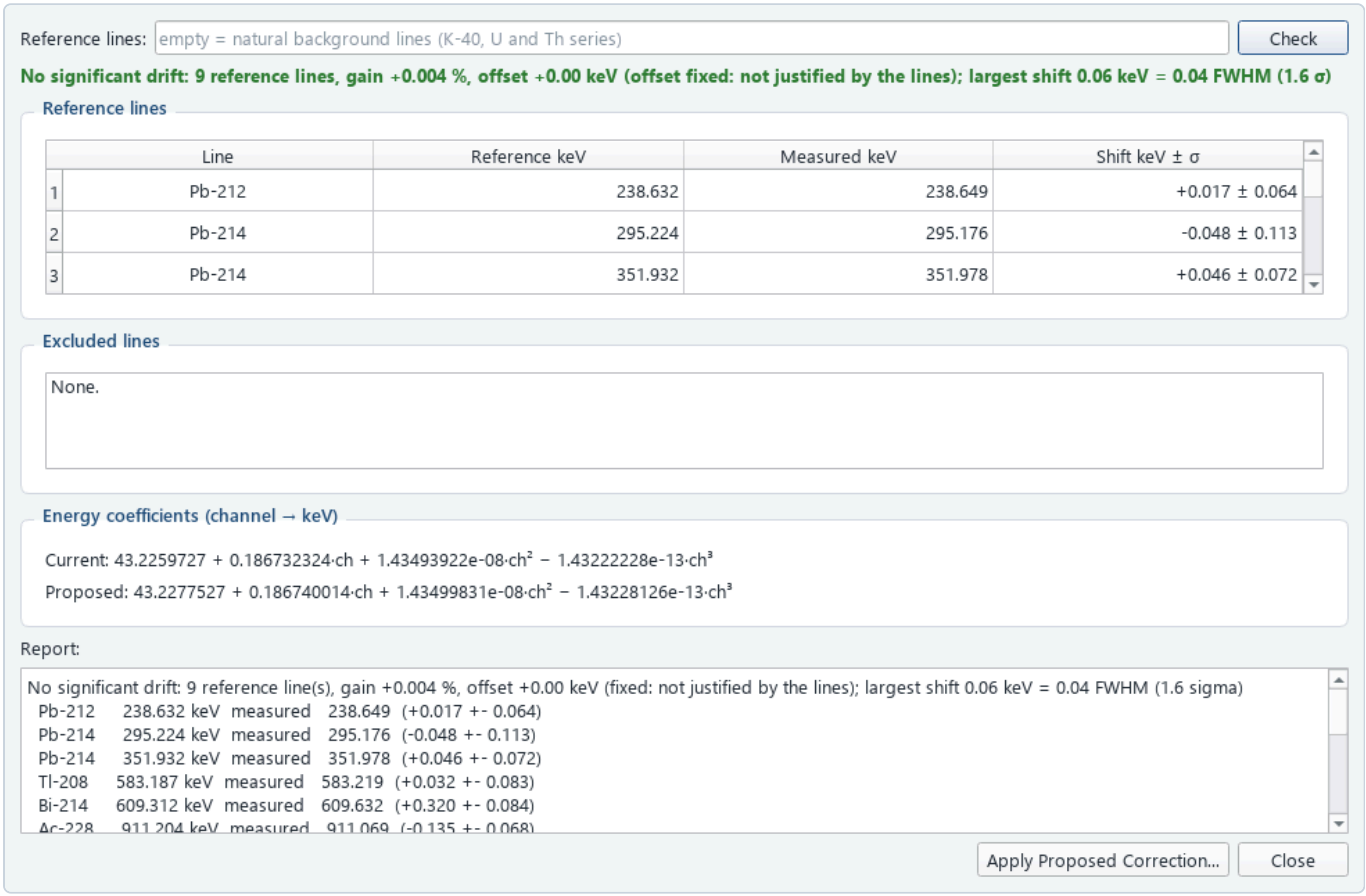


Figure 27 — Drift check of a soil spectrum against the natural lines: no significant drift (gain +0.004 %, largest shift 0.06 keV = 1.6 σ).

The result gives the gain and offset change, the largest shift in keV, in FWHM and in σ, the lines excluded and why, and the current and proposed coefficients. **Apply Proposed Correction...** replaces the energy calibration of the displayed spectrum after a confirmation, which says so when no significant drift was found; the change is recorded in the spectrum's calibration note. Apply it only when the drift is significant.

8.3 Calibration on the command line

```
smconvert calibrate Eu-152.S0 --source Eu-152                propose a calibration from source lines
smconvert calibrate *.S0 --source Co-60,Cs-137 -o D:\Recal     write recalibrated copies
smconvert calibrate X.spe --pairs 1893.7:661.66,3364:1173.23,3812:1332.49 --order 2 -o D:\Recal
smconvert calibrate X.spe --coeffs 0.12,0.3501,1.2e-8 -o D:\Recal  apply known coefficients
smconvert calibrate *.chn --source Eu-152 --only-uncalibrated -o D:\Recal
```

`--source` works as in section 8.1 (energy and measured-width FWHM calibration) and prints a table with channel, given and fitted energy and residuals. Without `-o` nothing is written. `--fwhm ch:keV,…` fits a FWHM calibration from given pairs.

```
Energy calibration, order 1: c0=3.01747241  c1=0.338212463
  #     channel       given      fitted   residual
  1     350.950     121.782     121.713    +0.0688  Eu-152 121.78
  ...
RMS residual 0.0608 keV, max |residual| 0.0882 keV, 9 degree(s) of freedom
Fwhm calibration, order 2: c0=1.16030193  c1=9.94072333e-05  c2=2.65962265e-08
```

8.4 Sum of spectra

```
smconvert sum A.S0 B.S0 C.S0 -o SUM.spe [--align auto|channel|energy]
```

Counts, live and real times are added. With different calibrations the spectra are aligned in energy (rebinned onto the first spectrum's calibration); `--align channel` forces a channel-by-channel sum.

8.5 Background subtraction

```
smconvert subtract CHERN.S0 CHERNBKG.S0 -o CHERN_net.spe [--scale K] [--clip] [--sigma-csv net.csv]
net = sample - 3.31379 x background; net counts 301801 +- 1071
```

The background is scaled by $LT_{sample}/LT_{background}$ unless `--scale` is given. `--clip` sets negative channels to zero; `--sigma-csv` also writes channel, energy, net counts and the propagated 1σ per channel. A subtracted spectrum is no longer Poisson: for activities, analyse the measured spectrum with its background instead (section 9.2).

8.6 Rebinning and truncation

```
smconvert rebin CHERN.S0 --factor 2 -o CHERN_4k.spe          8192 -> 4096 channels
smconvert rebin X.S0 --start 100 --stop 4196 --renumber -o part.spe
```

The calibration is transformed so that energies stay correct.

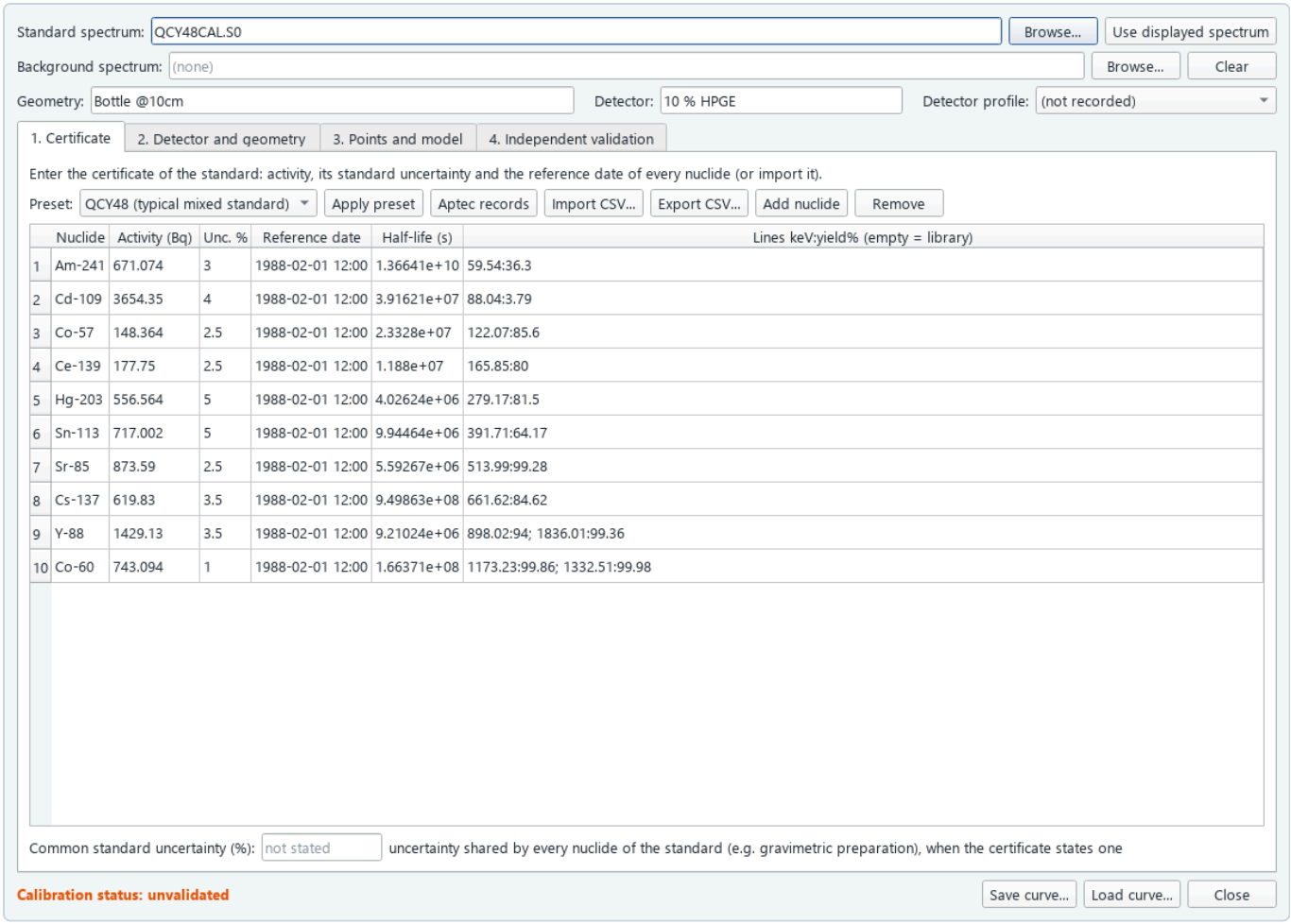
9. Efficiency calibration and activity

> **Before you start.** The efficiency curve is valid only for the **same detector, geometry (container, filling height, distance) and matrix density** as the standard. True-coincidence summing (TCS) is corrected only when you enable it and give the peak-to-total ratio of your detector and geometry (section 9.5); otherwise cascade emitters such as Co-60, Cs-134, Y-88 and Eu-152 are flagged. A density different from the standard's is corrected only on request (section 9.6). Results must be validated by the laboratory.

9.1 Efficiency calibration

Setup › Advanced Calibration Tools › Efficiency Calibration... works in four tabs.

1. **1. Certificate.** Open the standard spectrum (here `QCY48CAL.S0`, 10 ml bottle at 10 cm, HPGe) or **Use displayed spectrum**; optionally a **background spectrum**. One row per nuclide: activity (in the **Activity unit** chosen above the table: Bq, kBq, MBq, GBq, TBq, Ci, mCi, µCi, nCi or pCi; changing it converts the column, and the calibration always uses Bq), uncertainty %, reference date, half-life (s) and optionally the lines `keV:yield%` (empty = library values). Fill it from the **Aptec records** (the source records stored in Aptec `.S0` files), a **Preset** (QCY48, QCRB1241, Eu-152 point source), **Import CSV...** or by typing. A **common standard uncertainty** shared by all nuclides (e.g. gravimetric preparation) can be given once.
2. **2. Detector and geometry.** The detector profile and the geometry of the standard: kind (Marinelli, point source, cylinder / bottle, air filter), container, volumes, diameters, fill height, matrix, density, distance to the end cap, filter dimensions and position. Unknown values stay empty — they are never taken as zero. The geometry is stored with the curve and checked against the sample when activities are calculated.
3. **3. Points and model.** Choose the **model**: *two-branch log-log* (recommended for HPGe: low- and high-energy branches joined at a crossover energy) or a single *log-log polynomial*; orders and crossover *Auto (cross-validation)* or explicit. Click **Measure & Fit**: each line becomes a point (net area, efficiency, uncertainty, fitted value, residual). With a background spectrum the peaks are fitted on sample − k × background with its propagated variance. Untick **Use** for a doubtful point (a reason is asked and recorded) and press **Refit**; **Add manual point** adds a point by hand. Amber rows are cascade emitters (section 9.5).
4. **4. Independent validation.** Measure a second source that was **not** used in the fit, with the same detector and geometry: **Measure and compare** lists measured and curve efficiency, deviation and z for each line. Only when the check is consistent can the curve be **Marked as validated** (with a note: who, which source, reference). The activity dialog shows whether the curve used is validated.
5. Check χ^2/ν and the residual plot (points should scatter within their error bars), then **Save curve...** The curve (JSON) is stored in `%APPDATA%\SpectraMaster\efficiency\` and appears in the activity dialog.



	Nuclide	Activity (Bq)	Unc. %	Reference date	Half-life (s)	Lines keV:yield% (empty = library)
1	Am-241	671.074	3	1988-02-01 12:00	1.36641e+10	59.54:36.3
2	Cd-109	3654.35	4	1988-02-01 12:00	3.91621e+07	88.04:3.79
3	Co-57	148.364	2.5	1988-02-01 12:00	2.3328e+07	122.07:85.6
4	Ce-139	177.75	2.5	1988-02-01 12:00	1.188e+07	165.85:80
5	Hg-203	556.564	5	1988-02-01 12:00	4.02624e+06	279.17:81.5
6	Sn-113	717.002	5	1988-02-01 12:00	9.94464e+06	391.71:64.17
7	Sr-85	873.59	2.5	1988-02-01 12:00	5.59267e+06	513.99:99.28
8	Cs-137	619.83	3.5	1988-02-01 12:00	9.49863e+08	661.62:84.62
9	Y-88	1429.13	3.5	1988-02-01 12:00	9.21024e+06	898.02:94; 1836.01:99.36
10	Co-60	743.094	1	1988-02-01 12:00	1.66371e+08	1173.23:99.86; 1332.51:99.98

Figure 28 — Efficiency calibration of QCY48CAL.S0 from the Aptec source records: two-branch model, $\chi^2/\nu = 0.41$.

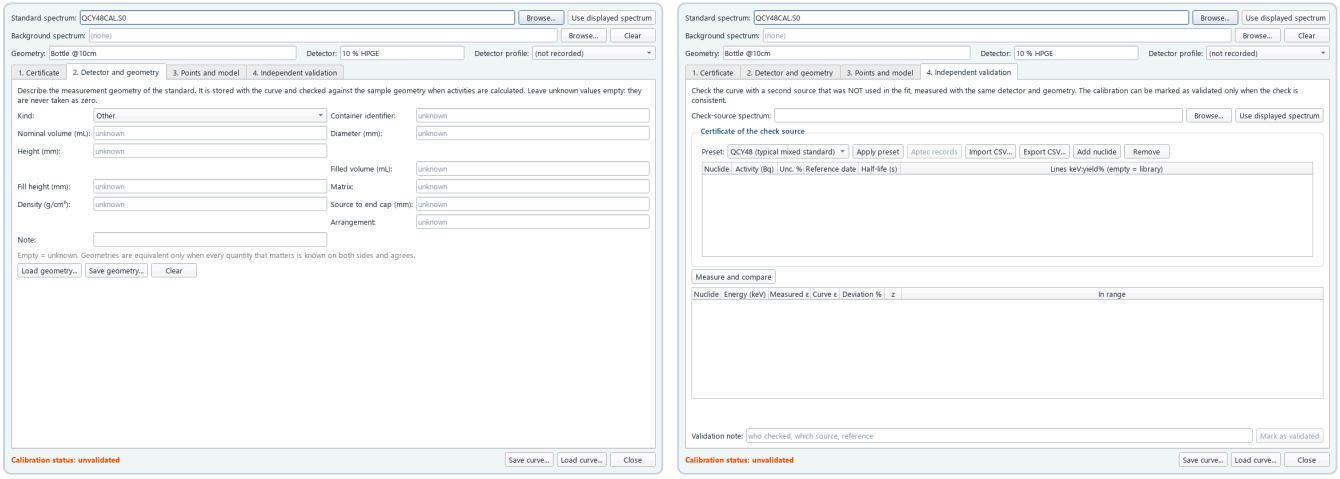
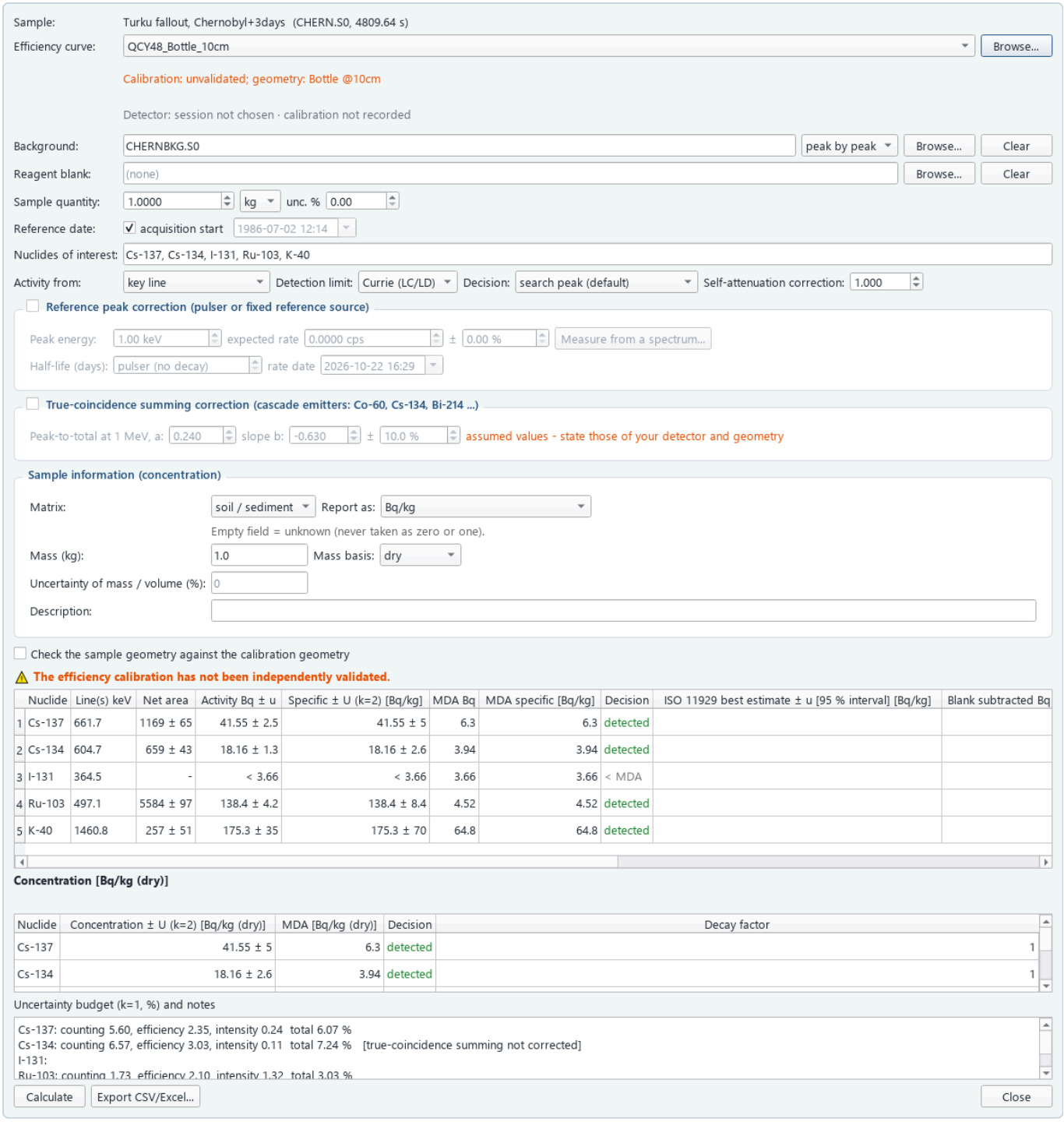


Figure 29 — Tabs 2 (detector and geometry) and 4 (independent validation).

Point efficiency: $\varepsilon = N / (t_{live} \cdot A(t_0) \cdot C \cdot I)$, with the activity decay-corrected from the reference date to the start of counting and C the decay during counting. The curve uncertainty combines the fit covariance (scaled by χ^2/ν when > 1), the part of the certificate uncertainty common to all lines and a **model error** from leave-one-out fits. A single outlying point is limited to 3 times the curve's scatter in this model error and named in the curve warnings, so it cannot inflate every result.

9.2 Activity calculation

1. Open the sample spectrum (calibrated in energy) and choose **Analyze** › **Activity Calculation**….
2. Select the **efficiency curve** (its validation status and geometry are shown). Optionally load a **background** spectrum: *peak by peak* or *full spectrum*. In both modes the peaks are fitted on sample − k × background (k = ratio of live times) with the variance sample + k^2 × background; in peak mode the background under each line is also reported.
3. Enter the **sample quantity** and unit (kg, g, L, mL) with its uncertainty, and the **reference date** (sampling date; default = acquisition start). All activities are decay-corrected to this date.
4. List the **nuclides of interest**; choose *key line* or *weighted mean of lines*, the **detection limit** (Currie L_C/L_D or ISO 11929), the **decision** rule (below) and, if needed, a manual **self-attenuation correction** factor (default 1.000; section 9.6).
5. Optional corrections: a **reagent blank** (9.3), a **reference peak** (9.4) and **true-coincidence summing** (9.5). Fill in the **sample information** to get concentrations (9.7) and, if the curve has a geometry, tick **Check the sample geometry against the calibration geometry**: a calibration of another geometry is then refused unless you accept it explicitly.
6. Click **Calculate**. The calculation runs in the background (the dialog shows *Calculating …*). **Export CSV/Excel…** saves the table and the uncertainty budget.



	Nuclide	Line(s) keV	Net area	Activity Bq ± u	Specific ± U (k=2) [Bq/kg]	MDA Bq	MDA specific [Bq/kg]	Decision	ISO 11929 best estimate ± u [95 % interval] [Bq/kg]	Blank subtracted Bq
1	Cs-137	661.7	1169 ± 65	41.55 ± 2.5	41.55 ± 5	6.3	6.3	detected		
2	Cs-134	604.7	659 ± 43	18.16 ± 1.3	18.16 ± 2.6	3.94	3.94	detected		
3	I-131	364.5	-	< 3.66	< 3.66	3.66	3.66	< MDA		
4	Ru-103	497.1	5584 ± 97	138.4 ± 4.2	138.4 ± 8.4	4.52	4.52	detected		
5	K-40	1460.8	257 ± 51	175.3 ± 35	175.3 ± 70	64.8	64.8	detected		

Nuclide	Concentration ± U (k=2) [Bq/kg (dry)]	MDA [Bq/kg (dry)]	Decision	Decay factor
Cs-137	41.55 ± 5	6.3	detected	1
Cs-134	18.16 ± 2.6	3.94	detected	1

Figure 30 — Activities of CHERN.S0 (Turku fallout, 1986) with the QCY48 curve, peak-by-peak background subtraction and concentrations in Bq/kg (dry).

Activity Bq ± u	Activity with its combined standard uncertainty (k = 1).
Specific ± U (k=2)	Activity per kg (or L …) with the **expanded** uncertainty U = 2u.
MDA	Minimum detectable activity (Bq and per unit quantity), from the continuum under the expected peak (Currie: $L_D = k^2 + 2L_C$, k = 1.645; ISO 11929: detection limit y#). With *weighted mean of lines*, the decision threshold and the MDA are those of the weighted mean itself, with the same weights as the activity.
Decision	*detected* when the net signal exceeds the decision threshold (Currie L_C or ISO y*); otherwise < *MDA*. A nuclide can be *detected* with an activity below its MDA (between L_C and L_D) — report it with its uncertainty.
ISO 11929 best estimate	The best estimate with its uncertainty and 95 % coverage interval, for every nuclide, also when it is not detected.
Primary result	For a nuclide **not detected**, the signed window result G − B (gross minus continuum in the line's window), which may be negative. It is given for information; the activity column still shows < *MDA*. Window results next to a found peak (*interfered*) or on a curved continuum where neither a straight nor a quadratic continuum fits (*curved*) are flagged and not used. The nuclide's primary result is combined like the activity: the key line, or with *weighted mean of lines* the weighted mean of the usable lines; when the key line is flagged, the usable line with the smallest MDA is used and named in the note.
Blank subtracted	The reagent-blank activity subtracted from the line (section 9.3).
Budget	Relative components (k = 1): counting, efficiency, emission probability, quantity, attenuation, and, when used, the reagent blank, the reference peak and the summing correction.

Decision rule

- **Search peak (default):** a nuclide is detected when the peak search found its line above the decision threshold.
- **ISO 11929 (window result too):** a line without a search peak is also detected when its window result exceeds the decision threshold. This finds weak lines the search missed, at a higher false-detection rate.

If the sample spectrum was already background subtracted, stripped, normalized or smoothed, its channels are no longer Poisson counts. The results then carry a note that their uncertainties, decision thresholds and MDA are not valid. Analyse the measured spectrum with its background instead. A spectrum that was already subtracted is not subtracted again. Peaks edited by hand (section 6.7) are used; with a background they are refitted on sample − k × background and the notes say so.

9.3 Reagent blank

A **reagent blank** is a blank sample prepared and counted like the samples, with the same reagents, container and geometry. Choose its spectrum in **Reagent blank**. It is analysed exactly like the sample, with the same curve, options and background, and its activity is subtracted line by line. The blank's uncertainty, including the part of its reference-peak correction it shares with the sample, is in the budget, and the decision threshold includes it. A blank analysed with other options (attenuation, summing correction, reference peak, background) is reported in the notes. The spectrum, background, blank and peaks are taken as they are when **Calculate** is pressed. The session and the report record the blank file with its SHA-256 fingerprint.

9.4 Reference peak correction

The live time corrects dead time but not every loss: random summing and pile-up remove counts from every peak. A **reference peak**, from a pulser or a fixed reference source counted with every sample, measures this loss. Each result is multiplied by K = expected rate / measured rate of that peak in the sample spectrum.

- Enter the **peak energy** and its **expected rate** (counts per live second) with its uncertainty. **Measure from a spectrum…** takes the rate from a low-rate reference measurement, and its date becomes the rate date.
- For a radioactive reference source, give its **half-life** in days so the expected rate is decay corrected. Use 0 for a pulser.
- The peak is measured in the **measured** sample spectrum (before any background subtraction), as its expected rate is, so a pulser or source also present in the background is still found.
- If the peak is not found, the calculation is refused. The reagent blank, when given, is corrected the same way.

9.5 True-coincidence summing correction

Nuclides that emit several photons in cascade (Co-60, Cs-134, Bi-214, Y-88, Ba-133, Eu-152 …) lose counts from their peaks when two photons are detected together; positron emitters (Na-22, Co-58 …) also lose counts by summing with the 511 keV annihilation photons. Some lines gain counts this way. The loss grows as the source gets closer to the detector. With **True-coincidence summing correction** enabled, each line of a cascade emitter is divided by its first-order coincidence summing factor (COI), computed from coincidence data and the efficiency curve.

- **Coincidence data.** The probabilities come from the ENSDF decay schemes of the IAEA (LiveChart of Nuclides): for each pair of gamma rays, the probability that the second follows the first in the same decay, from the level scheme with the conversion coefficients, isomers ending the cascade; and for each gamma the probability that a positron (two 511 keV photons) comes with it. Where a level of the scheme receives more gamma transitions than it emits (the listed intensities do not balance), the probability that a gamma was preceded by another is taken from the level populations of the scheme, so it never exceeds 1. They were checked against an independent simulation of the decays (appendix E). Short-lived daughters folded into a nuclide (Bi-207 → Pb-207m, Ce-144 → Pr-144 …) are included.
- **Peak-to-total.** The total efficiency is the peak efficiency divided by the **peak-to-total ratio** PT(E) = min(1, $a \cdot (E / 1000\ \text{keV})^{b}$). PT depends on the detector **and** the geometry. The values shown are assumptions; enter those of your detector, measured with single-line sources or taken from a detector model. Their uncertainty (default 10 %) is propagated.
- **The calibration curve.** If the efficiency curve was measured with cascade emitters in the same geometry (for example a mixed standard with Ba-133 and Y-88), its points already contain summing losses. They are corrected with the same PT and the curve is refitted until it no longer changes; an automatically chosen model (orders, crossover) is chosen again on the corrected points. A curve without points (an imported Aptec ln-ln polynomial) is refused unless you declare it free of summing; the declaration is recorded in the notes.
- **Limits.** A line that the coincidence data do not describe is not corrected, and the note says why. A partner photon far below the calibrated range (under 0.8 × its lowest energy) is left out: its loss is not corrected, because its efficiency is unknown. When such partners make up more than 1 % of the decays, or when the decay scheme's level intensities disagree with the line intensities enough to change the factor by 1 % or more, the correction is marked **partial**: the line's note gives the reason, the nuclide's note names the lines,

and a calibration curve corrected with such points names them in its note and warnings. A partial correction is applied as computed; it is not a complete one. A line whose first-order factor is outside its validity gives no result; a summing-out loss above 10 % is noted, because the first-order correction degrades in very close geometries.

- **Not included:** X-ray – gamma summing (EC nuclides such as Ba-133 and Eu-152: the coincidence table has no X-rays, so this loss is not corrected whatever the range of the efficiency curve) and angular correlation. The session records the coincidence table with its SHA-256 fingerprint.

Command line: `--tcs-pt A,B`, `--tcs-pt-unc` and `--tcs-curve-summing-free`; `--blank FILE`; `--ref-peak keV --ref-rate cps` (or `--ref-spectrum FILE`, `--ref-half-life s`).

9.6 Self-attenuation

A sample denser than the calibration standard absorbs more of its own gamma rays, mostly at low energies.

- **In the activity dialog** a single manual factor (default 1.000) multiplies every result; enter it from your own determination.
- **On the command line** an energy-dependent correction is computed from the NIST mass attenuation coefficients (SRD 126) for a slab of the sample and of the standard: `--self-att-matrix` and `--self-att-density` (sample), `--cal-matrix` and `--cal-density` (standard), `--self-att-height-mm` (sample height) and `--self-att-unc` (model uncertainty, %). Matrices: `water`, `organic` (resins, organic matter) and `mineral` (soil, concrete). The factor of every line is recorded, and its uncertainty enters the budget. With true-coincidence summing, the summing of the sample is computed with its self-attenuated efficiency.

```
smconvert activity SAMPLE.chn --eff curve.json --self-att-matrix organic --self-att-density 1.6 ^
          --cal-matrix organic --cal-density 1.15 --self-att-height-mm 35
```

> The slab model is an approximation; the matrices differ by up to 1.5 % below 100 keV (mineral against organic). Choose the matrix that matches the sample. The manual factor and the energy-dependent model correct the same effect: the program refuses both together.

9.7 Sample information and concentrations

In **Sample information**, choose the **matrix** (soil / sediment, water, food, air filter, other) and the unit to report: Bq, Bq/kg, Bq/L or Bq/m³.

- **Bq/kg** needs the mass and its basis (dry or wet). **Bq/L** needs the volume.
- **Bq/m³** needs the air volume, either given directly or as flow rate × sampling duration. It also uses the measured fraction of the filter and the collection efficiency. The collection efficiency is an assumption, so state its source.
- An empty field means *unknown*: it is never taken as zero or one. A missing value is listed instead of a concentration.
- Non-detects are shown as < MDA, never as zero.

9.8 Worked example: QCY48CAL → CHERN (command line)

```
smconvert effcal QCY48CAL.S0 -o qcy48.json --model two_branch --compare
model: two-branch log-log, Ec = 583.2 keV, orders 3/2   chi2/dof = 0.41
nuclide       keV        net        eps    u %        fit  resid %
Am-241      59.54       3614  0.0050063   4.46  0.0049738     0.65
Cs-137     661.62      10908  0.0070282   3.71  0.0068801     2.15
Co-60     1332.51       6885  0.0031553   2.03  0.0031223     1.06  cascade emitter ...
comparison with Aptec's stored efficiencies:  Cs-137 661.62  Aptec 0.0071482  ours 0.0070282  -1.68 %

smconvert activity CHERN.S0 --eff qcy48.json --bkg CHERNBKG.S0 --nuclides Cs-137,Cs-134,I-131,Ru-103,K-
40 ^
          --mass 1 --unit kg -o chern.xlsx
Nuclide  Line(s) keV  Net area   Activity Bq (k=1)  Bq/kg (k=2)  MDA Bq  Decision
Cs-137   661.7        1169 ± 65  41.53 ± 2.5        41.53 ± 5.1  6.3     detected
Cs-134   604.7        659 ± 43   18.16 ± 1.3        18.16 ± 2.6  3.94    detected
I-131    364.5        -          < 3.66             < 3.66       3.66    < MDA
Ru-103   497.1        5584 ± 97  138.4 ± 4.2        138.4 ± 8.4  4.52    detected
K-40     1460.8       257 ± 51   175.6 ± 35         175.6 ± 70   64.9    detected
  Cs-134: true-coincidence summing not corrected
```

The last line appears because the summing correction was not requested; with `--tcs-pt A,B` the Cs-134 lines are corrected (section 9.5). Further notes list the background re-binning, its live-time ratio (k = 3.31) and the decision rule.

Without `--cert` the certificate is taken from the Aptec records in the `.S0` file; `--compare` lists the difference to Aptec's own stored efficiencies (here within ±6 %). A certificate CSV has the columns `nuclide,activity_bq,unc_percent,ref_date[,half_life_s,energy_kev,intensity_percent]`.

> **About this example.** The two files were measured in different set-ups (a 10 ml bottle on a 10 % HPGe and a rag wipe in another geometry); the example only shows the workflow. In practice the standard and the sample must share detector and geometry.

10. Reports, review and batch analysis

10.1 Report of the displayed spectrum

File › Print Report… (Ctrl + P), **Print Preview…** and **Export Report as PDF…** (Ctrl + Shift + P) first open the **Report Options** dialog. If the report should contain peaks or nuclides and the spectrum has not been analysed yet, the program offers to run the analysis; it runs in the background and the report continues by itself when it has finished (if the analysis fails, its error is shown and no report is written).

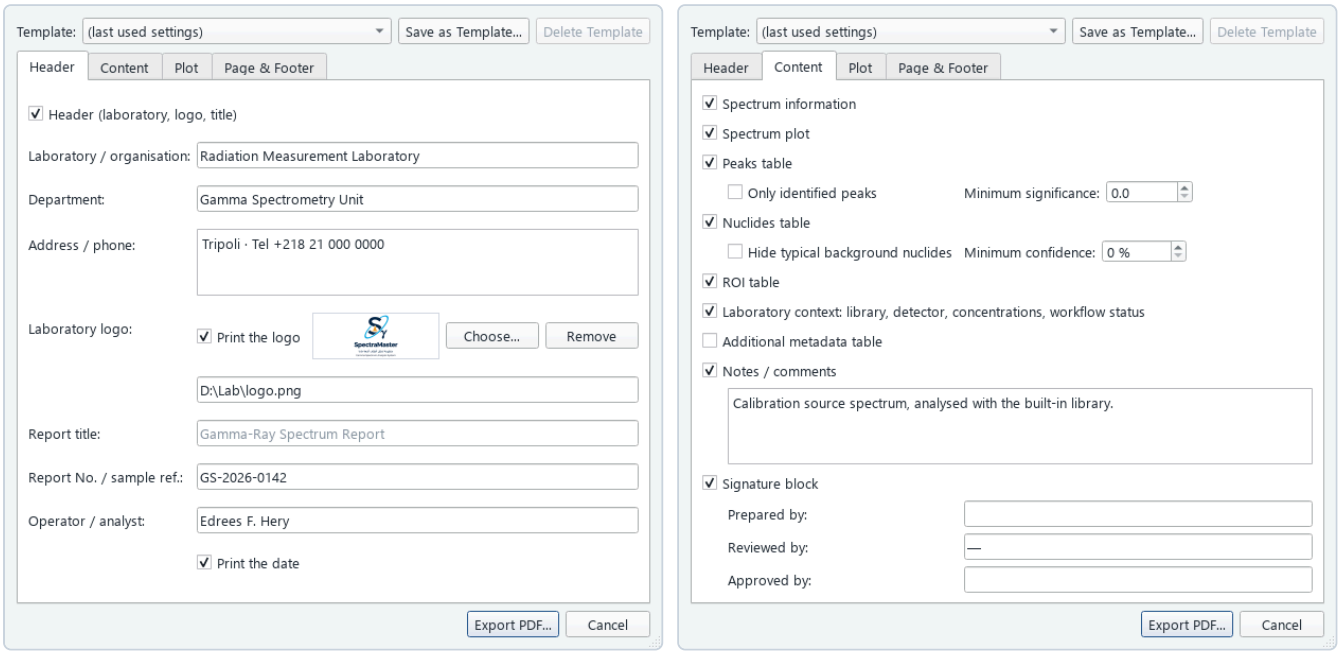
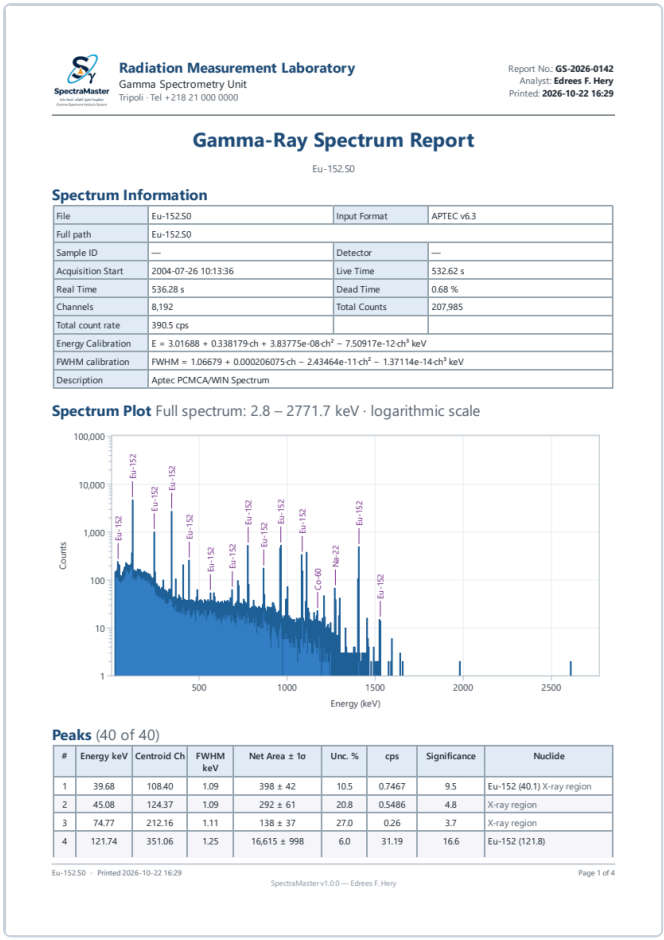


Figure 31 — Report options: Header and Content tabs.

Header	Laboratory, department, address/phone, laboratory logo, report title, report number / sample reference, analyst, date.
Content	Spectrum information, plot, peaks table (only identified peaks, minimum significance), nuclides table (minimum confidence, hide typical background nuclides), ROI table, additional metadata, the library, detector, geometry, concentrations and workflow statuses, notes, signature block (prepared / reviewed / approved).
Plot	Current zoom or full spectrum, counts scale, energy or channel axis, peak labels, ROI shading, overlays, plot height, vector graphics or 300 dpi image.
Page & Footer	Paper, orientation, margins, report language (Arabic, English or as the interface), footer items (page X of Y, file name, program credit, print date and time).



Radiation Measurement Laboratory
Gamma Spectrometry Unit
Tripoli · Tel +218 21 000 0000

Report No: **GS-2026-0142**
Analyst: **Edrees F. Hery**
Printed: **2026-10-22 16:29**

Gamma-Ray Spectrum Report

Eu-152.S0

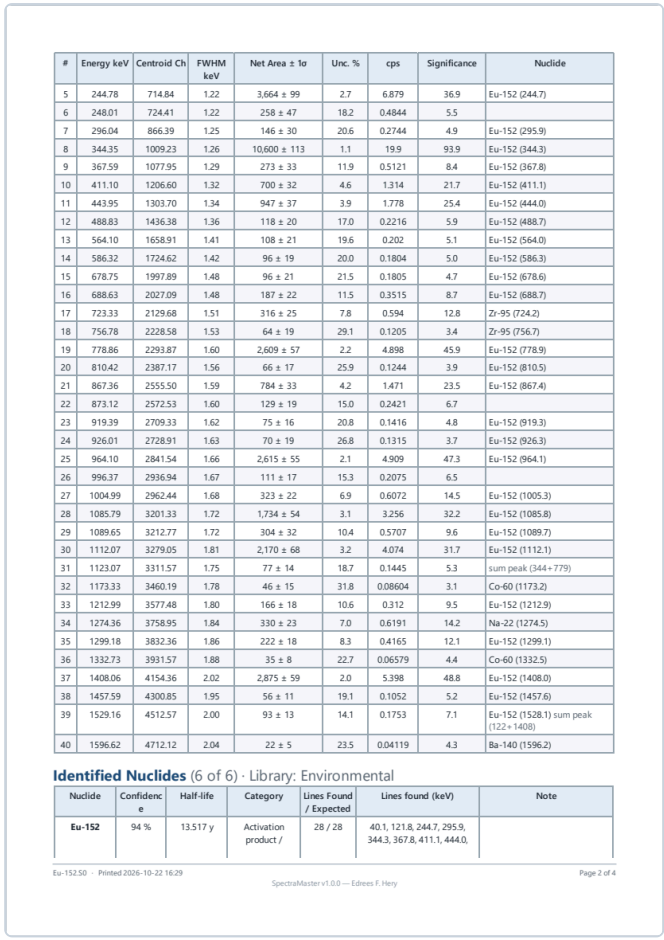
Spectrum Information

File	Eu-152.S0	Input Format	APTEC v6.3
Full path	Eu-152.S0		
Sample ID	—	Detector	—
Acquisition Start	2004-07-26 10:13:36	Live Time	532.62 s
Real Time	536.28 s	Dead Time	0.68 %
Channels	8,192	Total Counts	207,985
Total count rate	390.5 cps		
Energy Calibration	E = 3.01688 + 0.338179·ch + 3.83775e-08·ch² − 7.50917e-12·ch³ keV		
FWHM calibration	FWHM = 1.06679 + 0.000206075·ch − 2.43464e-11·ch² − 1.37114e-14·ch³ keV		
Description	Aptec PCMCA/WIN Spectrum		

Spectrum Plot Full spectrum: 2.8 – 2771.7 keV · logarithmic scale

Peaks (40 of 40)

#	Energy keV	Centroid Ch	FWHM keV	Net Area ± 1σ	Unc. %	cps	Significance	Nuclide
1	39.68	108.40	1.09	398 ± 42	10.5	0.7467	9.5	Eu-152 (40.1) X-ray region
2	45.08	124.37	1.09	292 ± 61	20.8	0.5486	4.8	X-ray region
3	74.77	212.16	1.11	138 ± 37	27.0	0.26	3.7	X-ray region
4	121.74	351.06	1.25	16,615 ± 998	6.0	31.19	16.6	Eu-152 (121.8)



#	Energy keV	Centroid Ch	FWHM keV	Net Area ± 1σ	Unc. %	cps	Significance	Nuclide
5	244.78	714.84	1.22	3,664 ± 99	2.7	6.879	36.9	Eu-152 (244.7)
6	248.01	724.41	1.22	258 ± 47	18.2	0.4844	5.5	
7	296.04	866.39	1.25	146 ± 30	20.6	0.2744	4.9	Eu-152 (295.9)
8	344.35	1009.23	1.26	10,600 ± 113	1.1	19.9	93.9	Eu-152 (344.3)
9	367.59	1077.95	1.29	273 ± 33	11.9	0.5121	8.4	Eu-152 (367.8)
10	411.10	1206.60	1.32	700 ± 32	4.6	1.314	21.7	Eu-152 (411.1)
11	443.95	1303.70	1.34	947 ± 37	3.9	1.778	25.4	Eu-152 (444.0)
12	488.83	1436.38	1.36	118 ± 20	17.0	0.2216	5.9	Eu-152 (488.7)
13	564.10	1658.91	1.41	108 ± 21	19.6	0.202	5.1	Eu-152 (564.0)
14	586.32	1724.62	1.42	96 ± 19	20.0	0.1804	5.0	Eu-152 (586.3)
15	678.75	1997.89	1.48	96 ± 21	21.5	0.1805	4.7	Eu-152 (678.6)
16	688.63	2027.09	1.48	187 ± 22	11.5	0.3515	8.7	Eu-152 (688.7)
17	723.33	2129.68	1.51	316 ± 25	7.8	0.594	12.8	Zr-95 (724.2)
18	756.78	2228.58	1.53	64 ± 19	29.1	0.1205	3.4	Zr-95 (756.7)
19	778.86	2293.87	1.60	2,609 ± 57	2.2	4.898	45.9	Eu-152 (778.9)
20	810.42	2387.17	1.56	66 ± 17	25.9	0.1244	3.9	Eu-152 (810.5)
21	867.36	2555.50	1.59	784 ± 33	4.2	1.471	23.5	Eu-152 (867.4)
22	873.12	2572.53	1.60	129 ± 19	15.0	0.2421	6.7	
23	919.39	2709.33	1.62	75 ± 16	20.8	0.1416	4.8	Eu-152 (919.3)
24	926.01	2728.91	1.63	70 ± 19	26.8	0.1315	3.7	Eu-152 (926.3)
25	964.10	2841.54	1.66	2,615 ± 55	2.1	4.909	47.3	Eu-152 (964.1)
26	996.37	2936.94	1.67	111 ± 17	15.3	0.2075	6.5	
27	1004.99	2962.44	1.68	323 ± 22	6.9	0.6072	14.5	Eu-152 (1005.3)
28	1085.79	3201.33	1.72	1,734 ± 54	3.1	3.256	32.2	Eu-152 (1085.8)
29	1089.65	3212.77	1.72	304 ± 32	10.4	0.5707	9.6	Eu-152 (1089.7)
30	1112.07	3279.05	1.81	2,170 ± 68	3.2	4.074	31.7	Eu-152 (1112.1)
31	1123.07	3311.57	1.75	77 ± 14	18.7	0.1445	5.3	sum peak (344+779)
32	1173.33	3460.19	1.78	46 ± 15	31.8	0.08604	3.1	Co-60 (1173.2)
33	1212.99	3577.48	1.80	166 ± 18	10.6	0.312	9.5	Eu-152 (1212.9)
34	1274.36	3758.95	1.84	330 ± 23	7.0	0.6191	14.2	Na-22 (1274.5)
35	1299.18	3832.36	1.86	222 ± 18	8.3	0.4165	12.1	Eu-152 (1299.1)
36	1332.73	3931.57	1.88	35 ± 8	22.7	0.06579	4.4	Co-60 (1332.5)
37	1408.06	4154.36	2.02	2,875 ± 59	2.0	5.398	48.8	Eu-152 (1408.0)
38	1457.59	4300.85	1.95	56 ± 11	19.1	0.1052	5.2	Eu-152 (1457.6)
39	1529.16	4512.57	2.00	93 ± 13	14.1	0.1753	7.1	Eu-152 (1528.1) sum peak (122+1408)
40	1596.62	4712.12	2.04	22 ± 5	23.5	0.04119	4.3	Ba-140 (1596.2)

Identified Nuclides (6 of 6) · Library: Environmental

Nuclide	Confidence	Half-life	Category	Lines Found / Expected	Lines found (keV)	Note
Eu-152	94 %	13.517 y	Activation product /	28 / 28	40.1, 121.8, 244.7, 295.9, 344.3, 367.8, 411.1, 444.0,	

Figure 32 — First two pages of a full report (Eu-152.S0).

10.2 Templates

The dialog always starts with the last used settings. **Save as Template…** stores the current options under a name (e.g. "Food samples – Arabic"); choose it later in the **Template** list. Built-in presets: *Full report*, *Plot only*, *Nuclides summary*. **Delete Template** removes a user template.

10.3 Reports for many files

File › Print Reports for Multiple Files…: add files or a whole folder, set the **Report Options…** once, choose **One PDF per spectrum file** (into an output folder) or **One combined PDF**, and optionally **Run peak search & identification for every file**. Click **Create PDFs**; the list shows the pages written or the error for each file.

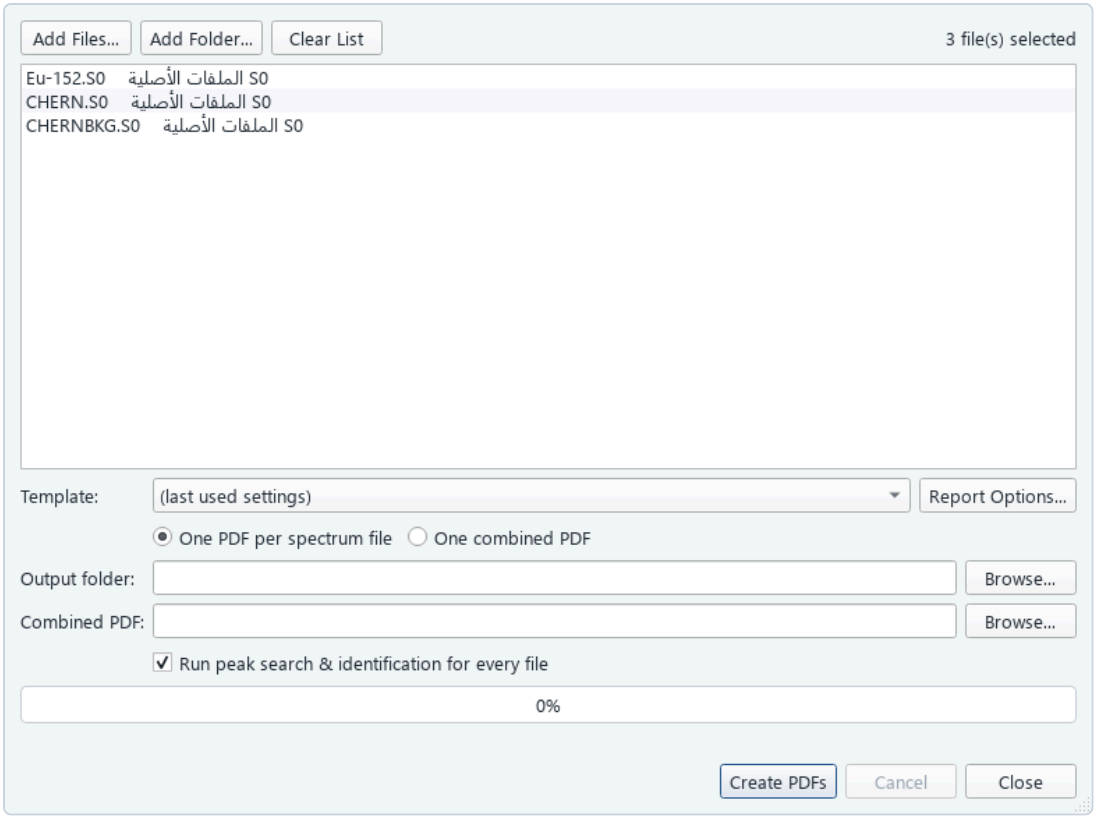


Figure 33 — Batch reports.

10.4 Review and the report record (JSON)

When the analysis of a spectrum is complete, the reviewer records it with **Analyze › Mark as Reviewed...**: name (remembered) and an optional note. A stale analysis (changed after the results were computed) cannot be marked as reviewed.



Figure 34 — Mark as Reviewed.

File › Export Report Record (JSON)... writes the complete record of the analysis: the spectrum and its fingerprint, calibrations, background, blank, coincidence table, options, results, workflow statuses, review and edit history. A record that is not complete for reporting (for example activities without an efficiency calibration) is exported only after a question and is marked `"reportable": false`. The record is meant for the laboratory's LIMS or archive.

10.5 Batch analysis

Analyze › Batch Analysis... analyses many spectra with the same setup, safely.

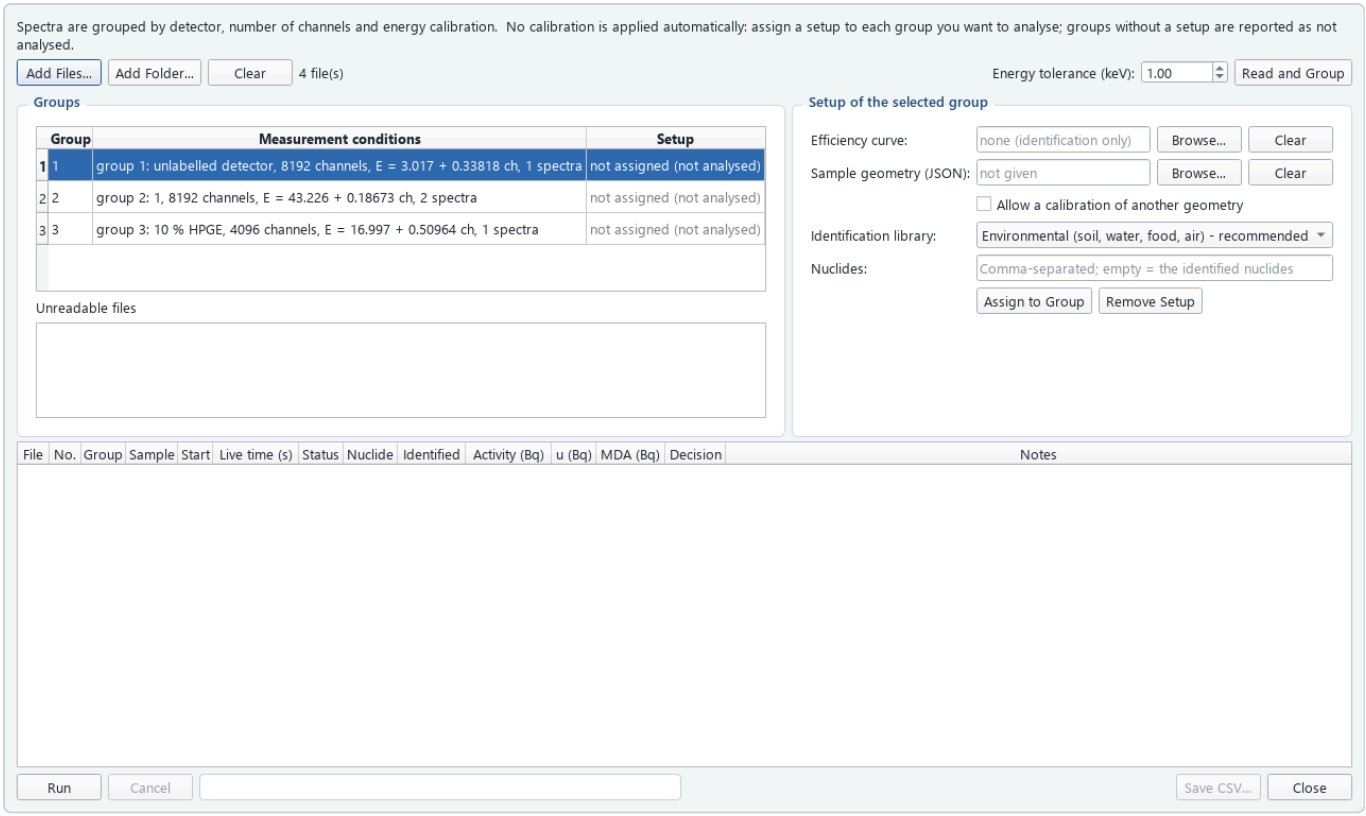


Figure 35 — Batch analysis: four files read and grouped by measurement conditions.

1. **Add Files...** or **Add Folder...**, then **Read and Group**. The spectra are grouped by detector, number of channels and energy calibration (within the energy tolerance); unreadable files are listed apart.
2. Select a group and give its **setup**: efficiency curve (none = identification only), sample geometry (JSON), identification library and the nuclides to quantify. **Assign to Group**. No calibration is applied automatically: a group without a setup is reported as *not analysed*.
3. **Run**. The analysis runs in the background with progress and **Cancel**; the table gives, per file and nuclide, identification, activity, uncertainty, MDA, decision and notes. **Save CSV...** writes the table.

11. Archive search

File › Archive Search… (Ctrl + Shift + F) finds spectra in large archives — by sample name, words in the description, nuclides found, date, detector, folder or live time — without moving or changing the original files.

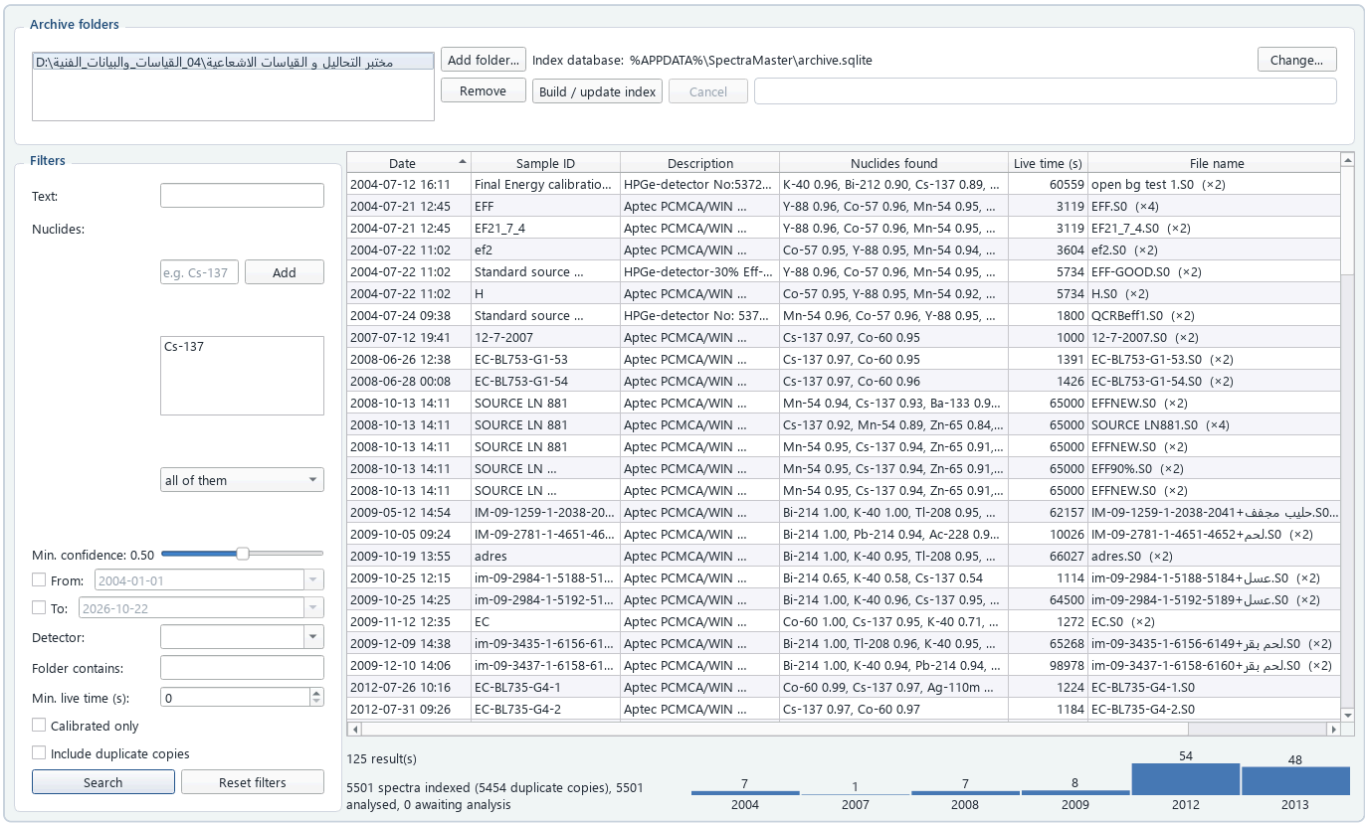


Figure 36 — Archive search on an index of the laboratory archive: all spectra with Cs-137 (confidence ≥ 0.50), results per year below.

11.1 Building and updating the index

1. **Add folder…** for each archive root (local disk or network share).
2. Click **Build / update index**. The program lists the files, reads the metadata of every spectrum and then analyses each unique spectrum (peak search + identification) in parallel. Progress and the remaining time are shown; **Cancel** stops safely and the next build resumes.
3. Later builds only process new or changed files and drop deleted ones, so updating is quick.

The index is a single SQLite file, by default `%APPDATA%\SpectraMaster\archive.sqlite` (**Change…** selects another file, e.g. a shared copy on the network). Byte-identical copies of a spectrum are recognised as duplicates and analysed once. The original files are only read.

11.2 Filters

Text	Words in the sample ID, description, comments, file or folder name. All words must match.
Nuclides	Add one or more nuclides (with auto-completion: `Cs-137`, `cs137`, `137Cs` are equivalent); match *all of them* or *any of them*, with a **minimum confidence** slider.
Date, detector, folder	From / to date, detector name, part of the folder path.
Other	Minimum live time, calibrated spectra only, include duplicate copies.

> **Arabic normalisation.** The text search ignores the differences أ إ آ → ا, ة → ه, ى → ي, diacritics (tashkeel) and tatweel, and treats Arabic-Indic digits like Latin digits — `شاى` finds `شاي`. The definite article is removed from the search words (also وال، بال، كال، فال، لل) when at least three letters remain, so `الشاي` finds `شاي`, `الشاي` and `بالشاي` alike.

11.3 Working with the results

Double-click a row to open the spectrum in the main window. The right-click menu offers **Open in main window**, **Add as overlay**, **Open containing folder** and **Copy path**. Columns can be sorted; the bar chart shows the results per year. **Export results to Excel / CSV…** saves the list with date, sample, description, nuclides, times, detector, path and number of copies.

The same index can be built from the command line for very large archives with `smconvert index` (Excel catalogue) — the window's index is built from the window itself.

12. Detector quality control

Tools › Detector QC… keeps control charts of the check-source and background measurements of each detector, as required by ISO/IEC 17025 (7.7). For every measurement it records the **centroid energy** (gain/offset drift), the **FWHM** (resolution), the **net count rate** decay-corrected to a reference date (efficiency stability), the **background rate** in an energy window, and live/dead time.

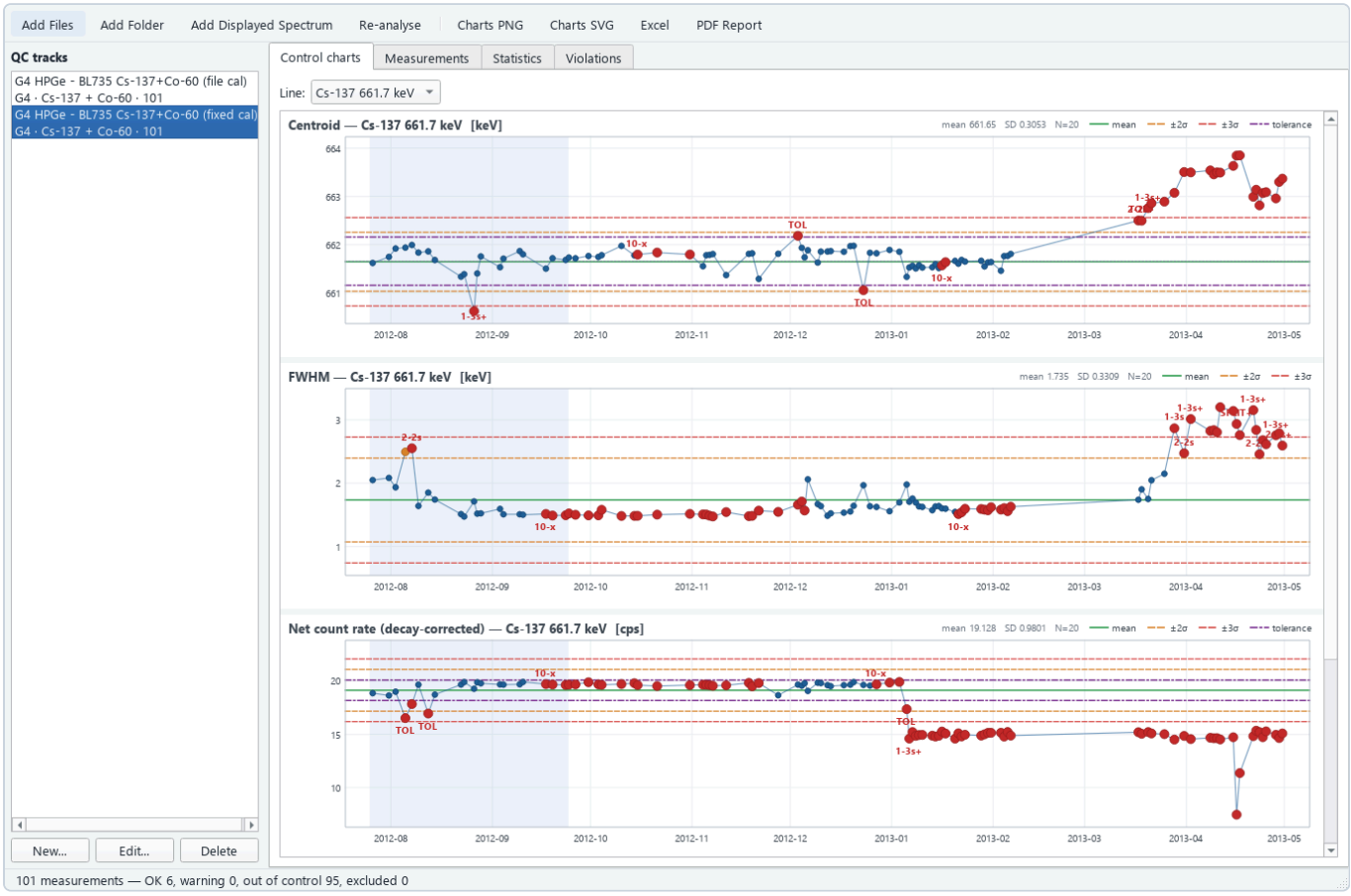


Figure 37 — Control charts of a Cs-137 + Co-60 check source on detector G4 (101 measurements, 2012–2013): baseline (shaded), mean, ±2σ, ±3σ and tolerance lines, violations labelled.

12.1 Creating a QC track

A **track** is one detector + one check source (or a background measurement). Click **New…** and fill in:

- name, detector, detector type (HPGe, NaI, LaBr3, CZT), type (check source / background);
- **monitored lines** — from a preset (Cs-137, Co-60, Cs-137 + Co-60, Eu-152, Ba-133, Am-241, Na-22, Background) or typed as `keV[:nuclide[:FWHM max]]`, e.g. `661.657:Cs-137, 1332.492:Co-60:2.0`;
- reference date and reference activity (for decay correction; a unit may follow the number, e.g. `Cs-137=1 µCi`, Bq when none; nuclides separated by `;`, a new line or a comma; a point for decimals and no thousands separator, so `1,000` is refused), peak match tolerance;
- **centroid energy from** each file's calibration, or a **fixed reference calibration** — the latter shows real gain drift even when each spectrum was recalibrated;
- background window, **baseline** (first N points or a period), the **Westgard rules** to apply and fixed tolerances (centroid ± keV, rate ± %, background ± %).

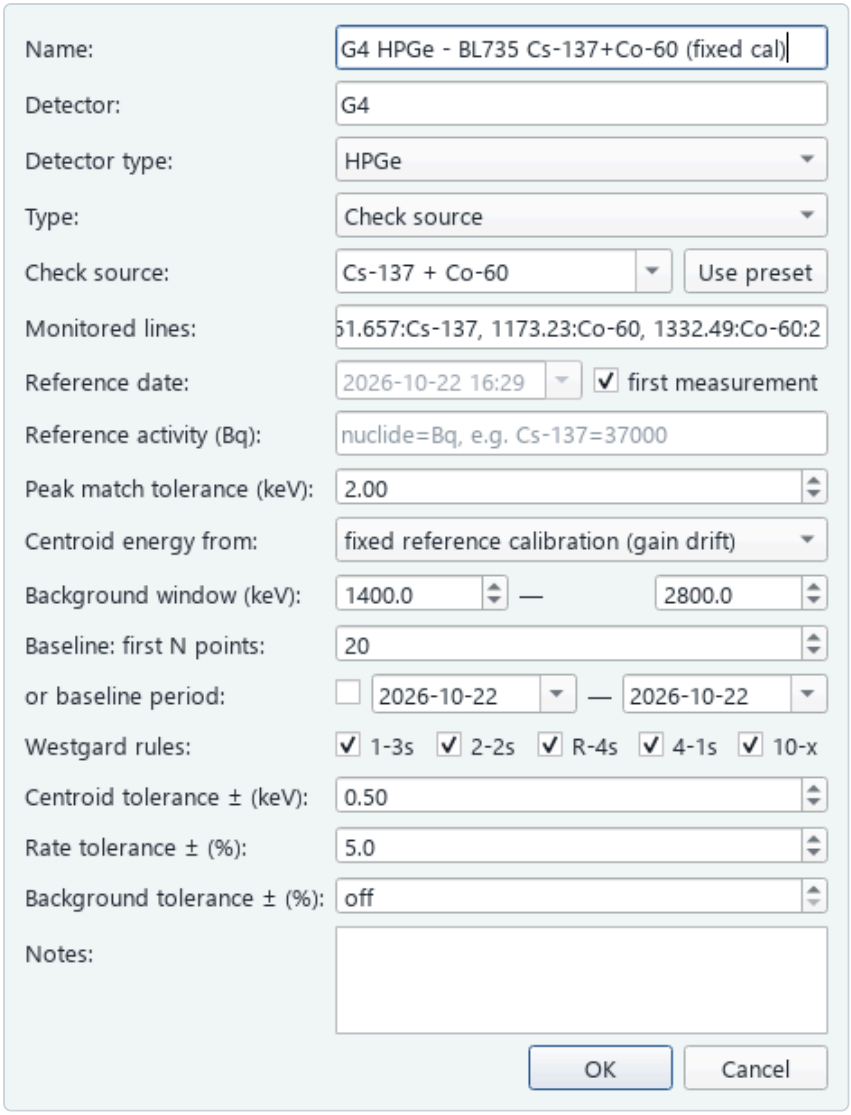


Figure 38 — QC track settings.

12.2 Adding measurements

Add Files…, **Add Folder…** (e.g. the whole year's QC folder) or **Add Displayed Spectrum**. Each spectrum is analysed once and its peak list stored, so lines and tolerances can be changed later without re-reading the files; duplicates are ignored. **Re-analyse** re-reads all source files. In the *Measurements* table a right click can **exclude** a point from the statistics (with a note, e.g. "source repositioned"), remove it, or show the spectrum in the main window.

12.3 Control charts and Westgard rules

Mean and standard deviation are computed from the baseline points; warning limits are ±2σ and action limits ±3σ. The rules are evaluated on all non-excluded points in date order:

Rule	Meaning	Severity
`1-2s`	one point beyond ±2σ	warning
`1-3s`	one point beyond ±3σ	out of control
`2-2s`	two consecutive points beyond 2σ on the same side	out of control

Rule	Meaning	Severity
`R-4s`	two consecutive points beyond 2σ on opposite sides	out of control
`4-1s`	four consecutive points beyond 1σ on the same side	out of control
`10-x`	ten consecutive points on the same side of the mean	out of control
`TOL`	fixed tolerance exceeded (centroid, FWHM max, rate, background)	out of control
`SPLIT`	line found split into two peaks (gain or zero jump during the count)	out of control

The *Violations* tab lists every rule hit with date, file, series and value; the *Statistics* tab gives N, mean, SD, CV % and the trend per year for each line and quantity.

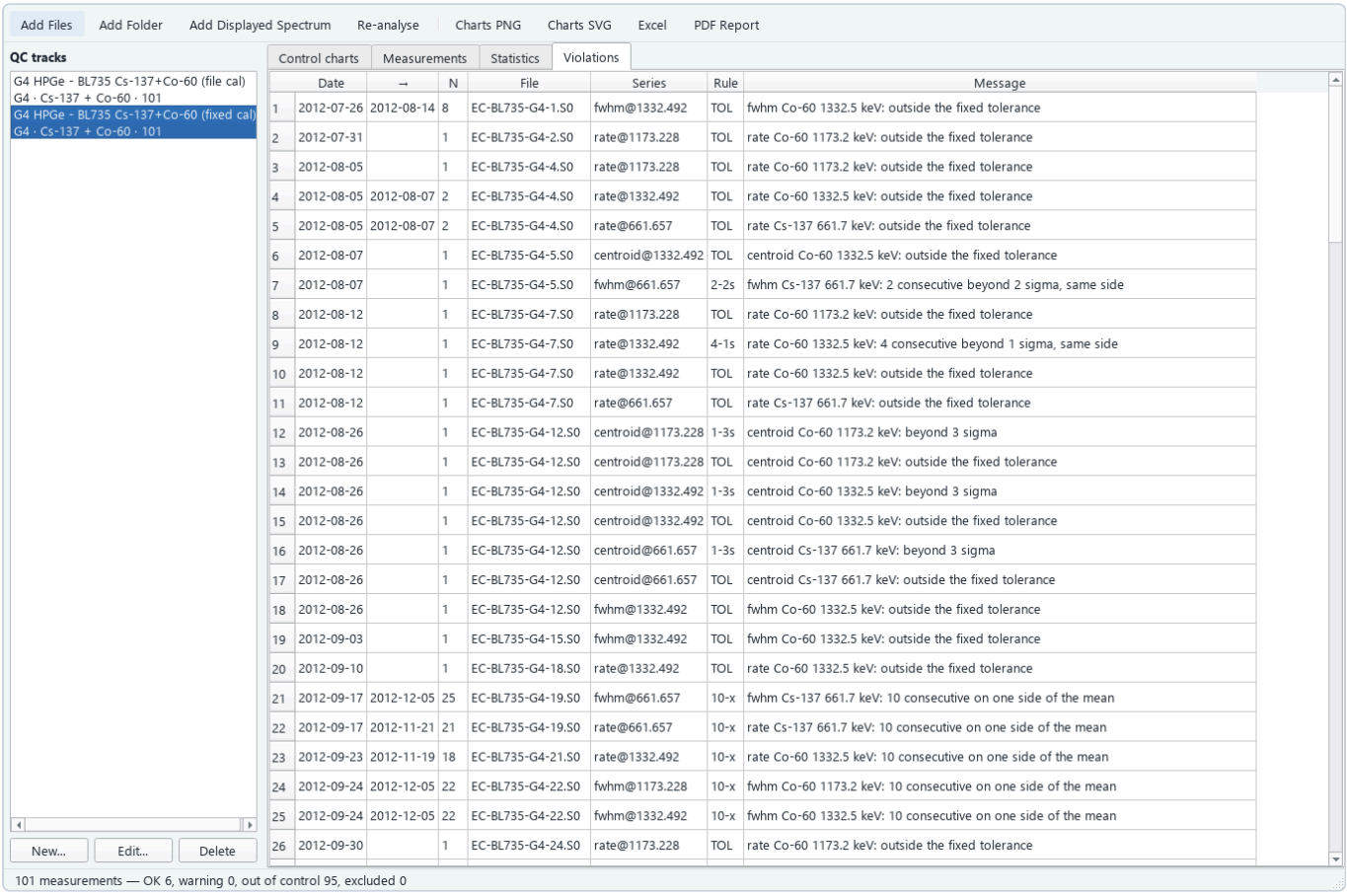


Figure 39 — Violations tab.

12.4 Exporting and reports

Charts PNG… / SVG… save the charts, **Excel…** a workbook with measurements, statistics and violations, and **PDF Report…** a complete QC report (track settings, summary, charts, statistics, measurements, violations and a signature line) in the interface language. Tracks are stored as JSON files in `%APPDATA%\SpectraMaster\qc\`.

12.5 Detector acceptance test (Co-60)

Tools › Detector Acceptance Test… uses the displayed spectrum of a Co-60 point source at 25 cm on the detector axis (IEEE Std 325, the test of the detector's data sheet):

Result	How it is obtained

FWHM, FWTM and FWTM / FWHM at 1332.5 keV	Net counts of a ±3 FWHM region (straight background between its end points); half and tenth maximum located by a parabola through the logarithms of three channels, exact for a Gaussian peak. A Gaussian gives FWTM / FWHM = 1.82; a larger value shows a tail. With a Co-57 spectrum the same is given at 122.1 keV.
Peak-to-Compton ratio	The highest channel of the 1332.5 keV peak divided by the mean counts per channel between 1040 and 1096 keV.
Relative efficiency	With the source activity (any unit of the list, Bq to TBq or Ci to pCi; the activity at the start is shown in Bq and in that unit), its uncertainty and reference date: net area / (live time × activity at the start × 0.9998), divided by 1.2×10^{-3} (a 3×3 inch NaI detector in the same geometry), in percent. No coincidence-summing correction, as in the standard test.

The widths need at least 3 channels in the FWHM (a note is shown otherwise). **Copy** puts the table on the clipboard.

12.6 Chi-square counting test

Tools › Chi-Square Counting Test… checks that repeated counts of the same source scatter as Poisson statistics predict (IAEA-TECDOC-1599). Type or paste the counts, one per line (10 or 20). $\chi^2 = \Sigma(x - \bar{x})^2 / \bar{x}$ is accepted between the 5 % and 95 % points of the chi-square distribution with n − 1 degrees of freedom: 3.325 – 16.919 for 10 counts, 10.117 – 30.144 for 20. A value above the range shows extra scatter (unstable electronics, high voltage, source position); a value below it shows too little scatter (for example counts that are not independent).

13. Troubleshooting and FAQ

The file is not recognised ("cannot recognise the format")

Check that the file is a spectrum (not a report or a calibration file) and not empty. Try **File › Open Input Files As…** with the expected format; the `smconvert info FILE --from CODE` command shows the reader's exact error. If *Auto-detect* is not active, a wrong fixed format is the most common cause — the status bar shows the active input mode. If the message names several possible formats, choose the right one by hand.

Peaks are at the wrong energies, or no nuclide is identified

- The file may be uncalibrated or carry an old calibration: check *Energy Calibration* in the information panel and recalibrate (chapter 8). CHN, CNF and Amptek files with the placeholder E = channel are read as uncalibrated (section 3.2).
- After a long measurement or a change of the electronics, run the drift check (section 8.2).
- After conversion to CHN/CNF/SPC, read the refit warning: the deviation of the refitted calibration is printed.
- Genie-2000 shows channel numbers one higher than SpectraMaster — energies are the same.

Very few peaks are found, or peaks are split in two

The width model may be wrong (a FWHM calibration from another detector, or of one channel). Look for the note *FWHM model implausible*; recalibrate the FWHM from a source (section 8.1), which fits the measured widths, and inspect the fits (section 6.7).

"uncertainties are not valid" / "not Poisson"

The spectrum was already background subtracted, stripped, normalized or smoothed. Open the measured spectrum and give the background in the activity dialog instead.

The main window does not respond

A modal window (Quick Energy, Full Calibration, Fit Inspection) is open, perhaps behind another window: finish or cancel it. Long calculations show a busy message and keep the window responsive.

A window is larger than my screen

Windows are fitted to the screen when they open; scroll the contents. The buttons of the last row stay visible. Maximize with the title-bar button or a double click on the title.

Arabic sample names became "????" in Genie or MAESTRO

CNF, CHN and SPC store only Latin text (section 3.4). Keep the Aptec S0, SPE, N42 or Excel file as the reference copy. Saving or converting to a format that stores only ASCII text (CHN, SPC, CNF, S100, ND66) prints a warning when the sample ID, description or detector contains Arabic (non-ASCII) characters.

"stores only 3 calibration terms: energy calibration refitted"

Not an error: the target format cannot hold the source polynomial. Accept small deviations ($\lesssim 0.05$ keV); for larger ones choose a format that keeps the full calibration (Aptec S0, IAEA SPE, N42, JSON, Excel).

"The peak-search engine is not available" / P is disabled

No spectrum is open, the spectrum is empty, or (SpectraMaster method) the analysis engine is missing. Nuclides are not identified in uncalibrated spectra.

GammaVision does not call SpectraMaster

Copy Program and Arguments again from **Help › GammaVision / MAESTRO Integration…**; put quotes around paths with spaces when not using the macros.

Where are settings and data stored?

Preferences	Language, last folders, input/output formats, window layout, view options and report templates: Windows registry `HKEY_CURRENT_USER\Software\SpectraMaster\SpectraMaster`.
Archive index	`%APPDATA%\SpectraMaster\archive.sqlite` (changeable in the window).
Efficiency curves	`%APPDATA%\SpectraMaster\efficiency\*.json`
Detector profiles	`%APPDATA%\SpectraMaster\detectors\*.json`
QC tracks	`%APPDATA%\SpectraMaster\qc\*.json`
Analysis sessions	Where you save them (`.smsession`).

`%APPDATA%` is usually `C:\Users\<name>\AppData\Roaming`. Back up the `SpectraMaster` folder together with the laboratory data. Deleting the registry key resets all preferences.

14. Appendices

A. Supported formats

Codes 1–16 are the DataMaster numbers; use them with `-i` / `-j`, `--to` and `--from`.

60 formats — 59 readable, 35 writable

Code	Key	Format	Extensions	Read	Write
1	`ortec_chn`	ORTEC ADCAM / MAESTRO	*.chn	✓	✓
2	`ortec_spc`	ORTEC INFORM (integer SPC)	*.spc	✓	✓
5	`ascii1`	ASCII-1 (one count per line)	*.txt	✓	✓
6	`ascii2`	ASCII-2 (channel, counts)	*.txt	✓	✓
7	`iaea_spe`	IAEA / Rossendorf ASCII SPE	*.spe	✓	✓
8	`iec1455`	IEC 1455 ASCII	*.iec	✓	✓
9	`ieee_asc`	IEEE ASCII	*.asc	✓	✓
10	`rms_phd`	RMS / IMS ASCII PHD	*.phd	✓	—
13	`aptec_s0`	APTEC v6.3	*.s0	✓	✓
14	`ci_s100`	Canberra S100	*.mca	✓	✓
15	`ci_accuspec`	Canberra AccuSpec	*.dat	✓	—
16	`ci_cnf`	Canberra CAM (Genie-2000)	*.cnf	✓	✓
17	`eml_sanderson`	EML/Sanderson ASCII test spectra	*.dat	✓	—
20	`n42_2006`	ANSI N42.42-2006 XML	*.n42 *.xml	✓	✓
21	`n42_2012`	ANSI N42.42-2012 XML	*.n42 *.xml	✓	✓
22	`gadras_pcf`	GADRAS PCF	*.pcf	✓	✓
23	`specutils_csv`	SpecUtils CSV (Energy, Data)	*.csv	✓	✓
24	`specutils_txt`	SpecUtils TXT	*.txt	✓	✓
25	`tka`	Canberra/Toolkit TKA	*.tka	✓	✓
26	`html_d3`	HTML page (interactive D3 chart)	*.html *.htm	—	✓
27	`spc_ascii`	ORTEC/IAEA ASCII SPC	*.spc	✓	✓
28	`spc_float`	ORTEC SPC (floating point)	*.spc	✓	✓
29	`exploranium_gr130`	Exploranium GR-130 binary	*.dat	✓	✓
30	`exploranium_gr135`	Exploranium GR-135 v2 binary	*.dat	✓	✓
31	`amptek_mca`	Amptek MCA (read only)	*.mca	✓	—
32	`radiacode_xml`	RadiaCode / BecqMoni XML (read only)	*.xml *.rcspg	✓	—
33	`lsrm_spe`	LSRM / SpectraLine SPE (read only)	*.spe	✓	—
34	`multiact`	MultiAct (read only)	*.spm	✓	—
35	`labzy_lzs`	LabZY nanoMCA LZS (read only)	*.lzs	✓	—
36	`caen_gxml`	CAEN Hexagon GXML (read only)	*.gxml	✓	—
37	`specutils_json`	SpecUtils JSON (read only)	*.json	✓	—
38	`tracs_mps`	TRACS MPS (read only)	*.mps	✓	—
39	`aram`	ARAM (read only)	*.xml	✓	—
40	`microraider`	ICx MicroRaider XML (read only)	*.xml	✓	—
41	`spm_daily`	Spectroscopic portal daily file (read only)	*.txt	✓	—
42	`scandata_xml`	ScanData XML (read only)	*.xml	✓	—

Code	Key	Format	Extensions	Read	Write
43	`ortec_listmode`	ORTEC list mode (read only, summed)	*.lis	✓	—
50	`excel_xlsx`	Microsoft Excel workbook	*.xlsx	✓	✓
51	`json`	SpectraMaster JSON	*.json	✓	✓
60	`aptec_ascii`	Aptec ASCII	*.apt	✓	✓
61	`bti_ascii`	BTI ASCII (read only)	*.bti	✓	—
62	`chandata_csv`	Aptec Channel Data CSV	*.csv	✓	✓
63	`ci90_ascii`	Canberra Series 90 ASCII (read only)	*.o	✓	—
64	`davidson_ascii`	Davidson ASCII	*.dav	✓	✓
65	`iaea_ascii_dat`	IAEA ASCII (Aptec, EML layout)	*.dat	✓	✓
66	`npl_ascii`	NPL ASCII	*.txt	✓	✓
67	`rump_ascii`	Rump ASCII (RBS)	*.mac	✓	✓
68	`wallac_ascii`	Wallac ASCII (1024 ch)	*.001	✓	✓
69	`laval_ascii`	Laval ASCII	*.de1	✓	✓
70	`jcac_vax`	JCAC VAX	*.spc	✓	✓
71	`nd66`	Nuclear Data ND66	*.001	✓	✓
72	`norland5000`	Norland PHA-5000	*.dat	✓	✓
73	`oxford_ans`	Oxford ANS	*.ans	✓	✓
74	`quantrad_scout`	Quantrad Scout (read only)	*.sct	✓	—
75	`spectran_f`	Spectran F (read only)	*.spc	✓	—
76	`nucleus_pca`	Nucleus PCA-I/PCA-II (read only)	*.spm	✓	—
77	`pca_multiport`	PCA-Multiport (read only)	*.spt	✓	—
78	`aptec_v30`	APTEC v3.0 (read only)	*.s0	✓	—
79	`aptec_v43`	APTEC v4.3 (read only)	*.s0	✓	—
80	`lab_dat1670`	EMCA2000 binary .dat (read only)	*.dat	✓	—

B. Command-line reference

Output of `smconvert --help` and `smconvert COMMAND --help`. Format arguments accept a code, key or alias. `smconvert` without arguments prints the usage summary and a reminder that the graphical program is `SpectraMaster.exe`.

```
usage: smconvert [-h] COMMAND ...

Gamma-spectrum file converter (DataMaster compatible).

positional arguments:
  COMMAND
    convert    convert files, folders or glob patterns
    info       print the metadata of spectrum files
    formats    list supported formats and their codes
    peaks      search and fit peaks, print a peak table
    identify   identify nuclides (peak search + library matching)
    gui        start the graphical interface
    index      catalogue a spectrum archive into an Excel workbook
    rescue     convert + verify a whole archive (resumable)
    sum        sum spectra (energy-aligned if calibrations differ)
    subtract   subtract a live-time normalised background
    rebin      compress channels (x2, x4 ...) and/or truncate the range
    calibrate  fit / propose / apply an energy (and FWHM) calibration
    batch      group spectra by detector/channels/calibration; analyse one
               group
    effcal     efficiency calibration from a certified standard spectrum
    activity   activity, specific activity and MDA of a sample spectrum

options:
  -h, --help   show this help message and exit

DataMaster form: spectramaster -i INPUT -j OUTPUT (i, j = format codes from
'spectramaster formats'; i = 0 auto-detects)
```

convert

```
usage: smconvert convert [-h] --to TO [--from IN_FMT] [--out OUT]
                         [--recursive] [--keep-tree] [--no-overwrite]
                         [--quiet]
                         inputs [inputs ...]

positional arguments:
  inputs             input files, folders or glob patterns

options:
  -h, --help         show this help message and exit
  --to, -t TO        output format key, alias or code
  --from, -f IN_FMT  input format key/code, or 'auto' (default)
  --out, -o OUT      output folder (default: next to each input)
  --recursive, -r    walk sub-folders
  --keep-tree        recreate the input sub-folder structure inside --out
  --no-overwrite     skip existing outputs (reported as failures)
  --quiet, -q        only print failures and the summary
```

info

```
usage: smconvert info [-h] [--from IN_FMT] files [files ...]

positional arguments:
  files

options:
  -h, --help         show this help message and exit
  --from, -f IN_FMT
```

formats

```
usage: smconvert formats [-h]

options:
  -h, --help  show this help message and exit
```

peaks

```
usage: smconvert peaks [-h] [--from IN_FMT] [--min-sig MIN_SIG]
                       files [files ...]

positional arguments:
  files

options:
  -h, --help         show this help message and exit
  --from, -f IN_FMT
  --min-sig MIN_SIG  significance threshold (default 3)
```

identify

```
usage: smconvert identify [-h] [--from IN_FMT] [--min-sig MIN_SIG]
                          [--lib LIB] [--min-conf MIN_CONF] [--peaks]
                          [--sampled SAMPLED]
                          files [files ...]

positional arguments:
  files

options:
  -h, --help           show this help message and exit
  --from, -f IN_FMT
  --min-sig MIN_SIG    peak significance threshold (default 3)
  --lib LIB            nuclide library (JSON like the bundled one, or CSV
                       nuclide,energy_keV,intensity_%[,half_life,category])
  --min-conf MIN_CONF  minimum confidence 0..1 (default 0.25)
  --peaks              also print the peak table with assignments
  --sampled SAMPLED    end of sampling (ISO date/time): nuclides decayed by
                       more than 10 half-lives since then get a low
                       plausibility
```

gui

```
usage: smconvert gui [-h] [files ...]

positional arguments:
  files

options:
  -h, --help  show this help message and exit
```

index

```
usage: smconvert index [-h] -o OUT [--no-recursive] [--all-files]
                       [--quiet]
                       root

positional arguments:
  root            archive folder (read only)

options:
  -h, --help      show this help message and exit
  -o, --out OUT   catalogue .xlsx (outside the archive)
  --no-recursive  do not walk sub-folders
  --all-files     sniff every file, not only spectrum extensions
  --quiet, -q
```

rescue

```
usage: smconvert rescue [-h] -o OUT [--to TO] [--resume] [--flat]
                        [--no-verify] [--no-recursive] [--skip-duplicates]
                        [--limit LIMIT] [--quiet]
                        root

positional arguments:
  root               archive folder (read only)

options:
  -h, --help         show this help message and exit
  -o, --out OUT      output folder (outside the archive)
  --to, -t TO        output format (repeatable; default n42_2012 + ci_cnf)
  --resume           skip files already converted and verified
  --flat             do not recreate the folder tree
  --no-verify        skip read-back verification
  --no-recursive
  --skip-duplicates  convert byte-identical copies only once
  --limit LIMIT      only the first N candidate files (trial run)
  --quiet, -q
```

sum

```
usage: smconvert sum [-h] -o OUT [--to TO] [--align {auto,channel,energy}]
                     files [files ...]

positional arguments:
  files

options:
  -h, --help            show this help message and exit
  -o, --out OUT
  --to, -t TO           output format (default: from the extension / first
                        input)
  --align {auto,channel,energy}
```

subtract

```
usage: smconvert subtract [-h] -o OUT [--to TO] [--scale SCALE] [--clip]
                          [--sigma-csv SIGMA_CSV]
                          sample background

positional arguments:
  sample
  background

options:
  -h, --help            show this help message and exit
  -o, --out OUT
  --to, -t TO
  --scale SCALE         background factor (default LT_sample / LT_background)
  --clip                set negative net channels to 0
  --sigma-csv SIGMA_CSV
                        also write channel, energy, net, sigma to this CSV
```

rebin

```
usage: smconvert rebin [-h] -o OUT [--factor FACTOR] [--start START]
                       [--stop STOP] [--renumber] [--to TO]
                       file

positional arguments:
  file

options:
  -h, --help       show this help message and exit
  -o, --out OUT
  --factor FACTOR
  --start START    first channel to keep (truncate)
  --stop STOP      channel after the last one kept
  --renumber       renumber truncated channels from 0
  --to, -t TO
```

calibrate

```
usage: smconvert calibrate [-h] [--pairs PAIRS] [--source SOURCE]
                           [--coeffs COEFFS] [--order ORDER] [--fwhm FWHM]
                           [--fwhm-order FWHM_ORDER] [-o OUT] [--to TO]
                           [--only-uncalibrated]
                           files [files ...]

positional arguments:
  files

options:
  -h, --help            show this help message and exit
  --pairs PAIRS         channel:keV pairs, e.g. 1893.7:661.66,3364:1173.23
  --source SOURCE       calibration source nuclide(s), e.g. Eu-152 or
                        Co-60,Cs-137
  --coeffs COEFFS       apply these coefficients c0,c1[,c2,c3]
  --order ORDER         polynomial order 1-3
  --fwhm FWHM           channel:FWHM_keV pairs for a FWHM calibration
  --fwhm-order FWHM_ORDER
  -o, --out OUT         write recalibrated copies into this folder
  --to, -t TO           output format (default: input format if writable, else
                        IAEA SPE)
  --only-uncalibrated   leave already calibrated files alone
```

batch

```
usage: smconvert batch [-h] [--group GROUP] [--eff EFF]
                       [--geometry GEOMETRY] [--allow-geometry-mismatch]
                       [--application APPLICATION] [--nuclides NUCLIDES]
                       [--energy-tol ENERGY_TOL] [-o OUT]
                       paths [paths ...]

positional arguments:
  paths                 files, folders or glob patterns

options:
  -h, --help            show this help message and exit
  --group GROUP         group to analyse (see the list printed first)
  --eff EFF             efficiency curve JSON for this group (omit:
                        identification only)
  --geometry GEOMETRY   sample geometry JSON
                        (lab.detector.MeasurementGeometry)
  --allow-geometry-mismatch
                        use a calibration of another geometry anyway (reported
                        in every row)
  --application APPLICATION
                        identification library: all, environmental,
                        calibration, safeguards, medical
  --nuclides NUCLIDES   comma-separated nuclides for activities (default:
                        those identified)
  --energy-tol ENERGY_TOL
                        calibration agreement for grouping, keV
  -o, --out OUT
```

`effcal`

```
usage: smconvert effcal [-h] [--cert CERT] -o OUT
                        [--model {loglog,two_branch}] [--order ORDER]
                        [--order-high ORDER_HIGH] [--crossover CROSSOVER]
                        [--u-sys U_SYS] [--bkg BKG]
                        [--min-intensity MIN_INTENSITY]
                        [--geometry GEOMETRY] [--detector DETECTOR]
                        [--name NAME] [--compare]
                        file

positional arguments:
  file                  calibration (standard) spectrum

options:
  -h, --help            show this help message and exit
  --cert CERT           certificate CSV: nuclide,activity_bq,unc_percent,ref_d
                        ate[,half_life_s,energy_kev,intensity_percent]
                        (default: Aptec records in an .S0 file)
  -o, --out OUT         efficiency curve JSON
  --model {loglog,two_branch}
  --order ORDER         log-log order, or low-branch order (default 3)
  --order-high ORDER_HIGH
                        high-branch order (two_branch)
  --crossover CROSSOVER
                        crossover keV (two_branch; default: automatic)
  --u-sys U_SYS         correlated relative uncertainty (fraction)
  --bkg BKG             background spectrum (peak-by-peak subtraction)
  --min-intensity MIN_INTENSITY
                        library lines >= this % (default 2)
  --geometry GEOMETRY
  --detector DETECTOR
  --name NAME
  --compare             compare with Aptec's stored efficiencies
```

`activity`

```
usage: smconvert activity [-h] --eff EFF [--bkg BKG]
                          [--bkg-mode {peak,spectrum}]
                          [--ref-peak REF_PEAK] [--ref-rate REF_RATE]
                          [--ref-rate-unc REF_RATE_UNC]
                          [--ref-spectrum REF_SPECTRUM]
                          [--ref-half-life REF_HALF_LIFE] [--tcs-pt A,B]
                          [--tcs-pt-unc TCS_PT_UNC]
                          [--tcs-curve-summing-free] [--blank BLANK]
                          [--mass MASS] [--unit UNIT]
                          [--mass-unc MASS_UNC] [--ref-date REF_DATE]
                          [--nuclides NUCLIDES] [--method {key,mean}]
                          [--mda {currie,iso11929}]
                          [--decision {search,iso11929}]
                          [--attenuation ATTENUATION]
                          [--self-att-density SELF_ATT_DENSITY]
                          [--self-att-matrix {water,organic,mineral}]
                          [--cal-density CAL_DENSITY]
                          [--cal-matrix {water,organic,mineral}]
                          [--self-att-height-mm SELF_ATT_HEIGHT_MM]
                          [--self-att-unc SELF_ATT_UNC] [-o OUT]
                          file

positional arguments:
  file

options:
  -h, --help            show this help message and exit
  --eff EFF             efficiency curve JSON (from 'effcal')
  --bkg BKG             background spectrum
  --bkg-mode {peak,spectrum}
  --ref-peak REF_PEAK   reference peak correction: energy keV of a pulser or
                        fixed reference source peak
  --ref-rate REF_RATE   its expected rate, counts per live second
  --ref-rate-unc REF_RATE_UNC
                        its relative uncertainty %
  --ref-spectrum REF_SPECTRUM
                        reference (low-rate) measurement giving the expected
                        rate
  --ref-half-life REF_HALF_LIFE
                        half-life (s) of a reference source (decay correction)
  --tcs-pt A,B          true-coincidence summing correction with the peak-to-
                        total ratio PT(E) = min(1, A (E/1000 keV)^B) of this
                        detector and geometry (an assumption)
  --tcs-pt-unc TCS_PT_UNC
                        relative uncertainty of PT, % (default 10)
  --tcs-curve-summing-free
                        declare the efficiency curve free of true-coincidence
                        summing (needed for a curve without calibration
                        points; not checked)
  --blank BLANK         reagent blank spectrum (prepared and counted like the
                        samples); its activities are subtracted line by line
  --mass MASS           sample quantity
  --unit UNIT           kg, g, L, mL (default kg)
  --mass-unc MASS_UNC   quantity uncertainty %
  --ref-date REF_DATE   reference (sampling) date, default acquisition start
  --nuclides NUCLIDES   comma separated, default Cs-137,K-40,Ra-226,Th-232,Cs-
                        134,Co-60,I-131,Be-7,Am-241
  --method {key,mean}   key line or weighted mean
  --mda {currie,iso11929}
  --decision {search,iso11929}
                        detected: a search peak above the decision threshold
```

```
                      (default), or also a window result above it (ISO
                      11929, about 5 % false detections per line)
  --attenuation ATTENUATION
                      constant self-attenuation correction factor (not
                      together with --self-att-density)
  --self-att-density SELF_ATT_DENSITY
                      sample density g/cm3: energy-dependent correction to
                      the calibration density (slab model)
  --self-att-matrix {water,organic,mineral}
                      sample matrix (default water)
  --cal-density CAL_DENSITY
                      density of the calibration source g/cm3
  --cal-matrix {water,organic,mineral}
                      calibration matrix (default: the sample's)
  --self-att-height-mm SELF_ATT_HEIGHT_MM
                      fill height of sample and calibration source, mm
  --self-att-unc SELF_ATT_UNC
                      model uncertainty of the factor, %
  -o, --out OUT       write results .csv or .xlsx
```

C. Glossary (English – Arabic)

English	Arabic	Meaning
Live time / real time / dead time	الزمن الحي / الحقيقي / الميت	Counting time corrected for dead time / clock time / their difference in %.
Channel, centroid	القناة، مركز القمة	MCA bin; fitted peak position in channels.
Energy calibration	معايرة الطاقة	Polynomial E(channel) in keV.
FWHM	قدرة الفصل (العرض عند نصف الارتفاع)	Full width at half maximum of a peak.
Calibration drift	انجراف المعايرة	Change of the energy scale since the calibration.
Net / gross area	المساحة الصافية / الإجمالية	Peak counts above / including the background.
Significance	الدلالة الإحصائية	Net area divided by its 1σ uncertainty.
Region of interest (ROI)	منطقة الاهتمام	Marked channel range with gross/net counts.
Multiplet	قمة متعددة	Overlapping peaks fitted together.
Overlay	طيف متراكب	Second spectrum drawn on the same plot.
Nuclide identification	التعرّف على النويدات	Matching peaks to library lines.
Confidence	الثقة	Heuristic identification score 0–100 %.
Full-energy-peak efficiency	كفاءة قمة الطاقة الكاملة	Counts in the peak per emitted photon.
Peak-to-total ratio	نسبة القمة إلى الكلي	Peak efficiency / total efficiency.
Emission probability (yield)	احتمال الانبعاث (الشدة)	Photons per 100 decays.
Specific activity	النشاط النوعي	Bq per kg or per litre.
Expanded uncertainty (k = 2)	الارتياب الموسَّع	2 × combined standard uncertainty (≈ 95 %).
Decision threshold / detection limit	عتبة القرار / حد الكشف	L_C or y* / L_D or y#.
Best estimate (ISO 11929)	أفضل تقدير	Value and coverage interval also for a non-detect.

English	Arabic	Meaning
MDA	أدنى نشاط قابل للكشف	Minimum detectable activity.
Reagent blank	العينة الفارغة (الكاشف الفارغ)	Blank prepared and counted like the samples.
Reference peak	القمة المرجعية	Pulser or fixed source correcting count losses.
True-coincidence summing	جمع التوافق الحقيقي	Loss/gain of counts by cascade gamma rays.
Annihilation summing	جمع الفناء	Summing with the 511 keV photons of a positron.
Self-attenuation	الامتصاص الذاتي	Absorption of gamma rays inside the sample.
Decay correction	تصحيح الاضمحلال	Activity referred to the sampling date.
Analysis session	جلسة التحليل	Spectrum, calibrations, results and edits with fingerprints.
Check source, control chart	مصدر الفحص، مخطط التحكم	QC measurement and its chart.
Baseline	خط الأساس	Points that define mean and σ of a control chart.
Gain drift	انجراف الكسب	Slow change of the energy scale.

D. Credits, licences and version

Program	SpectraMaster 1.0.0 — designed & developed by **Edrees F. Hery**. The analysis engine version is shown in **Help › About** and recorded in every session.
Qt / PySide6	Graphical interface; © The Qt Company, LGPL v3.
SpecUtils	Reading/writing of many formats (N42, PCF, SPE, CNF, SPC …); © Sandia National Laboratories, LGPL 2.1.
IAEA nuclear data	ENSDF decay data from the IAEA Nuclear Data Section: the *ndlab* database (gamma library; © IAEA, MIT licence) and the LiveChart of Nuclides decay schemes (coincidence table). Source: IAEA; the IAEA does not endorse SpectraMaster.
SandiaDecay	Decay data used to cross-check the library; © Sandia National Laboratories, LGPL 2.1.
LNHB / DDEP	Recommended decay data used to check and correct library lines.
NIST SRD 126	Mass attenuation coefficients for the self-attenuation correction.
NumPy, SciPy, openpyxl	Numerics, fitting and Excel files (BSD / MIT licences).

DataMaster, GammaVision and MAESTRO are trademarks of ORTEC / AMETEK; Genie-2000 of Mirion (Canberra); Aptec of its owners. They are named only to describe compatibility.

E. Checks against reference data

Every change of the analysis engine is checked against reference data with acceptance limits fixed **before** the run; a criterion that fails is reported as failed, never relaxed. The current state:

Check	Data	Result
Synthetic benchmark	HPGe and NaI spectra with known peaks (search, areas, identification, summing, backgrounds)	the large majority of criteria pass; the remaining failures are documented
IAEA 2002 test spectra	certified activities, Marinelli and pillbox geometries (Arnold et al., 2005)	all 7 criteria pass when the sample density is corrected (section 9.6); without it 4 of 7
Certified reference materials	IAEA-312, 375, 434, 444, 447, SL-2 (Ra-226, Ra-228, Th-228, K-40, Cs-137)	all 19 results within 20 % of the certificate; 6 of 8 criteria pass (Ra-228 and Cs-137 bias above 10 %)
Coincidence table	independent simulation of the decay cascades of 18 nuclides	237 of 237 probabilities down the cascade, 215 of 215 upward and 448 of 448 in either order agree
Aptec	results stored by Aptec in the laboratory archive	section 7.8

The protocols and results are in `docs/baseline` of the program's repository. These checks describe the behaviour of the program on these data; they are not a validation of a laboratory method and not a statement of compliance.



Designed & developed by

Edrees F. Hery